



B.Tech.

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

**CURRICULUM
AND
SYLLABI**

(Applicable from 2026 Admission onwards)

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY KOTTAYAM

Pala – 686635, KERALA, INDIA

Table of Contents

Programme Educational Objectives (PEOs)	5
Programme Outcomes (POs) and PSOs	6
Curriculum	7
Programme Structure	12
Detailed Syllabus	17
MAC111 Calculus and Linear Algebra [3-1-0-4]	17
CSC111 Computer Programming & Problem Solving [3-1-0-4]	17
ECC111 Digital Electronics [3-1-0-4]	18
ECC112 Network Theory [3-1-0-4]	18
HMC111 Communication Skills for the Digital Age [2-0-2-3]	19
CSL111 Computer Programming Lab [0-0-2-1]	20
ECL111 Digital Electronics Lab [0-0-2-1]	20
MAC124 Vector Calculus and Applications [3-1-0-4]	21
CSC121 Data Structures [3-1-0-4]	21
ECC121 Electronic Devices and Circuits [3-1-0-4]	22
HMC121 Personality Development and Soft Skills [3-0-0-3]	22
CSC122 Computer Organization and Architecture [3-1-0-4]	23
ECC122 Solid State Devices [3-0-0-3]	24
CSL121 Data Structures lab (C, C++) [0-0-2-1]	25
ECL121 Electronic Devices and Circuits Lab [0-0-2-1]	25
MAC211 Probability, Statistics and Random Processes [3-1-0-4]	25
CSC211 Computer Networks [2-1-0-3]	26
HMC211 Engineering Management and Entrepreneurship [2-0-0-2]	27
ECC211 Analog Electronics [3-0-0-3]	28
ECC212 Digital Design using Hardware Description Language [2-1-0-3]	28
ECC213 Linear Integrated Circuits [3-0-0-3]	29
ECC214 Signals and Systems [3-1-0-4]	29
ECL211 Analog Electronics & Linear Integrated Circuits Lab [0-0-2-1]	30
ECL212 Digital Design using Hardware Description Language Lab [0-0-2-1]	31
MAC221 Differential Equations and Transforms [3-1-0-4]	31
HMC221 Engineering Economics, Finance and Business Analytics [2-0-0-2]	32
ECC221 VLSI Design [3-0-0-3]	33
ECC222 Embedded Systems Design [3-0-0-3]	33
ECC223 Electromagnetic Theory [2-1-0-3]	34
ECC224 Analog Communication [2-1-0-3]	35
CAC221 Machine Learning and Data Analysis [2-1-0-3]	36
ECL221 VLSI Design Lab [0-0-2-1]	36
ECL222 Embedded Systems Lab [0-0-2-1]	37
ECC311 Digital Signal Processing [3-0-0-3]	37
ECC312 Control Systems [3-1-0-4]	38
ECC313 Digital Communication [3-0-0-3]	39
ECC314 Microwave Engineering [3-0-0-3]	39
ECL311 Digital Signal Processing Lab [0-0-2-1]	40
ECL312 Analog and Digital Communication Lab [0-0-2-1]	40
ECC321 Antenna Theory and Design [3-0-0-3]	41
ECC322 FPGA Based System Design [3-0-0-3]	41
ECC323 Wireless and Cellular Communication [3-0-0-3]	42
IOECXXX Design Thinking and Innovation [3-0-0-3]	42
ECL321 Antenna Design Lab [0-0-2-1]	43
ECL322 Makerspace/Innovation Lab [1-0-2-2]	44
ECE301 Advanced Digital Signal Processing [3-0-0-3]	44
ECE302 Statistical Signal Processing [3-0-0-3]	45

ECE303 Time Frequency Analysis [3-0-0-3]	46
ECE304 Chip Designing [3-0-0-3]	46
ECE305 VLSI Testing and Verification [3-0-0-3]	47
ECE306 CAD for VLSI [3-0-0-3]	47
ECE307 Micro-Electro-Mechanical-Systems [3-0-0-3]	48
ECE308 Hardware for AI/ML [3-0-0-3]	49
ECE309 Electronic Measurements and Instrumentation [3-0-0-3]	49
ECE310 Quantum Circuits and Quantum Chips [3-0-0-3]	50
ECE311 Low Power VLSI [3-0-0-3]	50
ECE312 Semiconductor Memory Devices and Circuits [3-0-0-3]	51
ECE313 Analog IC Design [3-0-0-3]	51
ECE314 Sensors and Actuators [3-0-0-3]	52
ECE315 Nanoelectronics [3-0-0-3]	53
ECE316 Advanced Communication Networks [3-0-0-3]	53
ECE317 Error Control Coding [3-0-0-3]	54
ECE318 Information Theory and Coding [3-0-0-3]	54
ECE319 Multimedia Communication [3-0-0-3]	55
ECE320 Digital Signal Processing for Communication [3-0-0-3]	56
ECE321 RF and Microwave Circuit Design [3-0-0-3]	56
ECE401 Digital Control Systems [3-0-0-3]	57
ECE402 Data Compression [3-0-0-3]	58
ECE403 Optical Wireless Communication [3-0-0-3]	58
ECE404 Adaptive Signal Processing [3-0-0-3]	59
ECE405 Neuromorphic Computing [3-0-0-3]	59
ECE406 Mixed Signal Design [3-0-0-3]	60
ECE407 Quantum Photonic Integrated Circuits [3-0-0-3]	60
ECE408 VLSI Signal Processing [3-0-0-3]	61
ECE409 Characterization and functionalization of nanomaterials for electronic applications [3-0-0-3]	62
ECE410 Nano Electro Mechanical Systems (NEMS) [3-0-0-3]	62
ECE411 Optoelectronics [3-0-0-3]	63
ECE412 Quantum Optics [3-0-0-3]	64
ECE413 Detection and Estimation [3-0-0-3]	65
ECE414 Advanced Wireless Communication Techniques [3-0-0-3]	65
ECE415 Optical Communication [3-0-0-3]	66
ECE416 Satellite Communication [3-0-0-3]	67
ECE417 5G Wireless System Design [3-0-0-3]	68
ECE418 MIMO Wireless Communication [3-0-0-3]	69
ECE419 Radar Engineering [3-0-0-3]	70
ECE420 Communication System Design [3-0-0-3]	70
ECE421 Ocean Optics [3-0-0-3]	71
ECE422 Pattern Recognition [3-0-0-3]	72
ECE423 Quantum Transformers [3-0-0-3]	73
ECE424 Quantum Machine Learning [3-0-0-3]	73
ECE425 Quantum Sensing [3-0-0-3]	74
ECO301 Speech and Audio Processing [3-0-0-3]	75
ECO302 Engineering Foundations of Quantum Technologies [3-0-0-3]	75
ECO303 Biomedical Signal Processing [3-0-0-3]	76
ECO304 Transform Theory [3-0-0-3]	77
ECO305 System-On-Chip Design [3-0-0-3]	77
ECO306 Hardware Security [3-0-0-3]	78
ECO307 Robotics and Automation [3-0-0-3]	78
ECO308 Internet of Things [3-0-0-3]	79
ECO309 Real Time Embedded Systems [3-0-0-3]	80
ECO310 Wireless Sensor Networks [3-0-0-3]	80
ECO311 Optimization Techniques for Communication and Machine Learning [3-0-0-3]	81
ECO312 Applied Linear Algebra for Signal Processing, Data analytics and Machine Learning [3-0-0-3]	82
ECO313 Digital Image Processing [3-0-0-3]	83

ECO314 Transformer Architectures [3-0-0-3]	83
ECO315 Tensor Computations for Data Science [3-0-0-3]	84
ECO401 Advanced Computer Architecture [3-0-0-3]	85
ECO402 Reconfigurable Computing [3-0-0-3]	85
ECO403 Computer Vision [3-0-0-3]	86
ECO404 Software Defined Radio [3-0-0-3]	86
ECO405 High Speed Networks [3-0-0-3]	87
ECO316 Natural Language Processing [3-0-0-3]	88
ECO406 Wireless Ad-hoc Networks [3-0-0-3]	89
ECP411 B.Tech Project Phase I [0-0-6-3]	89
ECP421 B.Tech Project Phase II [0-0-18-9]	89

Programme Educational Objectives (PEOs) of B.Tech. in ELECTRONICS AND COMMUNICATION ENGINEERING

After completion of this program successfully, the graduates will be able to

PEO1	Practice Electronics and Communication Engineering in a successful professional career and pursue lifelong learning.
PEO2	Demonstrate comprehensive knowledge of analytical foundations to Electronics and Communication Engineering in terms of founding principles of design, computing and communication.
PEO3	Demonstrate critical thinking and problem-solving abilities to handle the real-world problems by applying theoretical foundations and practical skills in different fields of Electronics and Communication Engineering.
PEO4	Recognize the place of Electronics and Communication in the ecosystem that enables Computing Technology
PEO5	Make the graduates responsible and sensitive to social, environmental and economic issues in their profession and inculcate the sense of ethics and professionalism in their approach.

Programme Outcomes (POs) and Programme Specific Outcomes (PSOs) of B.Tech. in ELECTRONICS AND COMMUNICATION ENGINEERING

After completion of B.Tech in Electronics and Communication Engineering, the Graduates will be able to

PO1	Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO3	Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions for complex problems
PO4	Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO5	Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PSO1	Possess sound theoretical and practical knowledge of Electronics and Communication Engineering.
PSO2	Apply the knowledge and skills to design and develop state-of-the-art hardware and software systems as well as products.

CURRICULUM

Total credits for completing B.Tech. in ELECTRONICS AND COMMUNICATION ENGINEERING is 160. (In addition, 10 mandatory Activity Credits must be earned; these are not counted towards the CGPA.)

COURSE CATEGORIES AND CREDIT REQUIREMENTS

The structure of the B.Tech. programme shall have the following course categories:

Sl. No.	Course Category	Number of Courses	Minimum Credits
1	Institute Core (IC)	17 (14 Theory + 3 Lab)	51
2	Programme Core (PC)	27 (18 Theory + 9 Lab)	68
3	Programme Electives (PE)	5	15
4	Open Electives (OE)	5	14
5	Projects (P)	2	12
Total credits (counted for CGPA)			160
6	Activity Credits (AC) – <i>mandatory, not counted for CGPA</i>	–	10

COURSE REQUIREMENTS. The effort to be put in by the student is indicated in the tables as follows:

L: Lecture (one unit is of 50 minutes) **T:** Tutorial (one unit is of 50 minutes) **P:** Practical (one unit is of one hour)

1. Institute Core (IC)

(a) Mathematics

Sl.	Code	Course Title	L	T	P	Credits
1	MAC111	Calculus and Linear Algebra	3	1	0	4
2	MAC124	Vector Calculus and Applications	3	1	0	4
3	MAC211	Probability, Statistics, and Random Processes	3	1	0	4
4	MAC221	Differential Equations and Transforms	3	1	0	4
Total			12	4	0	16

(b) Economics, Management & Entrepreneurship

Sl.	Code	Course Title	L	T	P	Credits
1	HMC211	Engg. Management, Entrepreneurship Development & Start-ups	2	0	0	2
2	HMC221	Engg. Economics, Finance & Business Analytics	2	0	0	2
Total			4	0	0	4

(c) Communication Skills and Personality Development

Sl.	Code	Course Title	L	T	P	Credits
1	HMC111	Communication Skills for the Digital Age	2	0	2	3
2	HMC121	Personality Development & Soft Skills	3	0	0	3
Total			5	1	0	6

(d) Engineering Foundation Courses

Sl.	Code	Course Title	L	T	P	Credits
1	CSC111	Computer Programming & Problem Solving	3	1	0	4
2	CSL111	Computer Programming Lab	0	0	2	1
3	CSC121	Data Structures	3	1	0	4
4	ECC121	Electronic Devices and Circuits	3	1	0	4
5	CSC122	Computer Organization and Architecture	3	1	0	4
6	CSL121	Data Structures Lab (C, C++)	0	0	2	1
7	ECL121	Electronic Devices and Circuits Lab	0	0	2	1
8	CSC211	Computer Networks	2	1	0	3
9	CAC221	Machine Learning and Data Analysis	2	1	0	3
Total			17	5	6	25

2. Programme Core (PC)

Sl.	Code	Course Title	L	T	P	Credits
1	ECC112	Network Theory	3	1	0	4
2	ECC111	Digital Electronics	3	1	0	4
3	ECL111	Digital Electronics Lab	0	0	2	1
4	ECC122	Solid State Devices	3	0	0	3
5	ECC211	Analog Electronics	3	0	0	3
6	ECC212	Digital Design using Hardware Description Language	2	1	0	3
7	ECC213	Linear Integrated Circuits	3	0	0	3
8	ECC214	Signals and Systems	3	1	0	4
9	ECL211	Analog Electronics and Linear Integrated Circuits Lab	0	0	2	1
10	ECL212	Digital Design using Hardware Description Language Lab	0	0	2	1
11	ECC221	VLSI Design	3	0	0	3
12	ECC222	Embedded Systems Design	3	0	0	3
13	ECC223	Electromagnetic Theory	2	1	0	3
14	ECC224	Analog Communication	2	1	0	3
15	ECL221	VLSI Design Lab	0	0	2	1
16	ECL222	Embedded Systems Lab	0	0	2	1
17	ECC311	Digital Signal Processing	3	0	0	3
18	ECC312	Control Systems	3	1	0	4
19	ECC313	Digital Communication	3	0	0	3
20	ECC314	Microwave Engineering	3	0	0	3
21	ECL311	Digital Signal Processing Lab	0	0	2	1
22	ECL312	Analog and Digital Communication Lab	0	0	2	1
23	ECC321	Antenna Theory and Design	3	0	0	3
24	ECC322	FPGA Based System Design	3	0	0	3
25	ECC323	Wireless and Cellular Communication	3	0	0	3
26	ECL321	Antenna Design Lab	0	0	2	1
27	ECL322	Makerspace/Innovation Lab	1	0	2	2
Total			52	7	18	68

3. Programme Electives (PE)

Sl.	Code	Course Title	L	T	P	Credits
1	ECE301	Advanced Digital Signal Processing	3	0	0	3
2	ECE302	Statistical Signal Processing	3	0	0	3
3	ECE303	Time frequency analysis	3	0	0	3
4	ECE304	Chip Designing	3	0	0	3
5	ECE305	VLSI Testing and Verification	3	0	0	3

6	ECE306	CAD for VLSI	3	0	0	3
7	ECE307	Micro-Electro-Mechanical-Systems (MEMS)	3	0	0	3
8	ECE308	Hardware for AI/ML	3	0	0	3
9	ECE309	Electronic Measurements and Instrumentation	3	0	0	3
10	ECE310	Quantum Circuits and Quantum Chips	3	0	0	3
11	ECE311	Low Power VLSI	3	0	0	3
12	ECE312	Semiconductor Memory Devices and Circuits	3	0	0	3
13	ECE313	Analog IC Design	3	0	0	3
14	ECE314	Sensors and Actuators	3	0	0	3
15	ECE315	Nanoelectronics	3	0	0	3
16	ECE316	Advanced Communication Networks	3	0	0	3
17	ECE317	Error Control Coding	3	0	0	3
18	ECE318	Information Theory and Coding	3	0	0	3
19	ECE319	Multimedia Communication	3	0	0	3
20	ECE320	Digital Signal Processing for Communication	3	0	0	3
21	ECE321	RF and Microwave Circuit Design	3	0	0	3
22	ECE401	Digital Control Systems	3	0	0	3
23	ECE402	Data Compression	3	0	0	3
24	ECE403	Optical Wireless Communication	3	0	0	3
25	ECE404	Adaptive Signal Processing	3	0	0	3
26	ECE405	Neuromorphic Computing	3	0	0	3
27	ECE406	Mixed Signal Design	3	0	0	3
28	ECE407	Quantum Photonic Integrated Circuits	3	0	0	3
29	ECE408	VLSI Signal Processing	3	0	0	3
30	ECE409	Characterization and functionalization of nanomaterials for electronic applications	3	0	0	3
31	ECE410	Nano Electro Mechanical Systems (NEMS)	3	0	0	3
32	ECE411	Optoelectronics	3	0	0	3
33	ECE412	Quantum Optics	3	0	0	3
34	ECE413	Detection and Estimation	3	0	0	3
35	ECE414	Advanced Wireless Communication Systems	3	0	0	3
36	ECE415	Optical Communication	3	0	0	3
37	ECE416	Satellite Communication	3	0	0	3
38	ECE417	5G Wireless System Design	3	0	0	3
39	ECE418	MIMO Wireless Communication	3	0	0	3
40	ECE419	Radar Engineering	3	0	0	3

41	ECE420	Communication System Design	3	0	0	3
42	ECE421	Ocean Optics	3	0	0	3
43	ECE422	Pattern Recognition	3	0	0	3
44	ECE423	Quantum Transformers	3	0	0	3
45	ECE424	Quantum Machine Learning	3	0	0	3
46	ECE425	Quantum Sensing	3	0	0	3

4. List of Open Electives (OE)

Courses offered by other departments or approved online platforms (e.g. NPTEL), subject to the maximum permitted under the B.Tech. Ordinances and Regulations. PE courses offered by the parent department are also available to its own students under this category.

Sl.	Code	Course Title	L	T	P	Credits
1	ECO301	Speech and Audio Processing	3	0	0	3
2	ECO302	Engineering Foundations of Quantum Technologies	3	0	0	3
3	ECO303	Biomedical Signal Processing	3	0	0	3
4	ECO304	Transform Theory	3	0	0	3
5	ECO305	System-On-Chip Design	3	0	0	3
6	ECO306	Hardware Security	3	0	0	3
7	ECO307	Robotics and Automation	3	0	0	3
8	ECO308	Internet of Things	3	0	0	3
9	ECO309	Real Time Embedded Systems	3	0	0	3
10	ECO310	Wireless Sensor Networks	3	0	0	3
11	ECO311	Optimization Techniques for Communication and Machine Learning	3	0	0	3
12	ECO312	Applied Linear Algebra for Signal Processing, Data Analytics and Machine Learning	3	0	0	3
13	ECO313	Digital Image Processing	3	0	0	3
14	ECO314	Transformer Architectures	3	0	0	3
15	ECO315	Tensor Computations for Data Science	3	0	0	3
16	ECO316	Natural Language Processing	3	0	0	3
17	ECO401	Advanced Computer Architecture	3	0	0	3
18	ECO402	Reconfigurable Computing	3	0	0	3
19	ECO403	Computer Vision	3	0	0	3
20	ECO404	Software Defined Radio	3	0	0	3
21	ECO405	High Speed Networks	3	0	0	3
22	ECO406	Wireless Ad-hoc Networks	3	0	0	3

5. Projects (P)

The major project represents the culmination of study towards the B.Tech. degree. It is offered in two parts – Project-1 and Project-2 – and gives students the opportunity to apply and extend the knowledge gained across the programme. The work may be analytical, simulation-based, hardware design, or a combination, in an emerging area of ELECTRONICS AND COMMUNICATION ENGINEERING, carried out under the supervision of a department

faculty member. Project work may be carried out individually or in groups (size decided by the department) and may build on an industrial problem identified during an internship.

6. Activity Credits (AC)

A minimum of 40 Activity Points must be earned to obtain the 10 credits required in the curriculum; these credits are not counted towards CGPA. The scheme of activity points, as approved by the Senate, is announced at the start of the first semester. A minimum of 20 points must be earned by the end of the 4th semester and 40 points by the end of the 7th semester.

PROGRAMME STRUCTURE

Semester-wise Course Structure (I–VIII)

Semester I						Semester II					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
MAC111	Calculus and Linear Algebra	3	1	0	4	MAC124	Vector Calculus and Applications	3	1	0	4
CSC111	Computer Programming & Problem Solving	3	1	0	4	CSC121	Data Structures	3	1	0	4
ECC111	Digital Electronics	3	1	0	4	ECC121	Electronic Devices and Circuits	3	1	0	4
ECC112	Network Theory	3	1	0	4	HMC121	Personality Development & Soft Skills	3	0	0	3
HMC111	Communication Skills for the Digital Age	2	0	2	3	CSC122	Computer Organization and Architecture	3	1	0	4
CSL111	Computer Programming Lab	0	0	2	1	ECC122	Solid State Devices	3	0	0	3
ECL111	Digital Electronics Lab	0	0	2	1	CSL121	Data Structures Lab (C, C++)	0	0	2	1
						ECL121	Electronic Devices and Circuits Lab	0	0	2	1
Total		14	5	4	21	Total		18	4	4	24
Cumulative Credits at the End of First Year: 45											
Semester III						Semester IV					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
MAC211	Probability, Statistics, and Random Processes	3	1	0	4	MAC221	Differential Equations and Transforms	3	1	0	4
CSC211	Computer Networks	2	1	0	3	HMC221	Engg. Economics, Finance & Business Analytics	2	0	0	2
HMC211	Engg. Management, Entrepreneurship Development & Start-ups	2	0	0	2	ECC221	VLSI Design	3	0	0	3
ECC211	Analog Electronics	3	0	0	3	ECC222	Embedded Systems Design	3	0	0	3
ECC212	Digital Design using Hardware Description Language	2	1	0	3	ECC223	Electromagnetic Theory	2	1	0	3
ECC213	Linear Integrated Circuits	3	0	0	3	ECC224	Analog Communication	2	1	0	3
ECC214	Signals and Systems	3	1	0	4	CAC221	Machine Learning and Data Analysis	2	1	0	3
ECL211	Analog Electronics & Linear Integrated Circuits Lab	0	0	2	1	ECL221	VLSI Design Lab	0	0	2	1
ECL212	Digital Design using Hardware Description Language Lab	0	0	2	1	ECL222	Embedded Systems Lab	0	0	2	1
Total		19	3	4	24	Total		17	4	4	23
Cumulative Credits at the End of Second Year: 92											
Semester V						Semester VI					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
ECC311	Digital Signal Processing	3	0	0	3	ECC321	Antenna Theory and Design	3	0	0	3
ECC312	Control Systems	3	1	0	4	ECC322	FPGA Based System Design	3	0	0	3
ECC313	Digital Communication	3	0	0	3	ECC323	Wireless and Cellular Communication	3	0	0	3
ECC314	Microwave Engineering	3	0	0	3	ECE3XX	EC Elective-3	3	0	0	3
ECE3XX	EC Elective-1	3	0	0	3	OE	Open Elective-2	3	0	0	3
ECE3XX	EC Elective-2	3	0	0	3	IOECXXX	Design Thinking & Innovation	3	0	0	3
OE	Open Elective-1	3	0	0	3	OE	Foreign Language (NPTEL)	2	0	0	2
ECL311	Digital Signal Processing Lab	0	0	2	1	ECL321	Antenna Design Lab	0	0	2	1
ECL312	Analog and Digital Communication Lab	0	0	2	1	ECL322	Makerspace/Innovation Lab	1	0	2	2
						Honours	Advanced course-1 for Hon.	3	0	0	3
Total		21	1	4	24	Total		21	0	4	23
Cumulative Credits at the End of Third Year: 139											
Semester VII						Semester VIII					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
ECE4XX	EC Elective-4	3	0	0	3	ECP421	B.Tech Project Phase II	0	0	18	9
ECE4XX	EC Elective-5	3	0	0	3	Honours	Hons. Project Phase 2	0	0	22	11
OE	Open Elective-5	3	0	0	3						
Honours	Advanced course-2 for Hon. (NPTEL)	3	0	0	3						
ECP411	B.Tech Project Phase I	0	0	6	3						
Honours	Hons. Project Phase- 1	0	0	6	3						
Total		19	0	12	12	Total					
Total Credits at the End of Fourth Year: 160 (excluding Activity Credits)											

Detailed Semester-wise Structure

Semester I

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC111	Calculus and Linear Algebra	3	1	0	4	IC
2	CSC111	Computer Programming & Problem Solving	3	1	0	4	IC
3	ECC111	Digital Electronics	3	1	0	4	PC
4	ECC112	Network Theory	3	1	0	4	PC
5	HMC111	Communication Skills for the Digital Age	2	0	2	3	IC
6	CSL111	Computer Programming Lab	0	0	2	1	IC
7	ECL111	Digital Electronics Lab	0	0	2	1	PC
Total						21	

Semester II

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC124	Vector Calculus and Applications	3	1	0	4	IC
2	CSC121	Data Structures	3	1	0	4	IC
3	ECC121	Electronic Devices and Circuits	3	1	0	4	IC
4	HMC121	Personality Development & Soft Skills	3	0	0	3	IC
5	CSC122	Computer Organization and Architecture	3	1	0	4	IC
6	ECC122	Solid State Devices	3	0	0	3	PC
7	CSL121	Data Structures lab (C, C++)	0	0	2	1	IC
8	ECL121	Electronic Devices and Circuits Lab	0	0	2	1	IC
Total						24	

Semester III

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC211	Probability, Statistics, and Random Processes	3	1	0	4	IC
2	CSC211	Computer Networks	2	1	0	3	IC
3	HMC211	Engg. Management, Entrepreneurship Development & Start-ups	2	0	0	2	IC
4	ECC211	Analog Electronics	3	0	0	3	PC
5	ECC212	Digital Design using Hardware Description Language	2	1	0	3	PC
6	ECC213	Linear Integrated Circuits	3	0	0	3	PC
7	ECC214	Signals and Systems	3	1	0	4	PC
8	ECL211	Analog Electronics & Linear Integrated Circuits Lab	0	0	2	1	PC
9	ECL212	Digital Design using Hardware Description Language Lab	0	0	2	1	PC
Total						24	

Semester IV

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC221	Differential Equations and Transforms	3	1	0	4	IC
2	HMC221	Engg. Economics, Finance & Business Analytics	2	0	0	2	IC
3	ECC221	VLSI Design	3	0	0	3	PC
4	ECC222	Embedded Systems Design	3	0	0	3	PC
5	ECC223	Electromagnetic Theory	2	1	0	3	PC
6	ECC224	Analog Communication	2	1	0	3	PC
7	CAC221	Machine Learning and Data Analysis	2	1	0	3	IC
8	ECL221	VLSI Design Lab	0	0	2	1	PC
9	ECL222	Embedded Systems Lab	0	0	2	1	PC
Total						23	

Semester V

Sl.	Code	Course Title	L	T	P	Credits	Category
1	ECC311	Digital Signal Processing	3	0	0	3	PC
2	ECC312	Control Systems	3	1	0	4	PC
3	ECC313	Digital Communication	3	0	0	3	PC
4	ECC314	Microwave Engineering	3	0	0	3	PC
5	ECE3XX	EC Elective-1	3	0	0	3	PE
6	ECE3XX	EC Elective-2	3	0	0	3	PE
7		Open Elective-1	3	0	0	3	OE
8	ECL311	Digital Signal Processing Lab	0	0	2	1	PC
9	ECL312	Analog and Digital Communication Lab	0	0	2	1	PC
Total						24	

Semester VI

Sl.	Code	Course Title	L	T	P	Credits	Category
1	ECC321	Antenna Theory and Design	3	0	0	3	PC
2	ECC322	FPGA Based System Design	3	0	0	3	PC
3	ECC323	Wireless and Cellular Communication	3	0	0	3	PC
4	ECE3XX	EC Elective-3	3	0	0	3	PE
5	OE	Open Elective-2	3	0	0	3	OE
6	IOECXXX	Design Thinking & Innovation	3	0	0	3	OE
7	OE	Foreign Language (NPTEL)				2	OE
8	ECL322	Makerspace/Innovation Lab	1	0	2	2	PC
9	ECL321	Antenna Design Lab	0	0	2	1	PC
10	Honours	Advanced course-1 for Hon.	3	0	0	3	
Total						23	

Semester VII

Sl.	Code	Course Title	L	T	P	Credits	Category
1	ECE4XX	EC Elective-4	3	0	0	3	PE
2	ECE4XX	EC Elective-5	3	0	0	3	PE
3	OE	Open Elective-5	3	0	0	3	OE
4	Honours	Advanced course-2 for Hon. (NPTEL)	3	0	0	3	
5	ECP411	B.Tech Project Phase I	0	0	6	3	P
6	Honours	Hons.Project Phase- 1	0	0	6	3	P
Total						12	

Semester VIII

Sl.	Code	Course Title	L	T	P	Credits	Category
1	ECP421	B.Tech Project Phase II	0	0	18	9	P
2	Honours	Hons. Project Phase 2	0	0	22	11	P
Total			0	0	18	9	

DETAILED SYLLABUS

MAC111 Calculus and Linear Algebra [3-1-0-4]

Course Objectives

- To study the basic topological properties of the real numbers.
- To develop knowledge of sequences of real numbers and their convergence.
- To study the notion of continuous functions and their properties.
- To gain an understanding of systems of linear equations.
- To introduce the fundamental concepts of vector spaces.
- To impart the basics of linear transformations, orthogonalization, basis, dimension, and eigenvalues.
- To provide the knowledge required to apply concepts of linear algebra in engineering applications.

Course Outcomes

- CO1: Understand the concepts of limits, continuity, differentiability, and classical theorems (Rolle's, Mean Value, Fundamental Theorem of Calculus) to model and solve real-world problems in science and engineering.
- CO2: Analyze convergence of sequences and series, and optimize multivariable functions using Hessian matrices to address practical problems in data modelling, optimization and artificial intelligence.
- CO3: Formulate and solve systems of linear equations using matrix methods, and interpret solutions in applied contexts such as network flows, resource allocation, and computational simulations.
- CO4: Apply vector space concepts, orthogonalization techniques, and inner product structures to design efficient representations in signal processing, machine learning, and numerical computation.
- CO5: Compute eigenvalues and eigenvectors, perform matrix diagonalization, and apply these techniques in stability analysis, principal component analysis, engineering design etc.

Syllabus

Calculus: Real Numbers, Properties of Real Numbers, Limit, Continuity, Discontinuity, Uniform continuity, Differentiability, Rolle's theorem, Mean Value theorem, Riemann Integration, fundamental theorem of calculus. Sequences and Series: Convergence and limit laws, Finite and infinite series, Sums of non-negative numbers, Absolute and conditional convergence of an infinite series, tests of convergence; Function of several variables: Limit, Continuity, Partial derivatives, Local maxima and local minima, Saddle point, Hessian Matrix.

Linear Algebra: System of linear equations, Row reduced echelon matrices, Rank of a matrix. Definition of a linear vector space and examples; linear independence of vectors, basis and dimension, Subspaces; Linear transformations, Matrix representation of Linear transformation, Inner product, norms, orthogonal basis, Gram-Schmidt orthogonalization process, Eigen vectors of a matrix and matrix diagonalization, Applications.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	2	2	2	2
CO2	3	2	2	2	2	2	2
CO3	3	3	2	2	2	2	2
CO4	3	2	2	2	2	2	2
CO5	3	2	2	2	2	2	2

Textbooks / References

1. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th Edition, Wiley, 2011.
2. T. M. Apostol, *Calculus*, Volume I, 2nd Edition, Wiley, 2007.
3. G. Strang, *Linear Algebra and Its Applications*, 5th Edition, Wellesley-Cambridge Press/SIAM, 2016.
4. K. Hoffman and R. Kunze, *Linear Algebra*, 2nd Edition, PHI, 2009.
5. E. Kreyszig, *Advanced Engineering Mathematics*, 10th Edition, Wiley, 2015.
6. Thomas and Finney, *Calculus*, 9th Edition, Addison Wesley Publishing, 1998.
7. Anton, Biven, *Calculus*, 11th Edition, Wiley, 2015
8. Gilbert Strang, *Linear Algebra and Learning from Data*, Wellesley-Cambridge Press, 2019.

CSC111 Computer Programming & Problem Solving [3-1-0-4]

Course Objectives

- To introduce problem solving techniques and their representation using algorithms, flowcharts, and pseudocode.
- To teach core concepts of C programming including data types, operators, control flow constructs, and input/output mechanisms.
- To familiarize students with arrays, character arrays, and user-defined functions with parameter passing mechanisms in C.
- To introduce the concepts of recursion and pointer-based programming in C.
- To familiarize students with derived data types such as typedef, enumerations, unions, and structures in C.

Course Outcomes

- CO1: Gain knowledge of problem solving techniques using algorithmic thinking, flowcharts, and pseudocode.
- CO2: Students learn to apply core C programming concepts including data types, control flow, arrays, functions, recursion, and pointers in problem solving.
- CO3: Students learn to use derived data types and structured programming constructs for developing efficient solutions in C.

Syllabus

Introduction to C Programming: Introduction to algorithm, flowchart, pseudocode. Introduction to C programming - Character set of C, basic data types, operators, scope of variables, formatted input and output using scanf and printf. Writing and compiling simple C programs.

Control Flow Constructs: Conditional statements, if-else and switch constructs, nested conditions. Entry tested loops, exit tested loops, nested loop programs.

Arrays and Functions: Single-dimensional arrays, multidimensional arrays, character arrays. User-defined functions, function declaration, definition, call, return types, actual and formal parameters, parameter passing by value.

Recursion and Pointers: Recursive problem definitions, base case and recursive case. Example programs: factorial using recursion, Fibonacci series, Printing patterns. Introduction to pointers, address and dereferencing, pass-by-reference using pointers.

Derived Data Types: Use of typedef for user-defined type names, enumeration, unions and structures in C. Defining and accessing union members and structure members.

Textbooks / References

1. Brian W. Kernighan and Dennis M. Ritchie, *The C Programming Language*, Second Edition, Prentice Hall, 1988.
2. E. Balagurusamy, *Programming in ANSI C*, Ninth Edition, McGraw Hill, 2024.
3. Paul Deitel and Harvey Deitel, *C: How to Program*, Ninth Edition, Pearson Education, 2022.
4. V. Rajaraman, *Computer Programming in C*, Second Edition, Prentice Hall India, 2004.
5. R. G. Dromey, *How to Solve It by Computer*, Pearson Education India, 2007.
6. Reema Thareja, *Computer Fundamentals and Programming in C*, Second Edition, Oxford University Press, 2016.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2				3	1
CO2	3	3		2		3	3
CO3	2	3		2		3	3

ECC111 Digital Electronics [3-1-0-4]

Course Objectives

- To develop a strong foundation in number systems, Boolean algebra, logic gates, and simplification techniques used in digital electronic systems.
- To enable students to analyze and design combinational and sequential logic circuits using standard digital building blocks such as multiplexers, counters, flip-flops, and shift registers.
- To introduce memory devices, programmable logic devices.

Course Outcomes

- CO1: Apply number system conversions, Boolean algebra, Karnaugh maps, and logic gate principles to analyze and simplify digital circuits.
- CO2: Design and evaluate combinational and sequential circuits including adders, multiplexers, counters, flip-flops, and shift registers with appropriate timing analysis.
- CO3: Explain the operation of memory and programmable logic devices and develop simple digital hardware modules.

Syllabus

Number Systems and Conversion (base- m to base- n), Binary Arithmetic, Representation of Negative Numbers: sign and magnitude, 2's complement, 1's complement forms, Arithmetic Operations with Signed Numbers, Hexadecimal Numbers, Octal Numbers, Binary Coded Decimal (BCD) Logic gates-NOT, AND, OR, XOR, XNOR, universal gates, timing diagrams. Boolean algebra-Boolean Expressions and Truth Tables, Basic Theorems, Commutative, Associative, and Distributive Laws, DeMorgan's Laws, Consensus Theorem

Combinational Logic Design Using Truth Table, Minterm and Maxterm Expansions, Incompletely Specified Functions, Don't-Care Conditions, Karnaugh Map, Karnaugh Map SOP Minimization,

Karnaugh Map POS Minimization, (Two, Three, Four and Five-variable Karnaugh maps), Determination of Prime Implicants, Determination of Minimum Expressions Using Essential Prime Implicants, Multi-Level Gate Circuits, Design of Multi-Level NAND and NOR Gate Circuits, Combinational Logic Circuits-Half and Full Adders, Parallel Binary Adders, Ripple Carry and Look-Ahead Carry Adders, Binary Subtractor, adder-subtractor (with overflow detection), Magnitude Comparators, Decoders, Encoders, Code Converters, Multiplexers, Demultiplexers.

Sequential Logic Circuits – Latches and Flip-Flops, SR Latch, Gated D Latch, Edge-Triggered D Flip-Flop, SR Flip-Flop, JK Flip-Flop, T Flip-Flop, Flip-Flops with Asynchronous Clear and Preset inputs, Master-Slave Flip-Flops, Excitation table, Flip-Flop conversions, concept of frequency division using Flip-Flops. Counters – Asynchronous Counters, Synchronous Counters, Up/Down Synchronous Counters, Design of Synchronous Counters. Shift Registers – data movements in shift registers. SISO, SIPO, PISO, PIPO shift registers. Design of sequential circuits – Finite State Machines, Moore model, Mealy model, design of sequence detector circuit. Programmable logic – PAL and PLA

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2				3	2
CO2	2	3	2			3	3
CO3		3				3	3

Textbooks / References

1. M. Morris Mano and Michael D. Ciletti, *Digital Design*, 6th Edition, Pearson Education, 2018.
2. Thomas L. Floyd, *Digital Fundamentals*, 11th Edition, Pearson Education, 2015.
3. Stephen Brown and Zvonko Vranesic, *Fundamentals of Digital Logic with Verilog Design*, 3rd Edition, McGraw-Hill Education, 2014.
4. John F. Wakerly, *Digital Design: Principles and Practices*, 4th Edition, Pearson Education, 2005.
5. Donald D. Givone, *Digital Principles and Design*, Tata McGraw-Hill Education, 2002.
6. Charles H. Roth Jr. and Larry L. Kinney, *Fundamentals of Logic Design*, 7th Edition, Cengage Learning, 2013.

ECC112 Network Theory [3-1-0-4]

Course Objectives

- Describe basic network concepts emphasizing source transformation, source shifting, mesh and nodal techniques to solve for resistance, voltage, current and power.
- Explain Thevenin's, Millman's, Superposition, Maximum Power transfer and Norton's Theorems and apply them in solving the problems related to Electrical Circuits.
- Explain the behavior of networks subjected to transient conditions.
- Use applications of Laplace transforms to network problems.
- Study two port network parameters like Z, Y, T and h and their inter-relationships and applications.
- Study of RLC Series and parallel tuned circuit.

Course Outcomes

- CO1: Determine currents and voltages using source transformation/source shifting/mesh/nodal analysis and reduce given network using star-delta transformation/source transformation/source shifting.
- CO2: Solve network problems by applying Superposition/Reciprocity/Thevenin's/Norton's/Maximum Power Transfer/Millman's Network Theorems and electrical laws to reduce circuit complexities and to arrive at feasible solutions.
- CO3: Calculate current and voltages for the given circuit under transient and steady-state conditions.

Syllabus

Charge, current, voltage and power, voltage and current sources, Ohm's Law, voltage and current laws, Kirchhoff's Current Law, Kirchhoff's Voltage Law, Series and Parallel Connected Sources, Resistors in Series and Parallel, Voltage and Current Division, Nodal Analysis, Mesh Analysis, Linearity and Superposition, Source Transformations, Thévenin and Norton Equivalent Circuits, Maximum Power Transfer, Millman's Theorem, Reciprocity Theorem, Delta-Wye Conversion.

Capacitors and inductors, Inductance and capacitance combinations, Basic RL and RC circuits, Source-Free RL and RC circuits, Driven RL and RC circuits, Natural and Forced Response, RLC circuit, Source-Free Parallel and Series RLC Circuits, Overdamped, Critical damped, Underdamped RLC Circuits. Sinusoidal steady-state analysis, Forced Response to Sinusoidal Functions, Phasor, Impedance and Admittance, Nodal and Mesh Analysis, Superposition, Source Transformations and Thévenin's Theorem, Phasor Diagrams

AC circuit power analysis, Instantaneous Power, Average Power, Effective (rms) values of Current and Voltage, Apparent Power and Power Factor, Complex Power. Magnetically coupled circuits, Mutual Inductance, Linear Transformer, Ideal Transformer, Damped Sinusoidal Forcing Function

Complex Frequency, Laplace Transform, Laplace Transforms of Simple Time Functions, Circuit analysis in the s-domain, Laplace Transforms of Simple Time Functions, Inverse Transform Techniques, Basic Theorems for the Laplace Transform, Initial-Value and Final-Value Theorems, circuit analysis in the s-domain

Frequency Response, Parallel Resonance, Bandwidth and High-Q Circuits, Series Resonance.

Two-port networks, Admittance Parameters, Impedance Parameters, Hybrid Parameters, Transmission Parameters

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3				3	1
CO2	3	3				3	1
CO3	3	3				3	1

Textbooks / References

1. M.E. Van Valkenberg, Network analysis, Prentice Hall of India
2. Hayt, William Hart, and Jack Ellsworth Kemmerly. "Engineering circuit analysis", Tata McGraw-Hill
3. Charles K Alexander and Mathew N O Sadiku, Fundamentals of Electric Circuits, Tata McGraw-Hill
4. Franklin F. Kuo, *Network Analysis and Synthesis*, Wiley
5. J. David Irwin, R. Mark Nelms, Basic Engineering Circuit Analysis, John Wiley, 8th edition
6. Norman Balabanian, Theodore A. Bickart *Linear Network Theory: Analysis, Properties, Design and Synthesis*, Matrix Publishers
7. Kumar, K. Suresh. *Electric circuits and networks*. Pearson Education India, 2008.

HMC111 Communication Skills for the Digital Age [2-0-2-3]

Course Objectives

- Understand the fundamentals of communication and technical communication.
- Develop effective listening, speaking, reading, and writing skills for academic and professional contexts.
- Apply grammatical structures and vocabulary appropriately in oral and written communication.
- Analyze communication barriers, audience needs, and communication styles in different contexts.
- Create professional documents, presentations, and technical content using appropriate language and format.
- Understand the impact of Artificial Intelligence and Digital Humanities on contemporary communication practices.
- Practice ethical, responsible, and inclusive communication in digital and multilingual environments.

Course Outcomes

- CO1: Apply effective listening, speaking, reading, and writing strategies in academic and professional contexts.
- CO2: Analyze communication barriers, audience needs, and communication styles in different contexts.
- CO3: Create professional documents, presentations, and technical content using appropriate language and format.
- CO4: Evaluate the role of AI and digital technologies in communication with ethical awareness.
- CO5: Demonstrate responsible and effective communication in multilingual and digital environments.

Syllabus

Fundamentals of Communication: Process and types of communication - Technical communication and its importance - Barriers to communication - Non-verbal communication - Communication in academic and profes-

sional contexts - Shifts in communication in the digital age

Listening and Speaking Skills: Listening process and active listening - Barriers to listening and strategies to overcome them - Fluency and accuracy in speech - Self-expression and tonal variations - Group discussion and interpersonal communication - Public speaking and presentation skills - Interviews and professional interaction - Responsible use of AI in communication

Reading, Writing and Digital Communication: Reading comprehension and critical reading - Elements of effective writing - Professional emails and formal letters - Technical reports and presentations - Introduction to academic writing - AI in multilingual communication and translation - Ethical and legal considerations in AI-mediated communication - Introduction to Digital Humanities and digital communication practices

Grammar and Vocabulary for Communication: Parts of speech and sentence structures - Tenses and Subject-Verb Agreement - Reported Speech and Active-Passive Voice - Common errors in English usage - Punctuation and mechanics of writing - Vocabulary enhancement in context

Practical / Language Laboratory Component: Listening to recorded talks and comprehension activities - Active listening exercises - Group discussion and interpersonal conversation - Pronunciation and tonal variation practice - Extempore speech and presentation activities - Mock interviews and persuasive speaking exercises - Speech assessment and peer feedback - Technology-assisted listening and speaking practice.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	2				1	
CO2		3			2		
CO3		2	2	2		1	1
CO4		2		2	3		
CO5		2		1	3	1	

Textbooks / References

1. Anderson, Paul V., and Kerry Surman. *Technical Communication: A Reader-Centered Approach*. Thomson/ Wadsworth, 2007.
2. Berry, David M. "Introduction: Understanding the Digital Humanities." In *Understanding Digital Humanities*. Palgrave Macmillan UK, 2012.
3. Davis, Marjorie T. "Assessing Technical Communication within Engineering Contexts Tutorial." *IEEE Transactions on Professional Communication*, 53(1), 2010.
4. Duin, Ann Hill, and Isabel Pedersen. *Augmentation Technologies and Artificial Intelligence in Technical Communication: Designing Ethical Futures*. Routledge, 2023.
5. Raman, Meenakshi, and Sangeeta Sharma. *Technical Communication: Principles and Practice*. Oxford University Press, 2004.
6. Verma, Shalini. *Technical Communication for Engineers*. Vikas Publishing House, 2015.
7. Yule, George. *Oxford Practice Grammar Advanced*. Oxford University Press, 2015.

CSL111 Computer Programming Lab [0-0-2-1]

Course Objectives

- To enable students to write, compile, and execute C programs systematically.
- To familiarize students with core C programming concepts including data types, operators, control flow constructs, and input/output mechanisms.
- To develop programming skills using arrays, character arrays, and user-defined functions with parameter passing mechanisms.
- To introduce recursive and pointer-based programming through hands-on implementation in C.
- To enable students to implement programs using derived data types such as typedef, enumerations, unions, and structures in C.

Course Outcomes

- CO1: Students learn to translate programming concepts into C programs using structured programming techniques.
- CO2: Students learn to develop recursive and pointer-based solutions for computational problems.
- CO3: Students learn to use C programs for solving real world problems.

Syllabus

Introduction to C Programming: Basic data types, operators, scope of variables, formatted input and output using scanf and printf. Writing and executing simple C programs.

Control Flow Constructs: Implementation of programs using conditional statements and loop constructs.

Arrays and Functions: Implementation of programs using arrays and character arrays. Programs involving user-defined functions with parameter passing by value.

Recursion and Pointers: Implementation of recursive programs. Implementation of programs using pointers and pass-by-reference mechanisms.

Derived Data Types: Implementation of programs using typedef, enumerations, unions, and structures in C. Defining and accessing union members and structure members.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2		2		3	2
CO2	3	3	2	2		3	3
CO3	2	3	2	2		3	3

Textbooks / References

1. Brian W. Kernighan and Dennis M. Ritchie, *The C Programming Language*, Second Edition, Prentice Hall, 1988.
2. E. Balagurusamy, *Programming in ANSI C*, Ninth Edition, McGraw Hill, 2024.
3. Paul Deitel and Harvey Deitel, *C: How to Program*, Ninth Edition, Pearson Education, 2022.
4. Ashok N. Kamthane, *Programming in C*, Third Edition, Pearson India, 2015.

ECL111 Digital Electronics Lab [0-0-2-1]

Course Objectives

- To design and verify the truth tables of basic gates
- To implement combinational circuits using standard ICs .
- To design and implement counters.
- To design and implement shift registers.

Course Outcomes

- CO1: Identify and verify the operation of logic gates, and design combinational circuits
- CO2: Explain the behaviour of flip-flops, and analyse their characteristic equations and timing.

- CO3: Design and implement synchronous and asynchronous sequential circuits, counters, using state diagrams and excitation tables.
- CO4: Configure and shift registers and apply them in serial data transfer applications.

Experiments

- Familiarization of Logic Gates
- Design of Combinational Logic Circuits
- Comparators, Adders, Code Converters
- Multiplexers, Demultiplexers
- Familiarization of Flip-Flops and Latches. SR, D, and JK
- Flip-Flops. Edge triggered and Master-Slave Flip-Flops
- Design of Sequential Logic Circuits
- Design of asynchronous counters
- Design of synchronous counters
- Shift registers.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	3		3	3	3
CO2	3	3	3		3	3	2
CO3	3	3	3		3	3	3
CO4	3	3	3		3	3	2

MAC124 Vector Calculus and Applications [3-1-0-4]

Course Objectives

- To understand the basic concepts of vector calculus in space.
- To learn how to compute line, surface, and multiple integrals.
- To study gradient, divergence, curl, and important integral theorems.
- To understand the Jacobian, Hessian, Taylor series, and Laplacian for multivariable functions.
- To learn the basics of discrete vector calculus and finite difference approximations.

Course Outcomes

- CO1: Compute arc length, line integrals, surface integrals, and apply Green's, Stokes', and Gauss' theorems.
- CO2: Find gradient, directional derivatives, divergence, and curl of vector fields.
- CO3: Use polar, cylindrical, and spherical coordinates in solving integrals.
- CO4: Compute and interpret Jacobian matrices, Hessian matrices, multivariable Taylor expansions, and the Laplacian operator.
- CO5: Apply finite difference methods to approximate gradients and the discrete Laplacian.

Syllabus

Parameterised curves in space, Arc length, Tangent and normal vectors, Line integral, Gradient, Directional derivatives, Tangent plane and normal vector, Vector field, Divergence, Curl.

Parameterised surface, Double and triple integrals, Change of variables formula, Applications to area and volume, Surface integral, Integral theorems: Green's

Theorem, Stokes' theorem, Gauss' divergence theorem. Coordinate system: Polar, spherical, cylindrical.

The Jacobian Matrix (and its geometric interpretation). The Hessian Matrix and Multivariable Taylor Series. The Laplacian Operator and its physical significance. Conservative Vector Fields, Path Independence.

Discrete Vector Calculus: Finite difference approximations for gradients and the discrete Laplacian.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	2	2	2	2
CO2	3	2	2	2	2	2	2
CO3	3	3	2	2	2	2	2
CO4	3	2	2	2	2	2	2
CO5	3	2	2	2	2	2	2

Textbooks / References

1. M. R. Spiegel, *Vector Analysis*, Schaum's Outline Series, McGraw-Hill, 1959.
2. E. Kreyszig, *Advanced Engineering Mathematics*, 10th Edition, Wiley, 2011.
3. J. Stewart, *Calculus: Early Transcendentals*, 8th Edition, Cengage Learning, 2015.
4. G. Strang, *Computational Science and Engineering*, Wellesley-Cambridge Press, 2007.
5. H. Anton, I. C. Bivens, and S. Davis, *Calculus: Early Transcendentals*, 12th Edition, John Wiley & Sons, 2021.

CSC121 Data Structures [3-1-0-4]

Course Objectives

- Define and describe simple data structures like arrays, linked lists, trees and graphs.
- Design and specify algorithms for searching and sorting, and those associated with the above data structures.
- Analyse simple algorithms, like sorting and searching using mathematical tools, like formulation and solving of recurrences.
- Analyse application problems and abstract them to formulate solutions involving data structures and algorithms.

Course Outcomes

- CO1: Students learn to define operations of data structures like arrays, linked lists, trees and graphs.
- CO2: Students learn to design and specify algorithms involving above types of data structures.
- CO3: Students learn to analyze simple algorithms and solve recurrences, asymptotic analysis and probabilistic analysis.
- CO4: Students learn to analyze application problems and abstract them to formulate solutions involving data structures and algorithms

Syllabus

Introduction- Algorithm Analysis, Finding Complexity. Fundamental data structures - List, Sorted Lists, Single and Double Linked Lists, Skip list. Dynamic Arrays, Circular Linked Lists, Applications of Lists in Browser History and Text Editors.

Stack & Queue with applications.

Binary Trees- Insertion and Deletion of nodes, Tree Traversals, Polish Notations, AVL Trees, Heaps, Priority Queues. Optimal binary search tree, Application problems on Optimal binary search Tree.

Sorting -Bubble, Selection, Insertion, Merge Sort, Quick Sort, Radix Sort, Heap sort. Searching. Hashing- Application problems on hashing.

Graphs- Representations, types and properties of graphs, Shortest path algorithms, Minimum Spanning Trees, BFS, DFS.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2				3	2
CO2	3	3	2	2		3	3
CO3	3	3	3	2		3	2
CO4	2	3	2	2		3	3

Textbooks / References

1. Clifford A Shaffer, Data Structures and Algorithm Analysis, Edition 3.2 (Java Version), 2011.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser. Data Structures And Algorithms In Java TM Sixth Edition, Wiley Publishers, 2014.
3. Mark Allen Weiss Data Structures And Algorithm Analysis In Java, Third Edition, 2012.
4. Robert L. Kruse, Data Structures And Program Design In C++, Pearson Education, Second Edition, 2006.
5. Ellis Horowitz, Fundamentals of Data Structures in C++, University Press, 2015.
6. Ajay Agarwal, Data Structure through C, A Complete Reference Guide, Cyber Tech Publications, 2005.

ECC121 Electronic Devices and Circuits [3-1-0-4]

Course Objectives

- To impart the basic concepts of semiconductor devices.
- To analyse the operation of fundamental electronic circuits.
- To design simple analog circuits using discrete and integrated components.

Course Outcomes

- CO1: Analyse PN-junction behaviour under various operating conditions.
- CO2: Design and analyse rectifier, clipper and clamper circuits.
- CO3: Design and analyse amplifier circuits using BJTs and MOSFETs.
- CO4: Simulate simple electronic circuits using SPICE-class tools.

Syllabus

Semiconductors: Intrinsic semiconductors, doped semiconductors, current flow in semiconductors, drift current, diffusion current, the pn junction, physical structure, semiconductor diode, diode equation, diode characteristics, static and dynamic resistances, diode equivalent models: ideal diode model, piecewise-linear model, constant voltage-drop model, diode applications: rectifiers, clipping and clamping circuits, zener diodes as regulators.

Bipolar Junction Transistors (BJTs): Introduction to transistors, transistor construction, operation, NPN and PNP transistors, transistor voltages and currents, transistor characteristics. Common-base and common-emitter transistor configurations, biasing BJTs: fixed-bias, emitter-stabilized bias, voltage-divider bias, dc bias with voltage feedback, emitter follower bias, current mirror, BJT as amplifiers, voltage gain, current gain, input resistance and output resistance using r_e model, BJT as a switch.

Field Effect Transistors: Introduction, MOSFET, device structure and physical operation, regions of operation, current-voltage characteristics, body effect, biasing of MOSFET.

Optoelectronic devices: photodiodes - current and voltage in an illuminated junction, photodetectors. Light Emitting Diode (LED).

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2				2	
CO2	3	3	2			2	1
CO3	3	3	2	2	3	2	
CO4	2	2	3		2	3	2

Textbooks / References

1. Sedra A. and Smith K. C., *Microelectronic Circuits*, Oxford University Press, Seventh Edition, 2015.
2. Robert L. Boylestad and Louis Nashelsky, *Electronic Devices and Circuit Theory*, Pearson Education, Eleventh Edition, 2015.
3. David A. Bell, *Electronic Devices and Circuits*, Oxford University Press, Fifth Edition, 2008.
4. Jacob Millman and Christos C. Halkias, *Electronic Devices and Circuits*, McGraw Hill, Fourth Edition, 2015.
5. Ben G. Streetman and Sanjay Kumar Banerjee, *Solid State Electronic Devices*, Seventh Edition, Pearson Education, 2016.
6. Donald A. Neamen and Dhrubus Biswas, *Semiconductor Physics and Devices*, Fourth Edition, McGraw Hill Education, 2012.
7. S. M. Sze and Kwok K. Ng, *Physics of Semiconductor Devices*, Third Edition, Wiley, 2018.

HMC121 Personality Development and Soft Skills [3-0-0-3]

Course Objectives

- To understand the significance of soft skills, professional etiquette, self-awareness, and ethical behaviour in personal and professional development
- To help students understand and explain individual behaviour and personality in interpersonal and professional contexts.
- To apply communication, leadership, teamwork, presentation, and interpersonal skills in academic and workplace contexts
- To create awareness regarding common psychological concerns and the importance of psychological wellbeing in academic and personal life.
- To analyze workplace situations and develop problem-solving, emotional intelligence, adaptability, and stress management skills for professional

- To familiarize students with strategies for emotional regulation, coping, and healthy personality development.

Course Outcomes

- CO1: Apply self-assessment and goal-setting techniques for personal and professional growth
- CO2: Understand one's own personality and that of others, appreciate individual differences, and adapt effectively to people and situations.
- CO3: Analyze workplace situations using communication, leadership, and emotional intelligence skills
- CO4: Develop the ability to cope with personal, academic, and professional challenges through an understanding of human behaviour.
- CO5: Demonstrate effective speaking, presentation, and interpersonal communication skills
- CO6: Evaluate professional challenges related to stress, ethics, teamwork, and work-life balance

Syllabus

Foundations of soft skills and personality development: Introduction to soft skills - Importance and aspects of soft skills - Self-esteem and self-awareness - Professional etiquette and social etiquette - Professional ethics and workplace behavior - Professional body language - SMART goals and personal goal setting - SWOT analysis for personality development.

From Campus to Corporate: Leadership skills and teamwork - Group discussion and meeting management - Adaptability and work ethics - Time management and stress management - Advanced speaking skills - Oral presentations, speeches, and debates - Combating nervousness and stage fear - Planning and preparation of presentations - Making effective presentations - Speeches for various occasions - Interviews and facing job interviews - Emotional intelligence and critical thinking - Work-life balance - Applied grammar for professional communication.

Understanding Personality and Behaviour: Personal-ity: Meaning and Assessment. Psychoanalytic and Neo Psychoanalytic Approach; Behavioural Approach; Cognitive Approach; Social- Cognitive Approach; Humanistic Approach; The Traits Approach; Models of healthy personality: the notion of the mature person, the self-actualizing personality; Indian perspective on personality.

Mental Health, Institutions, and Social Contexts: Mental health in social, cultural, and institutional contexts; campus and workplace mental health. Institutional violence, bullying, harassment, and discrimination based on caste, religion, gender, sexuality, disability, body image, and other social locations. Structural inequalities, stigma, inclusion, and access to mental health support. Student mental health and suicide prevention, including institutional accountability and recommendations of the Supreme Court's National Task Force on Student Suicides. Critical perspectives on positive psychology and mainstream mental health interventions; indigenous and community-based approaches to well being. Help-seeking,

collective care, peer support, and professional mental health services.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	1	2			2		
CO2		2			2		
CO3		3	2		2	1	
CO4		2			3		
CO5	1	2	2	1		1	
CO6		3	2		3		

Textbooks / References

1. Carnegie, Dale, *How to Win Friends and Influence People*, John Wiley & Sons, 2026.
2. Covey, Stephen R., *The 7 Habits of Highly Effective People*, Personal Finance Magazine, 2024.
3. Ghai, A., *Rethinking Disability in India*, Routledge, 2015.
4. Guru, G., and Sarukkai, S., *The Cracked Mirror: An Indian Debate on Experience and Theory*, Oxford University Press, 2012.
5. Goleman, Daniel, *Emotional Intelligence*, Bloomsbury Publishing, 2020.
6. Kumar, N., *Mental Health and Well-Being: An Indian Psychology Perspective*, Routledge, 2023.
7. Medden, Stephanie, et al., "From Public Speaking to Multimodal Communication: A Reevaluation of the Basic Communication Course," *Basic Communication Course Annual*, 37(1), 2025.
8. Patel, V., *Where There Is No Psychiatrist: A Mental Health Care Manual*, 2nd Edition, Royal College of Psychiatrists, 2014.
9. Patel, V., Saxena, S., Lund, C., Thornicroft, G., Baingana, F., Bolton, P., et al., "The Lancet Commission on Global Mental Health and Sustainable Development," *The Lancet*, 392(10157), 2018, pp. 1553–1598.
10. Pease, Barbara, and Allan Pease, *The Definitive Book of Body Language: The Hidden Meaning Behind People's Gestures and Expressions*, Bantam, 2008.
11. Rao, Manchanahalli Satyanarayana, *Soft Skills-Enhancing Employability: Connecting Campus with Corporate*, IK International Pvt. Ltd., 2010.
12. Tracy, Brian, *Eat That Frog!: 21 Great Ways to Stop Procrastinating and Get More Done in Less Time*, Berr-ett-Koehler Publishers, 2025.
13. Nevid, J. S., Rathus, S. A., and Greene, B., *Abnormal Psychology in a Changing World*, 11th Edition, Pearson, 2022.
14. Snyder, C. R., and Lopez, S. J. (Eds.), *Positive Psychology: The Scientific and Practical Explorations of Human Strengths*, 5th Edition, 2021.
15. Carr, A., *Positive Psychology: The Science of Happiness and Human Strengths*, 4th Edition, 2022.
16. Kabat-Zinn, J., *Mindfulness for Beginners: Reclaiming the Present Moment—and Your Life*, 2018.
17. Ryckman, R. M., *Theories of Personality*, 10th Edition, 2017.
18. Schultz, D. P., and Schultz, S. E., *Theories of Personality*, 12th Edition, 2020.

CSC122 Computer Organization and Architecture [3-1-0-4]

Course Objectives

- This course will introduce to students the fundamental concepts underlying modern computer organization and architecture.
- Students should be able to know the overall working of a computer.
- Students should be able to get a detailed understanding of the design principles involved in developing a computer.
- They should know the representation of data, how programs are represented, executed and how programs manipulate and operate on data.

- They should also be able to appreciate how the memory organization is done and how to organize memory for faster execution of programs.
- Students should also be able to appreciate the concepts in pipelining.

Course Outcomes

- CO1: Understand the basics of computer hardware and how software interacts with computer hardware.
- CO2: Analyse and evaluate the performance of computers.
- CO3: Understand basics of Instruction Set Architectures and their impact on processor design.
- CO4: Understand how computers represent and manipulate data.
- CO5: Understand how computers perform arithmetic operations, how they are optimized and made to run faster.
- CO6: Design a pipeline for consistent execution of instructions with minimum hazards.
- CO7: Understand how the memory management takes place in a computer system.
- CO8: Design a simple computer with hardware design including data format, instruction format, instruction set, addressing modes, bus structure, input/output, memory, Arithmetic/Logic Unit, control unit, and data, instruction and address flow.

Syllabus

Basic principles, hardware components Measuring performance: evaluating, comparing and summarizing performance.

Instructions: operations and operands of the computer hardware, representing instructions, making decision, supporting procedures, character manipulation, styles of addressing, starting a program.

Computer Arithmetic: signed and unsigned numbers, addition and subtraction, logical operations, constructing an ALU, multiplication and division, floating point representation and arithmetic, Parallelism and computer arithmetic.

The processor: building a data path, simple and multi-cycle implementations, microprogramming, exceptions, Pipelining, pipeline Data path and Control, Hazards in pipelined processors

Memory hierarchy: caches, cache performance, virtual memory, common framework for memory hierarchies

Input/output: I/O performance measures, types and characteristics of I/O devices, buses, interfaces in I/O devices, design of an I/O system, parallelism and I/O. Introduction to multicores and multiprocessors.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2				3	2
CO2	2	3	2			3	2
CO3	3	2				3	2
CO4	3	2				3	2
CO5	3	3	2	2		3	3
CO6	3	3	3	2		3	3
CO7	3	2	2	2		3	3
CO8	3	3	3	2		3	3

Textbooks / References

1. D. A. Patterson and J. L. Hennessy, Computer Organisation and Design: The Hardware/ Software Interface, Fourth Edition, Morgan Kaufman, 2009.
2. V. P. Heuring and H. F. Jordan, Computer System Design and Architecture, Prentice Hall, 2003.
3. Computer Architecture: A Quantitative Approach, Fifth Edition, Morgan Kaufman, 2011.
4. Carl Hamazher, Zvonko Vranesic and Safwat Zaky, Computer Organization, Fifth Edition, McGraw Hill, 2002.

ECC122 Solid State Devices [3-0-0-3]

Course Objectives

- To develop a fundamental understanding of semiconductor physics, carrier transport mechanisms, and energy band theory governing the operation of semiconductor materials and devices.
- To analyze the electrical characteristics and operating principles of semiconductor junction devices, including p-n junctions, Schottky diodes, and bipolar junction transistors under various biasing and dynamic conditions.
- To study the operation and scaling behavior of MOS-based devices, including MOS capacitors and MOSFETs, with emphasis on short-channel effects, device modeling, and performance limitations in modern integrated circuits.

Course Outcomes

- CO1: Analyze carrier concentration, transport phenomena, recombination-generation processes, and electrical properties of intrinsic and extrinsic semiconductors using semiconductor physics principles.
- CO2: Evaluate the behavior and characteristics of semiconductor junction devices and bipolar junction transistors under equilibrium, steady-state, switching, and high-frequency operating conditions.
- CO3: Interpret and analyze the operation of MOS capacitors and MOSFETs, including current-voltage characteristics, threshold control, short-channel effects, and non-ideal device phenomena relevant to VLSI and nanoelectronic applications.

Syllabus

Energy Bands and Charge Carriers in Semiconductors: Bonding Forces in Solids, Energy Bands, Direct and Indirect Semiconductors, Charge Carriers in Semiconductors: Electrons and Holes, Effective Mass, Intrinsic Material, Extrinsic Material, Carrier Concentrations: The Fermi Level, Electron and Hole Concentrations at Equilibrium, Temperature Dependence of Carrier Concentrations, Compensation and Space Charge Neutrality, Conductivity and Mobility, Drift and Resistance, Effects of Temperature and Doping on Mobility, High-Field Effects, The Hall Effect, Invariance of the Fermi Level at Equilibrium, Excess Carriers in Semiconductors: Optical absorption, Luminescence, Carrier lifetime and Photoconductivity: Diffusion of carriers.

Junctions: Fabrication of p-n Junctions, Equilibrium Conditions, Forward- and Reverse-Biased Junctions; Steady state conditions, Reverse-Bias Breakdown, Transient and A-C Conditions, Metal-Semiconductor Junctions: Schottky Barriers, Rectifying Contacts, Ohmic Contacts, Typical Schottky Barriers.

Bipolar Junction Transistors: Fundamentals of BJT Operation, Minority carrier distributions and terminal currents, Generalized biasing, Switching, Drift in the Base Region, Base Narrowing, Avalanche Breakdown, Injection Level; Thermal Effects, Base Resistance and Emitter Crowding, Kirk Effect, Frequency limitations of Transistors. MOS Capacitor, MOS Field-Effect Transistor, Operation, Current-Voltage Characteristics, Short Channel Effects, Control of Threshold Voltage, body effect, Subthreshold Characteristics, Equivalent Circuit for the MOSFET, Drain-Induced Barrier Lowering, Short Channel Effect and Narrow Width Effect, Gate-Induced Drain Leakage.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3		2	2	3	2
CO2	3	3	2	2	2	3	3
CO3	3	3	2	2	2	3	3

Textbooks / References

1. Ben G. Streetman and Sanjay Banerjee, *Solid State Electronic Devices*, 7th ed., Pearson Education India.
2. Donald A. Neamen, *Semiconductor Physics and Devices*, 4th ed., McGraw Hill Education.
3. S. M. Sze and Kwok K. Ng, *Physics of Semiconductor Devices*, 3rd ed., John Wiley & Sons.
4. Robert F. Pierret, *Semiconductor Device Fundamentals*, Pearson Education India.

CSL121 Data Structures lab (C, C++) [0-0-2-1]

Course Objectives

- Develop proficiency in implementing fundamental data structures and algorithms.
- Analyze the time and space complexity of data structure operations.
- Apply appropriate data structures to solve computational problems efficiently.
- Gain practical experience with tree, graph, sorting, searching, and hashing techniques.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Implement and evaluate linear and non-linear data structures.
- CO2: Analyze the performance of data structure operations and associated algorithms.
- CO3: Apply trees, graphs, hashing, sorting, and searching techniques to solve computational problems.
- CO4: Design efficient solutions using suitable data structures and justify their selection.

Syllabus

Linear Data Structures: Implementation and analysis of Lists, Sorted Lists, Singly Linked Lists, Doubly Linked Lists, Circular Linked Lists, Dynamic Arrays, and Skip Lists. Applications of lists in Browser History Management and Text Editors.

Stacks and Queues: Implementation using arrays and linked lists. Applications including Expression Evaluation, Celebrity Problem, Largest Rectangular Area in a Histogram, and Queue-based Scheduling Problems.

Trees: Binary Tree Operations, Insertion and Deletion of Nodes, Tree Traversals, Polish Notations, AVL Trees, Heaps, Priority Queues, Optimal Binary Search Trees, and Application Problems on Optimal Binary Search Trees.

Sorting and Searching: Implementation and performance analysis of Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort, Radix Sort, Heap Sort, Linear Search, and Binary Search.

Hashing: Hash Functions, Collision Resolution Techniques, and Application Problems on Hashing.

Graphs: Graph Representations, Types and Properties of Graphs, Breadth First Search (BFS), Depth First Search (DFS), Shortest Path Algorithms, and Minimum Spanning Tree Algorithms.

Mini Project / Case Study (optional): Design, implementation, analysis, and presentation of a data structure-based solution to a real-world problem.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2		2		3	2
CO2	2	3	2	2		3	2
CO3	3	3	2	2		3	3
CO4	3	3	3	2		3	3

Textbooks / References

1. Clifford A Shaffer, Data Structures and Algorithm Analysis, Edition 3.2 (Java Version), 2011.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser. Data Structures And Algorithms In Java TM Sixth Edition, Wiley Publishers, 2014.
3. Mark Allen Weiss Data Structures And Algorithm Analysis In Java, Third Edition, 2012.
4. Robert L. Kruse, Data Structures And Program Design In C++, Pearson Education, Second Edition, 2006.
5. Ellis Horowitz, Fundamentals of Data Structures in C++, University Press, 2015.
6. Ajay Agarwal, Data Structure through C, A Complete Reference Guide, Cyber Tech Publications, 2005.

ECL121 Electronic Devices and Circuits Lab [0-0-2-1]

Course Outcomes

- CO1: Identify and safely use basic electronic laboratory equipment, instruments, and electronic components for circuit implementation and testing.
- CO2: Analyze the characteristics of semiconductor devices such as PN junction diodes, Zener diodes, and bipolar junction transistors through experimental investigations.
- CO3: Design, implement, and evaluate diode-based circuits including clippers, clampers, and rectifiers for specified applications.
- CO4: Simulate and validate the behavior of diode and transistor circuits using SPICE-based simulation tools and compare the results with practical observations.

List of Experiments

- Familiarization of basic electronic lab equipment and components
- Diode characteristics
- Clippers
- Clampers
- Rectifiers
- Zener characteristics
- BJT input and output characteristics
- Simulation of diode and transistor circuits using SPICE

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	1	2	3	2			
CO2	3	3	2	2	3		
CO3		2		3	2	3	
CO4		2	3	2			2

MAC211 Probability, Statistics and Random Processes [3-1-0-4]

Course Objectives

- To expose the students to the modern theory of probability, concept of random variables and their expectations.

- To introduce various discrete and continuous distributions and concept of estimation theory, confidence interval.
- To develop the ability to perform statistical inference and data analysis using estimation, confidence intervals, and hypothesis testing techniques.
- To illustrate the concept of hypothesis testing, tests for means and variances, Goodness-of-fit tests.
- To introduce the concept of random processes, Markov chains, and Brownian Motion.
- To introduce stochastic modeling techniques and their applications in data-driven systems.

Course Outcomes

- CO1: Define and apply the concepts of probability and conditional probability.
- CO2: Define and illustrate discrete and continuous random variables, their probability mass functions and probability density functions.
- CO3: Understand the concepts and significance of estimation theory and hypothesis testing.
- CO4: Apply estimation and hypothesis testing techniques to construct confidence intervals for means and variances and conduct goodness-of-fit tests for statistical inference and data-driven decision making.
- CO5: Analyze random processes, Markov chains, and Brownian motion and explain their applications in stochastic modelling and computing systems.

Syllabus

Basics of Probability: Axiomatic construction of the theory of probability, independence, conditional probability, and basic formulas.

Random Variables and Distributions: Univariate, Bivariate and multivariate random variables, cumulative and marginal distribution functions, conditional and multivariate distributions, functions of random variables: sum, product, ratio, change of variables, mathematical expectations, moments, moment generating function, characteristic functions.

Special Probability Distributions: discrete and continuous distributions, Limit theorems, Binomial distribution, Geometric distribution, Poisson distribution, Normal distribution, Exponential distribution, Gamma distribution, Beta distribution, Central Limit Theorem, Tchebyshev's inequality, Law of Large Numbers.

Estimation Theory: Bias of estimates, confidence intervals, minimum variance unbiased estimation, Bayes' estimators, moment estimators, maximum likelihood estimators, Chi-square distribution, confidence intervals for parameters of normal distribution.

Hypothesis Testing: Tests for means and variances, hypothesis testing and confidence intervals, Bayes' decision rules, power of tests, Goodness-of-fit tests, Kolmogorov-Smirnov Goodness-of-fit test.

Random Processes: Definition and classification of random processes, discrete-time Markov chains, Poisson pro-

cess, continuous-time Markov chains, stationary processes, Gaussian process, Brownian motion.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	2	2	2	2
CO2	3	2	2	2	2	2	2
CO3	3	2	2	2	2	2	2
CO4	3	2	2	2	2	2	2
CO5	3	3	2	2	2	2	2

Textbooks / References

1. S. Ross, *Introduction to Probability and Statistics for Engineers and Scientists*, 3rd ed., Elsevier, 2004.
2. P. G. Hoel, S. C. Port, and C. J. Stone, *Introduction to Probability Theory*, Universal Book Stall, 2000.
3. S. M. Ross, *Introductory Statistics*, 2nd ed., Academic Press, 2009.
4. J. Medhi, *Stochastic Processes*, 3rd ed., New Age International, 2009.
5. V. K. Rohatgi and A. K. Saleh, *An Introduction to Probability and Statistics*, 3rd ed., Wiley Student Edition, 2006.
6. G. R. Grimmett and D. R. Stirzaker, *Probability and Random Processes*, Oxford Univ. Press, 2001.
7. W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1, 3rd ed., Wiley, 1968.
8. S. M. Ross, *Stochastic Processes*, 2nd ed., Wiley, 1996.
9. C. M. Grinstead and J. L. Snell, *Introduction to Probability*, 2nd ed., Universities Press India, 2009.
10. S. Ross, *A First Course in Probability*, 10th ed., Pearson Education, Delhi, 2018.
11. K. P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022.
12. G. Casella and R. L. Berger, *Statistical Inference*, Chapman and Hall/CRC, 2024.
13. S. Ross, *Introduction to Probability Models*, 13th ed., Elsevier, 2023.
14. R. V. Hogg, J. W. McKean, and A. T. Craig, *Introduction to Mathematical Statistics*, 8th ed., Pearson, 2019.

CSC211 Computer Networks [2-1-0-3]

Course Objectives

- To understand the principles, architecture, and operation of modern computer networks.
- To study networking protocols and services across different layers of the TCP/IP model.
- To understand routing, switching, and transport mechanisms in computer networks.
- To explore wireless networking technologies and emerging networking paradigms.
- To analyze network performance, reliability, and security mechanisms.
- To provide the foundation required for network design, administration, and advanced networking studies.

Course Outcomes

- CO1: Explain networking architectures, communication models, and protocol layering principles.
- CO2: Analyze data transmission, error control, flow control, and medium access mechanisms in communication networks.
- CO3: Apply Internet addressing, routing, and transport protocols for network communication and management.
- CO4: Explain the operation of common application-layer protocols and network services.
- CO5: Assess wireless networking technologies, network security threats, and emerging networking paradigms.

Syllabus

Introduction to Computer Networks: Evolution of computer networks, network components and network types, network topologies, layered network architecture, OSI reference model, TCP/IP protocol suite, Internet standards and RFCs.

Data Communication and Data Link Layer: Data transmission fundamentals, transmission media, encoding techniques, framing, error detection and correction, flow control, sliding window protocols, HDLC and PPP protocols, medium access control protocols, switching concepts.

Network Layer and Routing: Internet Protocol (IP), addressing and subnetting, ARP, ICMP, DHCP, CIDR and NAT, routing principles, distance vector and link state routing, RIP, OSPF and BGP, introduction to Software Defined Networking (SDN).

Transport and Application Layers: UDP and TCP, reliable data transfer, flow control, congestion control, Quality of Service (QoS), DNS, HTTP and HTTPS, electronic mail protocols, multimedia networking basics.

Wireless, Network Security and Emerging Networks: Wireless LAN and Wi-Fi standards, network security fundamentals, firewalls and intrusion detection concepts, Internet of Things (IoT) networking, edge computing, and future Internet architectures.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2				3	2
CO2	3	3	2	2		3	2
CO3	3	3	2	2		3	3
CO4	2	2		2		3	2
CO5	2	3	2	2	2	3	3

Textbooks / References

1. Kurose, J. F. and Ross, K. W., *Computer Networking: A Top-Down Approach*, 8th Edition, Pearson, 2021.
2. Tanenbaum, A. S. and Wetherall, D. J., *Computer Networks*, 6th Edition, Pearson, 2021.
3. Peterson, L. L. and Davie, B. S., *Computer Networks: A Systems Approach*, 6th Edition, Morgan Kaufmann, 2021.
4. Forouzan, B. A., *Data Communications and Networking*, 5th Edition, McGraw-Hill Education, 2013.
5. Comer, D. E., *Internetworking with TCP/IP: Principles, Protocols, and Architecture*, 6th Edition, Pearson, 2013.

HMC211 Engineering Management and Entrepreneurship [2-0-0-2]

Course Objectives

- To introduce the principles and practices of engineering management and enable students to understand managerial functions in technology-oriented organizations.
- To provide students with tools and techniques for planning, organizing, leading, and controlling engineering projects and resources.
- To develop an entrepreneurial mindset by exposing students to innovation, opportunity identification, startup ecosystems, and venture creation.
- To equip students with the knowledge required to evaluate business opportunities and prepare sustainable technology based business models and ventures.

Course Outcomes

- Understand the functions, responsibilities, and practices of managers in engineering and technology organizations.
- Apply basic management, project management, and decision-making tools for effective management of engineering projects and teams.
- Analyze entrepreneurial opportunities, innovation processes, and startup ecosystems for technology-driven ventures.
- Develop a preliminary business plan incorporating market, financial, legal, and sustainability considerations.

Syllabus

Engineering Management: Meaning, definition, scope and significance of management in engineering organizations; Evolution of management thought; Contributions of F.W. Taylor, Henry Fayol, Elton Mayo and Behavioral School; Functions of management; Levels of management; Managerial roles and skills; Strategic thinking; Management lessons from Indian ethos and philosophy; Teamwork, productivity, effectiveness and efficiency.

Planning and Organizing: Planning process, objectives, policies, strategies and decision making; Management by Objectives (MBO); Organizing principles; Organizational structures and design; Departmentation; Centralization and decentralization; Delegation of authority; Span of control; Staffing and human resource management; Team building; Communication and coordination in engineering organizations.

Leadership, Operations and Project Management: Leadership concepts, leadership styles and motivation theories; Directing and controlling functions; Operations management fundamentals; Productivity and quality management; Lean concepts and continuous improvement; Engineering project management; Project life cycle; Work Breakdown Structure (WBS); PERT and CPM techniques; Resource allocation; Cost estimation; Risk management; Project monitoring and control.

Entrepreneurship and Venture Creation: Entrepreneurship concepts and characteristics; Entrepreneurial competencies; Creativity, innovation and design thinking; Opportunity identification and feasibility analysis; Startup ecosystem in India; Technology entrepreneurship; Incubation and acceleration; Business Model Canvas; Customer discovery and value proposition; Marketing fundamentals for startups; Sources of finance including bootstrapping, angel investment, venture capital and crowdfunding; Intellectual Property Rights (IPR); Business planning; Sustainable and social entrepreneurship; Case studies of successful technology ventures.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	1	2			2		
CO2	2	3	2	1		1	1
CO3	2	3	2		2	1	2
CO4	2	3	2	1	3	1	2

Textbooks / References

1. P. K. Wright, *Engineering Management*, Addison-Wesley, 1998.
2. H. Koontz and H. Weihrich, *Essentials of Management*, Tata McGraw-Hill, 1998.
3. S. P. Robbins and M. A. Coulter, *Management*, 6th Edition, Pearson Education, 2004.
4. S. P. Robbins and M. A. Coulter, *Management*, 15th Edition, Pearson Education, 2022.
5. P. C. Tripathi and P. N. Reddy, *Principles of Management*, Tata McGraw-Hill, 2017.
6. Y. K. Bhushan, *Fundamentals of Business Organisation and Management*, Sultan Chand & Co., 2019.
7. R. D. Hisrich, M. P. Peters, and D. A. Shepherd, *Entrepreneurship*, McGraw-Hill Education, 2023.
8. C. B. Gupta and N. P. Srinivasan, *Entrepreneurship Development*, S. Chand Publishing, 2022.
9. A. Osterwalder and Y. Pigneur, *Business Model Generation*, Wiley, 2010.
10. T. H. Byers, R. C. Dorf, and A. J. Nelson, *Technology Ventures: From Idea to Enterprise*, McGraw-Hill Education, 2020.
11. P. F. Drucker, *Innovation and Entrepreneurship*, Harper Business, 2006.
12. E. Ries, *The Lean Startup*, Crown Business, 2011.

ECC211 Analog Electronics [3-0-0-3]

Course Objectives

- Understand small-signal models of BJT and MOSFET devices for circuit analysis.
- Design and analyze BJT and MOS amplifier configurations and analyse frequency response.
- Apply feedback theory to evaluate gain, bandwidth, and impedance transformations in amplifiers.
- Classify and compare output stages on efficiency and linearity.

Course Outcomes

- CO1: Model BJT and MOSFET devices using small-signal equivalent.
- CO2: Design single and multi-stage amplifiers and determine the frequency response of amplifiers.
- CO3: Identify feedback topology and analyse its effect on gain, bandwidth, and terminal impedances.
- CO4: Design oscillator circuits using the Barkhausen stability criterion.
- CO5: Compare power amplifier classes on efficiency, linearity.

Syllabus

Bipolar Junction Transistors—Small signal operation and Models: Collector current and transconductance, Base current and input resistance, Emitter current and input resistance, voltage gain, The hybrid Π model. BJT based amplifier—Single stage amplifier—AC analysis for CE fixed bias, voltage divider bias with and without bypassed capacitor. Multistage Amplifier—Darlington pair. MOSFETs—VI characteristics, transconductance, channel length modulation, velocity saturation, body effect. MOS Device models—Large Signal models, small-signal model, MOS Amplifiers—Common source, Source follower, Common gate, cascode stages and current mirrors, Differential amplifier—VI characteristics and voltage gain. Frequency Response—Bode's Rule, Association of poles with nodes—Miller's theorem. High frequency model of MOS transistors. Frequency response of CS amplifier: Low-frequency response and high-frequency response. Feedback—Elements of feedback systems, Loop gain, Properties of negative feedback, Amplifiers—sense and return mechanisms, Analysis of feedback circuits. Four types of feedback circuits: shunt-series, series-shunt, shunt-shunt, series-series, effect of feedback on input and output impedance. Oscillators: Barkhausen Criteria, Oscillator operation, FET based RC Phase shift oscillator, Wien bridge oscillator, LC Oscillators: Hartley Oscillator, Colpitts Oscillator and Crystal Oscillators. Output Stages and Power Amplifiers: Introduction, Classification of

output stages, Class A output stage, Class B output stage: Transfer Characteristics, Power Dissipation, Power Conversion efficiency, Class AB output stage, Class C tuned Amplifier.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	3			3	3
CO2	2	3	2	3		3	3
CO3	2	3				3	3
CO4	3	3	3			3	3

Textbooks / References

1. Adel S. Sedra and Kenneth C. Smith, *Microelectronic Circuits: Theory and Applications*, 6th ed., Oxford University Press.
2. Behzad Razavi, *Fundamentals of Microelectronics*, 2nd ed., John Wiley, 2015.
3. Robert L. Boylestad and Louis Nashelsky, *Electronic Devices and Circuit Theory*, 11th ed., Pearson Education.
4. J. Millman and C. C. Halkias, *Integrated Electronics*, 2nd ed., TMH, 2010.

ECC212 Digital Design using Hardware Description Language [2-1-0-3]

Course Objectives

- Design digital systems in a larger system design context.
- Understand the concepts of Verilog HDL.
- Implementation of digital systems using Verilog HDL.

Course Outcomes

- CO1: Construct combinational circuits, using discrete gates.
- CO2: RTL design of sequential circuits and perform timing analysis.
- CO3: Describe Verilog model for combinational and sequential circuits and corresponding testbench for verification.

Syllabus

Revision of basic Digital systems: Combinational Circuits, Sequential Circuits, Timing, set-up and hold time, Power Dissipation, Mealy and Moore Machines, Equivalent States and Reduction of State Tables. Current state of the field: IP Design, Design Methodology, System modeling, Application Domains. Digital System Design: Top-Down approach to design, Datapath, Control-path, Pipelining, Resource sharing. Introduction to HDL (VHDL/Verilog): (Use either VHDL or Verilog for RTL). Computer Aided Design, Hardware Description Languages, Verilog description of combinational circuits, Verilog modules and assignments, Modeling flip flops using an Always Block, Delays in Verilog, Compilation, Simulation and Synthesis of Verilog code, Verilog datatypes and operators, Verilog models for multiplexers, Modeling Registers using Always statements, Behavioral

and Structural Verilog.

Introduction to Programmable Logic Devices: Overview of Programmable Logic Devices, Simple Programmable Logic Devices (SPLDs), Programmable Logic Arrays (PLAs), Programmable Logic Devices (PLDs), Complex Programmable Logic Devices (CPLDs), Field Programmable Gate Arrays (FPGAs), Design flow for FPGAs.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	2	2	3	3
CO2	3	3	2	2	2	3	3
CO3	3	3	2	2	2	3	3

Textbooks / References

1. Charles Roth, Lizy K. John, and Byeong Kil Lee, *Digital System Design Using Verilog*.
2. Stephen Brown and Zvonko Vranesic, *Fundamentals of Digital Logic with Verilog Design*.
3. Peter J. Ashenden, *Digital Design: An Embedded Systems Approach Using Verilog*, Elsevier, 2010.

ECC213 Linear Integrated Circuits [3-0-0-3]

Course Objectives

- Understand the basic principles of operational amplifier and its working.
- Develop skills to analyze and design various circuits using operational amplifiers.
- Understand the operation and applications of timer ICs, PLLs and VCOs.
- Develop skills to analyse and design active filters using operational amplifiers.
- Understand the principles of operation of ADCs and DACs.

Course Outcomes

- CO1: Understand the basic concepts of amplification and the significance of operational amplifiers.
- CO2: Analyze and design op-amp-based circuits for linear and nonlinear applications.
- CO3: Design waveform generators and active filters using op-amps.
- CO4: Explain the operation and applications of ADC, DAC, timer, PLL and VCO ICs.
- CO5: Simulate models of electronic systems with linear ICs and study their steady-state and transient behaviour using SPICE solutions.

Syllabus

Review of fundamentals: Amplifiers, Need for amplification, signal amplification, voltage gain, current gain, power gain, Gain in dB, Circuit Model of typical voltage amplifier, effect of loading, linearity, distortion, frequency response, bandwidth.

Introduction to operational amplifiers: concept of negative

feedback and virtual short, difference amplifier, ideal operational amplifier, op-amp terminals, function and characteristics of ideal op-amp, differential and common-mode signals, inverting and non-inverting configurations, closed-loop gain, effect of finite open-loop gain, input and output resistances, weighted summer, voltage follower, difference amplifiers, instrumentation amplifier, integrators and differentiators, inverting configuration with general impedances.

Log and antilog amplifiers, analog multipliers, voltage comparators—Schmitt trigger, square wave generator, monostable multivibrator, triangular wave generator, oscillators—RC phase shift oscillator, Wien Bridge oscillator. Linear half-wave rectifiers, precision rectifiers, peak detectors.

DC performance: Input bias current, input offset current, input offset voltage. AC performance: bandwidth, slew rate.

Active filters: Butterworth first order and second order function for LPF, HPF, BPF and notch filters.

Data converters: Definitions and specifications, Digital-to-Analog Converters (DAC): Different types and comparison, Analog-to-Digital Converters (ADC): Different types and comparison.

Timer IC 555: Internal diagram, working, multivibrators with timer IC 555. Phase locked loops: Introduction, basic principles, phase detector, VCO, low pass filter.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3			3	2
CO2	3	2	2	3	3	3	3
CO3	3	2	3			3	2
CO4	2	3	3	3		3	2

Textbooks / References

1. Adel S. Sedra and Kenneth C. Smith, *Microelectronic Circuits*, 8th ed., Oxford University Press.
2. Sergio Franco, *Design with Operational Amplifiers & Analog Integrated Circuits*, McGraw Hill.
3. J. M. Fiore, *Operational Amplifiers and Linear Integrated Circuits*, Jaico Publishing House.
4. R. F. Coughlin and F. F. Driscoll, *Operational Amplifiers and Linear Integrated Circuits*, Pearson Education.
5. Ramakant A. Gayakwad, *Op-amps and Linear Integrated Circuits*, 4th ed., Pearson Education.

ECC214 Signals and Systems [3-1-0-4]

Course Objectives

- Understand the mathematical description of continuous and discrete time signals and systems.
- Analyze the signals in the time domain using convolution sum and Integral.
- Classify signals into different categories based on their properties.

- Analyze Linear Time Invariant (LTI) systems in time and transform domains

Course Outcomes

- CO1: Analyze the different types of signals and systems.
- CO2: Determine the linearity, causality, time-invariance and stability properties of continuous and discrete time systems.
- CO3: Represent continuous and discrete systems in time and frequency domain using different transforms
- CO4: Analyse the stability of Linear Time Invariant systems

Syllabus

Definition of signals and systems, Classification of signals. Basic Operations on signals: Amplitude scaling, addition, multiplication, differentiation, integration, time scaling, time shift and reflection. Elementary signals: Exponential, sinusoidal, step, impulse and ramp signals. Systems viewed as interconnection of operations, Properties of systems: Stability, memory, causality, invertibility, time invariance, linearity. Time domain representation for linear time-invariant (LTI) systems: Impulse response, convolution sum, convolution integral. Properties of the impulse response representation of LTI systems: Interconnections of LTI systems, Memoryless, Causal, Stable, Invertible systems, step response.

Fourier series representation of continuous-time periodic signals: Trigonometric Fourier series, Exponential Fourier Series, properties, Fourier series representation of discrete-time periodic signals, properties, Fourier series and LTI systems, filtering, Fourier Representation of aperiodic signals: Continuous-Time Fourier Transform, properties, Fourier Transform for periodic signals, Discrete-Time Fourier Transform, properties, magnitude-phase representation of the frequency response of LTI systems

Sampling, representation of a continuous-time signal by its samples: the sampling theorem, reconstruction of a signal from its samples using interpolation.

Laplace Transform, Region of Convergence for Laplace Transform, Inverse Laplace Transform, properties of Laplace Transform, analysis and characterization of LTI systems using the Laplace transform (causality, stability), unilateral Laplace Transform.

z-Transform, Region of Convergence for z-Transform, Inverse z-Transform, properties of z-Transform, analysis and characterization of LTI systems using the z-Transform (causality, stability), unilateral z-Transform.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2			2	3	2
CO2	3	3			2	3	2
CO3	3	3	1	1	2	3	3
CO4	3	3	1	1	2	3	2

Textbooks / References

1. Simon Haykin and Barry Van Veen, "Signals and Systems", 2nd Edition, 2008, Wiley India.
2. Michael Roberts, "Fundamentals of Signals & Systems", 2nd edition, Tata McGraw-Hill, 2010
3. Alan V Oppenheim, Alan S, Willsky and A Hamid Nawab, "Signals and Systems" Pearson Education Asia / PHI
4. B. P. Lathi, "Linear Systems and Signals", Oxford University Press, 2005

ECL211 Analog Electronics & Linear Integrated Circuits Lab [0-0-2-1]

Course Objectives

- To provide understanding of BJT and MOSFET device characteristics, biasing methods, and small-signal amplifier design through hands-on hardware experimentation.
- To design and analyse feedback oscillator circuits.
- To develop proficiency in designing Op-Amp based circuits.
- To familiarise students with waveform generation and timing techniques using the 555 timer.

Course Outcomes

- CO1: Analyse BJT and MOSFET biasing techniques and design amplifier circuits.
- CO2: Design and implement feedback oscillator circuits (Hartley/Colpitts) and explain the conditions for sustained oscillation in terms of frequency, phase shift, and loop gain.
- CO3: Design op-amp based linear circuits and its application in wave shaping circuits.
- CO4: Simulate current mirror and differential amplifier in circuit design simulators.

List of Experiments

- Study of biasing techniques—Fixed bias and voltage divider bias.
- Study of BJT Amplifier Configuration—Design and set up the BJT common emitter voltage amplifier.
- Study of MOSFET based Amplifier—Design and setup the Common Source amplifier.
- Study of Oscillators—Design and implement Hartley/Colpitts Oscillator.
- Op-amp based basic circuits—Design and implement inverting amplifier, non-inverting amplifier, adder, subtractor using op-amp.
- Op-amp Application—Design and implement Integrator and Differentiator circuits using Op-Amp.
- Wave shaping circuits—Design and implement Monostable and Astable Multivibrator using 555 Timer.
- Software experiments—Design and implement current mirror and differential amplifier.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	3	3		2	3	3
CO2	3	3	3		2	3	2
CO3	3	3	3		2	3	3
CO4	3	3	3	3		3	2

ECL212 Digital Design using Hardware Description Language Lab [0-0-2-1]

Course Objectives

This course will enable students to

- To understand and implement combinational and sequential digital circuits using Verilog HDL through RTL modeling techniques.
- To develop and verify digital hardware designs by creating effective testbenches and performing functional simulations.
- To design and analyze finite state machines (FSMs), sequence detectors, counters, and shift registers using hardware description languages.
- To implement and validate digital designs on FPGA platforms, gaining hands-on experience with modern digital design workflows.

Course Outcomes

Upon successful completion of the course, students will be able to:

- 1: Design and model combinational and sequential digital circuits using Verilog HDL at the RTL level.
- 2: Develop testbenches, perform simulation, and verify the functionality of digital designs using industry-standard EDA tools.
- 3: Design and implement finite state machines, sequence detectors, counters, and shift registers to solve digital system design problems.
- 4: Synthesize, implement, and validate digital designs on FPGA boards while analyzing hardware resource utilization and timing behavior.

Syllabus

For all experiments, architecture design, RTL design and functional verification using a test bench is required.

- (a) Introduction to HDL and Simulation Environment
- (b) RTL Design and Verification (using Test bench) of Basic Logic Gates using Verilog HDL
- (c) RTL Design and Functional Verification of Combinational Circuits
- RTL implementation and test bench for Half Adder, Full Adder, Half Subtractor, and Full Subtractor, Ripple Carry Adder.
- RTL Design and functional verification of Multiplexers and Decoders.
- Architecture and RTL design of 2:1, 4:1, 8:1 Multiplexer and 2-to-4/3-to-8 Decoder.
- Development of test benches and timing verification.
- RTL Design and functional verification of Arithmetic and Logic Unit (ALU).
- Familiarization of Blocking and Non-Blocking Assignments.
- Design and simulation of sequential logic circuits demonstrating behavioral differences.
- Timing and race-condition analysis.
- RTL Design and Verification of Flip-Flops and latches.
- Architecture and RTL design of SR, JK, D, and T Flip-Flops.
- Development of clock-driven test benches and timing verification.
- Design of Counters using HDL.
- RTL implementation of synchronous and asynchronous counters (up/down counters).
- Functional verification and waveform analysis using test benches.
- Design of Shift Registers and Sequence Detectors.
- RTL design of Serial-In Serial-Out (SISO), Serial-In Parallel-Out (SIPO), and sequence detector circuits.
- Verification of sequential behavior through simulation.

- Finite State Machine (FSM) Design and Verification.
- Architecture, RTL coding, and test bench development for Moore and Mealy FSMs.
- Simulation-based verification of state transitions and output.
- RTL Design of Sequence Detectors.
- RTL design of simple sequence detector circuits using Mealy or Moore machine.
- Verification of sequential behavior through simulation.
- Familiarization of AMD Xilinx board for implementation of the experiments.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	3	1	3	3
CO2	3	3	2	3	1	3	3
CO3	3	3	2	3	1	3	3
CO4	3	3	3	3	1	3	3

MAC221 Differential Equations and Transforms [3-1-0-4]

Course Objectives

- Develop a comprehensive understanding of Fourier Series, Fourier Transform, and Z-Transform concepts with their fundamental properties and applications.
- Provide knowledge of analytical techniques for solving ordinary and partial differential equations arising in engineering and scientific applications.
- Enable students to apply transform methods for analyzing continuous and discrete systems and solving mathematical models.
- Introduce the concepts of convergence, harmonic analysis, and solution techniques for differential equations using appropriate mathematical approaches.
- Improve students' mathematical modelling and problem-solving capabilities by applying transform techniques and differential equation methods to real-world applications.

Course Outcomes

- CO1: Apply Fourier Series concepts, convergence criteria, and harmonic analysis techniques for representing periodic functions.
- CO2: Utilize Fourier Transform and Z-Transform techniques to analyze signals, systems, and solve engineering problems.
- CO3: Solve ordinary differential equations using appropriate analytical methods including variation of parameters and special forms of equations.
- CO4: Formulate and solve partial differential equations using standard methods and Fourier series-based solution techniques.
- CO5: Apply transformation methods and differential equation models to analyze and solve problems in science and engineering disciplines.

Syllabus

Fourier Series: Dirichlet's conditions - General Fourier series - Odd and even functions - Half-range sine and

cosine series - Complex form of Fourier series - Parseval's identity - Harmonic analysis. Convergence of Fourier series, differentiation and integration of Fourier series.

Fourier Transform: Fourier Integral Theorem - Fourier transform pair - Sine and cosine transforms - Properties - Transform of elementary functions - Convolution theorem - Parseval's identity.

Z-transforms: Linearity Property, Damping rule, Shifting property, Multiplication by n , Initial and Final value theorem, Inverse Z-transforms, Convolution theorem, Evaluation of Inverse Z-transform, Application to difference equations.

Ordinary Differential Equations: Method of variation of parameters - Method of undetermined coefficients - Homogeneous equations of Euler's and Legendre's type - System of simultaneous linear differential equations with constant coefficients.

Partial Differential Equations: Formation - Solutions of first-order equations - Standard types and equations reducible to standard types - Singular solutions - Lagrange's linear equation - Integral surface passing through a given curve - Classification of partial differential equations - Solution of linear equations of higher order with constant coefficients - Linear non-homogeneous equations.

Fourier Series Solutions of Partial Differential Equations: Method of separation of variables - Solutions of one-dimensional wave equation and one-dimensional heat equation - Steady-state solution of two-dimensional heat equation - Fourier series solutions in Cartesian coordinates.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	2	2	2	2
CO2	3	2	2	2	2	2	2
CO3	3	2	2	2	2	2	2
CO4	3	2	2	2	2	2	2
CO5	3	3	2	2	2	2	2

Textbooks / References

1. C. Edwards and D. Penney, *Elementary Differential Equations with Boundary Value Problems*, 6th Edition, Pearson, 2003.
2. W. E. Boyce and R. C. DiPrima, *Elementary Differential Equations*, 7th Edition, John Wiley & Sons, 2002.
3. Erwin Kreyszig, *Advanced Engineering Mathematics*, 10th Edition, Wiley, 2015.
4. Tyn Myint-U and L. Debnath, *Linear Partial Differential Equations for Scientists and Engineers*, 4th Edition, Birkhäuser, 2007.
5. B. S. Grewal, *Higher Engineering Mathematics*, 44th Edition, Khanna Publishers, 2021.
6. K. A. Stroud and Dexter J. Booth, *Engineering Mathematics*, 7th Edition, Palgrave Macmillan, 2013.

HMC221 Engineering Economics, Finance and Business Analytics [2-0-0-2]

Course Objectives

- To introduce the fundamental concepts of engineering economics and their application in engineering decision-making.
- To provide students with knowledge of financial management principles, financial markets, and investment evaluation techniques.
- To familiarize students with business analytics concepts, data-driven decision making, and analytical tools used in modern organizations.
- To develop the ability to evaluate economic, financial, and business alternatives for effective managerial and entrepreneurial decision making.

Course Outcomes

- Understand the principles of engineering economics, cost analysis, and economic decision-making relevant to engineering and technology sectors.
- Apply financial concepts and techniques for budgeting, investment appraisal, and financial planning.
- Analyze business data using descriptive, predictive, and prescriptive analytics approaches to support managerial decisions.
- Evaluate business opportunities and organizational performance using economic, financial, and analytical tools.

Syllabus

Engineering Economics: Nature and scope of engineering economics; Economic decision making in engineering; Demand, supply and market equilibrium; Cost concepts and cost analysis; Fixed and variable costs; Break-even analysis; Time value of money; Simple and compound interest; Cash flow diagrams; Economic comparison of alternatives; Present worth, annual worth and future worth methods; Depreciation concepts and methods; Inflation and its impact on economic decisions; Replacement and maintenance analysis.

Financial Management and Financial Markets: Introduction to finance and financial management; Objectives and functions of financial management; Financial statements and financial analysis; Ratio analysis; Working capital management; Capital budgeting; Sources of finance; Cost of capital; Risk and return concepts; Financial planning and budgeting; Overview of financial markets and institutions; Equity, debt and mutual funds; Basics of investment management and portfolio concepts; Venture capital and startup financing.

Business Analytics Fundamentals: Introduction to business analytics; Role of analytics in business decision making; Types of analytics—descriptive, diagnostic, predictive and prescriptive analytics; Data collection and data visualization; Measures of central tendency and dispersion; Probability concepts; Hypothesis Testing, Correlation and

regression analysis; Forecasting techniques; Data-driven decision making; Introduction to spreadsheet-based analytics and business intelligence tools.

Applications of Analytics in Business and Engineering: Analytics for operations, marketing, finance and human resources; Key Performance Indicators (KPIs); Dashboard reporting; Decision trees and risk analysis; Data ethics and governance; Artificial Intelligence and analytics in business; Case studies in engineering economics, financial decision making and business analytics; Applications in startups, technology ventures and digital enterprises.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	3	2		2	1	
CO2	2	3	2	1		1	1
CO3	2	3	3	2		1	2
CO4	2	3	3	2	2	1	2

Textbooks / References

1. W. G. Sullivan, E. M. Wicks, and C. P. Koelling, *Engineering Economy*, McGraw-Hill Education, 2018.
2. L. Blank and A. Tarquin, *Engineering Economy*, Pearson Education, 2020.
3. J. A. White, K. E. Case, and D. B. Pratt, *Principles of Engineering Economic Analysis*, Wiley, 2012.
4. E. F. Brigham and J. F. Houston, *Fundamentals of Financial Management*, Cengage Learning, 2021.
5. I. M. Pandey, *Financial Management*, Vikas Publishing House, 2021.
6. E. F. Brigham and M. C. Ehrhardt, *Financial Management: Theory and Practice*, McGraw-Hill Education, 2023.
7. J. Evans, *Business Analytics*, Pearson Education, 2020.
8. S. C. Albright and W. L. Winston, *Business Analytics: Data Analysis and Decision Making*, Cengage Learning, 2019.
9. F. Provost and T. Fawcett, *Data Science for Business*, O'Reilly Media, 2013.
10. R. Sharda, D. Delen, and E. Turban, *Business Intelligence, Analytics, and Data Science*, Pearson Education, 2018.
11. U. D. Kumar and D. Kumar, *Business Analytics: The Science of Data-Driven Decision Making*, Wiley India, 2017.
12. C. R. Thomas and S. C. Maurice, *Managerial Economics*, Oxford University Press, 2020.
13. M. Hirschey, *Managerial Economics*, McGraw-Hill Education, 2022.
14. D. R. Anderson, D. J. Sweeney, T. A. Williams, J. D. Camm, and J. J. Cochran, *Statistics for Business and Economics*, Pearson Education, 2020.
15. T. H. Davenport and J. G. Harris, *Competing on Analytics*, Harvard Business Review Press, 2017.

ECC221 VLSI Design [3-0-0-3]

Course Objectives

- Study the fundamentals of CMOS circuits and its characteristics.
- Learn the design and realization of combinational & sequential digital circuits using CMOS.
- Architectural choices and performance trade-offs involved in designing and realizing the circuits in CMOS technology.

Course Outcomes

- CO1: Be able to create models of moderately sized CMOS circuits that realize specified digital functions.
- CO2: Have an understanding of the characteristics of CMOS circuit construction and the comparison between different state-of-the-art CMOS technologies and processes.

- CO3: Be able to complete a significant VLSI design project having a set of objective criteria and design constraints.

Syllabus

Introduction to IC Technology—MOS, PMOS, NMOS, CMOS & BiCMOS Technologies. VLSI Design Flow, MOS Layers, Design Rules and Layout, CMOS Design rules for wires, Contacts and Transistors. Layout, Scaling of MOS circuits. CMOS logic: PMOS, NMOS and CMOS, MOS transistors as a switch, CMOS inverter, Power Consumption: Static and Dynamic. Designing Combinational Logic gates in CMOS: Static CMOS Design, Complementary CMOS, Ratioed Logic, Pass transistor logic, Dynamic CMOS design (Precharge and Evaluation), Implementation of gates and combinational circuits using CMOS, Transistor Sizing, Logical Effort. Designing Sequential Logic Circuits: Timing Metrics for sequential Circuits, Setup time, hold time, propagation delay, Latches vs. Registers, Static Timing Analysis, Classification of Memory Elements. Datapath subsystem design: Adders, ALU, multipliers. Implementation Strategies and testing for Digital ICs: Full Custom, Semi-custom, Gate Array based design approaches, Programmable Logic Devices, Field Programmable Gate Arrays, Standard Cell based design. Design for Testability: Ad Hoc Testing, Scan Design, BIST, IDDQ Testing, Boundary Scan.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	2	3	3	3
CO2	3	2	2	2	2	3	3
CO3	3	3	3	3	3	3	3

Textbooks / References

1. Jan M. Rabaey, Anantha Chandrakasan, and Borivoje Nikolić, *Digital Integrated Circuits: A Design Perspective*.
2. Michael John Sebastian Smith, *Application Specific Integrated Circuits*.
3. John P. Uyemura, *CMOS Logic Circuit Design*, Springer, 2007.

ECC222 Embedded Systems Design [3-0-0-3]

Course Objectives

- Understand the evolution of embedded computing platforms from microprocessor-based systems to modern ARM-based embedded systems.
- Analyze architectural concepts including processor organization, memory systems, interfacing techniques, and embedded peripherals.
- Apply ARM architecture and programming concepts for embedded system development.
- Design embedded solutions using communication interfaces, sensors, actuators, and firmware development methodologies.
- Evaluate embedded system design choices considering performance, power, cost, reliability, and real-time requirements.

Course Outcomes

- CO1: Explain the architecture and evolution of embedded computing platforms including 8086, 8051, and ARM processors.

- CO2: Analyze processor organizations, memory architectures, and interfacing mechanisms used in embedded systems.
- CO3: Develop ARM assembly and Embedded C programs for solving engineering problems.
- CO4: Design embedded systems using peripherals, communication interfaces, sensors, and actuators.
- CO5: Apply embedded software design methodologies and real-time concepts in embedded applications.

Syllabus

Introduction to Embedded Systems, characteristics and classification of embedded systems, design metrics and constraints, hardware-software partitioning, embedded hardware and firmware, evolution of embedded computing platforms. Processor architecture concepts: CISC versus RISC, Von Neumann versus Harvard Architecture, Little Endian versus Big Endian organization, performance, power and cost considerations in embedded design. 8086-Based Embedded Systems: Overview of 8086 architecture, register organization, memory segmentation, memory banking, bus architecture and system organization. Memory and I/O interfacing concepts: Memory-mapped I/O and I/O-mapped I/O. Role of peripheral controllers in embedded systems: 8255 Programmable Peripheral Interface (PPI), 8253/8254 Programmable Interval Timer, 8259 Programmable Interrupt Controller (PIC), 8237 DMA Controller and 8288 Bus Controller. 8051-Based Embedded Systems: Need for microcontrollers, 8051 architecture overview, memory organization, timers/counters, serial communication, interrupt structure, GPIO ports and embedded control applications. Comparison of microprocessor-based and microcontroller-based embedded systems.

Introduction to ARM architecture and RISC philosophy. ARM7TDMI-S architecture: Programmer's model, register organization, pipeline concept, processor operating modes, exception and interrupt handling, ARM and Thumb instruction sets, memory organization, stack operation and debugging support. Comparison of ARM architecture with conventional microprocessor and microcontroller architectures.

ARM assembly language fundamentals: Data transfer instructions, arithmetic and logical instructions, branch and control instructions, addressing modes and subroutines. Embedded C programming concepts. Firmware development process, cross-compilation, linking and loading, debugging and testing. Introduction to CMSIS and embedded software development tools.

Embedded system building blocks: Memory systems, sensors and actuators, ADC and DAC. Communication interfaces: UART, SPI, I²C and CAN. Introduction to Real-Time Embedded Systems: Tasks and processes, scheduling concepts, synchronization concepts, reliability and fault tolerance. Case studies in industrial, automotive, consumer and IoT embedded applications.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2				2	1
CO2	3	3	2	2		2	1
CO3	2	3	3	2		3	2
CO4	2	2	3	3	2	3	2
CO5	2	2	2	3	2	2	3

Textbooks / References

1. Marilyn Wolf, *Computers as Components: Principles of Embedded Computing System Design*, 4th ed., Morgan Kaufmann, 2017.
2. Raj Kamal, *Embedded Systems: Architecture, Programming and Design*, 3rd ed., McGraw-Hill Education, 2017.
3. A. K. Ray and K. M. Bhurchandi, *Advanced Microprocessors and Peripherals*, 3rd ed., Tata McGraw-Hill, 2012.
4. Douglas V. Hall and S. S. S. P. Rao, *Microprocessors and Interfacing*, 3rd ed., McGraw-Hill Education, 2012.
5. Kenneth J. Ayala, *The 8051 Microcontroller and Embedded Systems*, Cengage Learning, 2010.
6. Steve Furber, *ARM System Architecture*, Addison-Wesley, 1996.
7. William Hohl and Christopher Hinds, *ARM Assembly Language: Fundamentals and Techniques*, 2nd ed., CRC Press, 2014.
8. Joseph Yiu, *The Definitive Guide to ARM Cortex Processors*, Newnes (Elsevier), Latest Edition.

ECC223 Electromagnetic Theory [2-1-0-3]

Course Objectives

- To develop a strong foundation in electrostatic and magnetostatic field theory through the study of Coulomb's law, Gauss's law, magnetic field laws, and boundary conditions.
- To analyze electromagnetic phenomena using vector calculus, Maxwell's equations, and electromagnetic potentials for solving field and wave propagation problems in different media.
- To apply the principles of electromagnetic fields and waves to understand energy transfer, induction, transmission characteristics, and wave propagation.

Course Outcomes

- CO1: Analyze electric and magnetic field distributions using Coulomb's law, Gauss's law, Biot-Savart law, Ampere's law, and vector calculus techniques.
- CO2: Solve electrostatic and magnetostatic boundary-value problems using potential functions, Poisson's and Laplace's equations, and evaluate inductance, magnetic circuits, and electromagnetic forces.
- CO3: Interpret and apply Maxwell's equations to electromagnetic wave propagation problems.

Syllabus

Coulomb's Law, Electric Field Intensity and Flux density: Experimental law of Coulomb, Electric field intensity, Field due to continuous volume charge distribution, Field of a line charge, Field due to sheet of charge, Electric flux density. Gauss's law and Divergence: Gauss's law, Application of Gauss's law to volume charge, Point (differential) form of Gauss's law, Divergence. Maxwell's First equation (Electrostatics), Vector Operator ∇ and divergence theorem. Energy, Potential and Conductors: Energy expended or work done in moving a point charge in an electric field, The line integral, Definition of potential difference and potential, The potential field of point charge, Potential gradient. Poisson's and Laplace's Equations: Derivation of Poisson's and Laplace's Equations, Uniqueness theorem, Examples of the solution of Laplace's equation. Current and Current density, Continuity of current. Steady Magnetic Field: Biot-Savart Law, Ampere's circuital law, Stokes' theorem, Magnetic flux and magnetic flux density, Basic concepts of Scalar and Vector Magnetic Potentials. Magnetic Forces: Force on a moving charge, differential current elements, Force between differential current elements. Magnetic boundary conditions. Faraday's law of Electromagnetic Induction—Integral form and Point form. Maxwell's equations: Continuity equation, Inconsistency of Ampere's law with continuity equation, displacement current, Conduction current, Derivation of Maxwell's equations in point form and integral form, Maxwell's equations for different media. Plane wave, Uniform plane wave, Derivation of plane wave equations from Maxwell's equations, Solution of wave equation for perfect dielectric, Relation between E and H, Wave propagation in free space, Solution of wave equation for sinusoidal excitation, wave propagation in any conducting media (γ , α , β , η) and good conductors, Skin effect or Depth of penetration, Poynting's theorem and wave power.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3		2	2	3	2
CO2	3	3	2	2	2	3	2
CO3	3	3	2	2	2	3	3

Textbooks / References

1. W. H. Hayt and J. A. Buck, *Engineering Electromagnetics*, 8th ed., Tata McGraw Hill.
2. Matthew N. O. Sadiku, *Elements of Electromagnetics*, 4th ed., Oxford University Press.
3. E. C. Jordan and K. G. Balmain, *Electromagnetic Waves and Radiating Systems*, 2nd ed., PHI.
4. Joseph Edminister, *Electromagnetics*, Schaum Outline Series, McGraw Hill.

ECC224 Analog Communication [2-1-0-3]

Course Objectives

- Understand and analyse concepts of Analog Modulation schemes.
- Understand and analyse concepts of digitization of signals: sampling, quantizing and encoding.
- Evolve the concept of SNR in the presence of channel induced noise and study Demodulation of analog modulated signals.
- Evolve the concept of quantization noise for sampled and encoded signals and study the concepts of reconstruction from these samples at a receiver.

Course Outcomes

- CO1: Analyze and compute performance of AM and FM modulation in the presence of noise at the receiver.
- CO2: Analyze and compute performance of digital formatting processes with quantization noise.
- CO3: Multiplex digitally formatted signals at Transmitter and demultiplex the signals and reconstruct digitally formatted signals at the receiver.

Syllabus

Amplitude Modulation: Introduction, Amplitude Modulation: Time & Frequency Domain description, switching modulator, envelope detector. Double Side Band Suppressed Carrier Modulation: Time and Frequency Domain description, Ring modulator, Coherent detection, Costas Receiver, Quadrature Carrier Multiplexing, SSB Modulation, VSB Modulation, Frequency Translation, Frequency-Division Multiplexing. Pulse Amplitude Modulation, Pulse Position Modulation, Pulse Width Modulation.

Angle Modulation: Basic definitions, Frequency Modulation, Narrow Band FM, Wide Band FM, Transmission bandwidth of FM Signals, Generation of FM Signals, Demodulation of FM Signals, FM Stereo Multiplexing, Phase-Locked Loop: Nonlinear model of PLL, Linear model of PLL, Nonlinear Effects in FM Systems. Homodyne and Superheterodyne Receiver concepts.

Noise Equivalent Bandwidth, Noise in Analog Modulation: Introduction, Receiver Model, Noise in DSB-SC receivers, Noise in AM receivers, Threshold effect, Noise in FM receivers, Pre-emphasis and De-emphasis in FM. Random Processes, Mean, Correlation, and Covariance Functions, Transmission of a Random Process Through a Linear Filter, Power Spectral Density, Gaussian Process.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	3		3	
CO2	3	3	3	3		3	
CO3	3	3	3	2		3	

Textbooks / References

1. Simon Haykin and Moher, *Communication Systems*, 5th ed., John Wiley, India.
2. B. P. Lathi, *Modern Digital and Analog Communication Systems*, 4th ed., Oxford University Press.
3. Rodger E. Ziemer and William H. Tranter, *Principles of Communications*, John Wiley & Sons, 2014.

4. Simon Haykin, *An Introduction to Analog and Digital Communication*, John Wiley India.
5. H. Taub and D. L. Schilling, *Principles of Communication Systems*.
6. John G. Proakis and Masoud Salehi, *Fundamentals of Communication Systems*, Pearson Education India, 2007.
7. Harold P. E. Stern, Samy and A. Mahmoud, *Communication Systems*, Pearson Edition.

CAC221 Machine Learning and Data Analysis [2-1-0-3]

Course Objectives

- To introduce the mathematical and statistical foundations required for machine learning and data analysis.
- To develop skills in exploratory data analysis, preprocessing, and visualization techniques.
- To provide knowledge of supervised, unsupervised, and reinforcement learning methods.
- To enable students to design, implement, and evaluate machine learning models.
- To develop analytical and problem-solving skills for intelligent data-driven applications.

Course Outcomes

- CO1: Understand the mathematical and statistical foundations of machine learning.
- CO2: Perform exploratory data analysis and preprocessing on datasets.
- CO3: Apply supervised and unsupervised learning algorithms for predictive analytics.
- CO4: Evaluate and optimize machine learning models using appropriate metrics.
- CO5: Develop intelligent data-driven solutions for real-world applications.

Syllabus

Mathematical Foundations and Exploratory Data Analysis Overview of Linear Algebra: Scalars, vectors, and matrices, Matrix operations, Systems of linear equations, Eigenvalues and eigenvectors, Matrix decomposition. Overview of Probability and Statistics: Random variables, Probability distributions, Descriptive statistics, Measures of central tendency, Measures of dispersion, Sampling techniques. Overview of Optimization Techniques: Optimization fundamentals, Objective functions and constraints, Gradient and Hessian matrix, Least squares optimization, Gradient descent methods. Exploratory Data Analysis (EDA): Data collection and preprocessing, Handling missing values, Data transformation, Outlier detection, Correlation analysis, Data visualization techniques.

Statistical Learning and Model Selection Introduction to Machine Learning: Definitions and goals of machine learning, History and applications of machine learning, Types of machine learning. Statistical Learning: Lin-

ear classification, Classification errors, Regression techniques, Predictive analytics concepts. Model Selection Techniques: Empirical Risk Minimization (ERM), Structural Risk Minimization (SRM), Cross-validation techniques, Regularization methods, Bias-variance tradeoff, Performance evaluation metrics.

Mining Frequent Patterns, Associations, and Correlations Association Rule Mining: Frequent itemset mining, Apriori algorithm, FP-growth concepts, Association rules, Support and confidence measures, Market basket analysis applications.

Supervised Learning Fundamentals: Supervised learning concepts, Generative and discriminative learning, Parametric and non-parametric learning. Regression Techniques: Linear regression, Model fitting and prediction. Classification Techniques: Decision trees, k-Nearest Neighbors (KNN), Naïve Bayes classifier with Laplace smoothing, Logistic regression with Maximum Likelihood Estimation Model Evaluation: Regression Metrics and Confusion matrix, Precision, Recall, F1-score, ROC curve.

Unsupervised Learning: Unsupervised learning concepts, Clustering techniques, K-Means clustering, Hierarchical clustering. Dimensionality Reduction: Principal Component Analysis (PCA), Feature selection and feature extraction techniques.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2				3	2
CO2	2	3	2	2		3	2
CO3	3	3	2	2		3	3
CO4	3	3	3	2		3	3
CO5	2	3	3	3		3	3

Textbooks / References

1. Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer.
2. Tom M. Mitchell, *Machine Learning*, McGraw-Hill.
3. Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow*, O'Reilly Media.
4. Gareth James et al., *An Introduction to Statistical Learning*, Springer.
5. Trevor Hastie, Robert Tibshirani, Jerome Friedman, *The Elements of Statistical Learning*, Springer.
6. Gilbert Strang, *Introduction to Linear Algebra*, Wellesley-Cambridge Press.
7. 3. Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*, Cambridge University Press.

ECL221 VLSI Design Lab [0-0-2-1]

Course Objectives

- To familiarize students with the VLSI design flow, from RTL design to ASIC implementation, using industry-standard EDA tools from Synopsys and Cadence.
- To develop proficiency in CMOS circuit design, simulation, and analysis of fundamental digital logic gates and subsystems.
- To apply optimization techniques such as static timing analysis, pipelining, and parallelism to improve performance, area, and power characteristics of digital circuits.

- To provide hands-on experience in designing, synthesizing, verifying, and implementing complex digital systems and mini-projects using a complete RTL-to-GDSII design methodology.

Course Outcomes

Upon successful completion of the course, students will be able to:

- Design, simulate, and analyze CMOS-based digital circuits and verify their functionality using industry-standard VLSI design tools.
- Perform RTL design, logic synthesis, and functional verification of digital systems using Synopsys or Cadence EDA environments.
- Apply static timing analysis, pipelining, and parallel processing techniques to optimize digital designs for performance and resource utilization.
- Execute a complete RTL-to-ASIC design flow for a digital system (e.g., ALU, MAC unit, FIR filter), including synthesis, implementation, verification, and design reporting.

List of Experiments

- Introduction to VLSI Design Flow and EDA Tools
- Familiarization with RTL design flow using Synopsys and Cadence Design Systems tools
- Compilation, simulation, synthesis, and report generation
- Design and Simulation of CMOS Inverter characteristics
- Design and Simulation of Basic CMOS Logic Gates
- Design and transient simulation of CMOS NAND, NOR, and XOR gates
- RTL Design and Synthesis of Combinational Circuits
- RTL coding of adders, multiplexers, and encoders
- Logic synthesis using Synopsys Design Compiler or Cadence Genus
- Area, timing, and power analysis
- RTL Design and Synthesis of Sequential Circuits
- RTL implementation of flip-flops, counters, and shift registers
- Gate-level netlist generation and timing verification
- RTL Design and Implementation of Pipelined Arithmetic Circuits
- RTL design of pipelined adder or multiplier architectures
- Performance comparison between pipelined and non-pipelined implementations
- Timing and throughput analysis after synthesis
- RTL Design and Synthesis using Parallelism Techniques
- Design of parallel arithmetic structures such as Carry Look-Ahead Adder or Parallel Multiplier
- Synthesis and performance evaluation for speed optimization
- Static Timing Analysis and Optimization
- Timing constraint definition and analysis using synthesis tools
- Identification and optimization of setup and hold violations
- RTL design of a small digital system such as ALU, MAC unit, or FIR filter
- Logic synthesis, timing analysis
- Evaluation of area, power, and performance trade-offs
- 10. Mini projects for RTL to ASIC Design Flow: Synopsys or Cadence EDA Tools

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	3	1	3	3
CO2	3	3	2	3	1	3	3
CO3	3	3	2	3	1	3	3
CO4	3	3	3	3	1	3	3

ECL222 Embedded Systems Lab [0-0-2-1]

Course Objectives

- Provide hands-on experience in programming and interfacing microprocessor, microcontroller and ARM-based embedded platforms.

- Develop assembly language and embedded C programming skills for embedded applications.
- Expose students to peripheral interfacing, communication protocols and embedded system integration.
- Enable students to design, test and debug real-time embedded applications.

Course Outcomes

- CO1: Develop assembly language programs using 8086 and perform basic interfacing operations.
- CO2: Implement embedded control applications using 8051 peripherals, timers and serial communication.
- CO3: Develop ARM-based embedded applications using GPIO, display and communication interfaces.
- CO4: Design and integrate embedded hardware and software modules to realize real-time applications.

List of Experiments

- Study of Embedded System Development Tools.
- 8086 Assembly Language Programming for Data Processing.
- Temperature Monitoring and Control using 8086 and 8255.
- Study of Interrupt and DMA Controllers in 8086 Systems.
- 8051 GPIO Programming.
- 8051 Timer and UART Communication.
- ARM GPIO Programming and LED Control.
- ARM LCD/Seven-Segment Display Interfacing.
- ARM I²C and UART Communication.
- ARM-Based Real-Time Embedded Application with IPC.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	2		2	1
CO2	2	2	3	2		3	1
CO3	2	2	3	2	2	3	2
CO4	2	2	3	3	2	3	3

ECC311 Digital Signal Processing [3-0-0-3]

Course Objectives

- To develop a rigorous understanding of discrete-time spectral analysis techniques, including discrete Fourier transforms and fast Fourier transform algorithms, for efficient analysis and processing of digital signals.
- To analyze and design FIR and IIR digital filters using classical approximation techniques, frequency-domain methods, and realization structures suitable for practical signal processing applications.
- To study the architecture and computational principles of digital signal processors, enabling efficient implementation of real-time DSP algorithms in embedded and communication systems.

Course Outcomes

- CO1: Analyze discrete-time signals and systems using DFS, DFT, and FFT techniques, and apply transform-domain methods for efficient spectral analysis and digital filtering.
- CO2: Design and realize FIR and IIR digital filters using windowing, frequency sampling, analog-to-digital transformation methods, and appropriate realization structures based on system requirements.
- CO3: Evaluate DSP hardware architectures and processor characteristics for implementing real-time digital signal processing algorithms in practical engineering applications.

Syllabus

Discrete Fourier Series & Fourier Transforms: DFS representation of periodic sequences, Properties of discrete Fourier series, Discrete Fourier transforms: Development of DFT from DTFT, DFT as a linear transformation, Properties of DFT, linear filtering methods based on DFT, Fast Fourier transforms (FFT)—Radix-2 decimation in time and decimation in frequency FFT Algorithms, Inverse FFT.

Design of FIR Digital Filters & Realizations: Characteristics of linear phase FIR filters and frequency response. Design of linear phase FIR filters using windows—Rectangular, Bartlett, Hanning and Hamming windows. Design of FIR filters using frequency sampling method. Basic structures of FIR systems—Direct form, Cascade form and Lattice structures.

Design of IIR Digital Filters & Realizations: Analog filter approximations—Butterworth and Chebyshev, Design of IIR Digital filters—Impulse Invariant Transformation, Bilinear Transformation—Frequency Warping, Design of Lowpass Digital Butterworth Filter. Analog and Digital frequency transformations. Basic structures of IIR systems in Direct form I and II.

Digital Signal Processors: DSP Architecture, DSP Hardware Units, Digital Signal Processors and Manufacturers, Floating point and Fixed point digital signal processors.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	1	2	3	3	3
CO2	3	3	3	2	3	3	3
CO3	2	2	2	1	3	3	3

Textbooks / References

1. Proakis & Monolakis, *Digital Signal Processing: Principles, Algorithms & Applications*, 4th ed., Pearson Education.
2. Sanjit K. Mitra, *Digital Signal Processing: A Computer Based Approach*, 4th ed., McGraw Hill Education.
3. Oppenheim & Schaffer, *Discrete Time Signal Processing*, PHI.
4. Li Tan and Jean Jiang, *Digital Signal Processing: Fundamentals and Applications*, Academic Press.

ECC312 Control Systems [3-1-0-4]

Course Objectives

- To understand the modeling and analysis of linear control systems using transfer functions and state-space methods.
- To study the time-domain and frequency-domain response characteristics of control systems.
- To analyze stability and design suitable controllers and compensators for control systems.

Course Outcomes

- CO1: Analyze control systems using transfer functions, block diagrams, and signal-flow graphs.
- CO2: Evaluate stability and performance of control systems in time and frequency domains.
- CO3: Design controllers and compensators using root-locus, frequency-response, and state-space techniques.

Syllabus

Introduction to Control Systems, Closed-Loop and Open-Loop Control Systems, Mathematical Modeling of Control Systems, Transfer Function and Impulse Response Function, Block diagrams, Signal-Flow Graphs, Mason's Rule.

Transient and Steady-State Response Analyses, Typical Test Signals, First-Order Systems, Response of First-Order Systems to Unit step, unit ramp and unit impulse inputs, Second-Order Systems and Transient-Response Specifications, Underdamped Second-Order Systems. Steady-State Errors in Unity-Feedback Control Systems, Static Position Error Constant, Static Velocity Error Constant, Static Acceleration Error Constant, Routh's Stability Criterion, Proportional Control Action, Integral Control Action, Proportional-Plus-Integral Control Action, Proportional-Plus-Derivative Control Action, Proportional-Plus-Derivative Control Action.

Control Systems Analysis and Design by the Root-Locus Method, Root-Locus Plots, Root-Locus Plots of Positive Feedback Systems, Lead Compensation, Lag Compensation, Lag-Lead Compensation.

Control Systems Analysis and Design by the Frequency-Response Method: Bode diagram, Polar plot, Nyquist plot, All-pass and Minimum-phase systems, Log-Magnitude-versus-Phase Plots, Stability in frequency domain, Nyquist stability criterion, Gain Margin and Phase Margin.

Control Systems Analysis in State Space, State-Space Representations of Transfer-Function Systems, Solving the Time-Invariant State Equation, Matrix Exponential, Laplace Transform Approach to the Solution of Homogeneous State Equations, State-Transition Matrix, Properties of State-Transition Matrices, Solution of Nonhomogeneous State Equations, Controllability, Observability.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3				3	
CO2	3	3				3	1
CO3	3	3				3	2

Textbooks / References

1. Katsuhiko Ogata, *Modern Control Engineering*, Pearson Education.
2. Norman S. Nise, *Control Systems Engineering*, John Wiley & Sons.
3. I. J. Nagrath and M. Gopal, *Control Systems Engineering*, New Age International Publishers.
4. Gene F. Franklin, J. David Powell, and Abbas Emami-Naeini, *Feedback Control of Dynamic Systems*, Pearson/Prentice Hall.
5. Richard C. Dorf and Robert H. Bishop, *Modern Control Systems*, Pearson/Prentice Hall.

ECC313 Digital Communication [3-0-0-3]

Course Objectives

- To develop a rigorous understanding of the mathematical foundations of digital communication systems.
- To analyze the performance of baseband and passband digital transmission techniques in the presence of noise, bandwidth limitations, intersymbol interference, and synchronization constraints using probabilistic and geometric methods.
- To investigate advanced modulation and spread-spectrum techniques with emphasis on spectral efficiency, error performance, robustness, and design trade-offs relevant to modern wireless and communication systems research.

Course Outcomes

- CO1: Formulate and analyze equivalent low-pass representations of bandpass communication signals, line coding schemes, and pulse modulation techniques.
- CO2: Evaluate digital communication system performance under noisy and bandwidth-constrained channels through matched filter detection, signal-space methods, equalization techniques, and maximum-likelihood decoding frameworks.
- CO3: Analyze coherent and non-coherent digital modulation schemes, M-ary signaling methods, and spread-spectrum communication systems.

Syllabus

Baseband and passband signal representations, Band-pass Signal to Equivalent Low pass: Hilbert Transform, Complex envelopes, Canonical representation of bandpass signals, Complex representation of bandpass signals. Time Division Multiplexing, Pulse Code Modulation—sampling, quantization, and encoding, Differential PCM, Delta Modulation, Adaptive Delta Modulation, Line codes: unipolar NRZ, polar NRZ, unipolar RZ, bipolar

RZ, Manchester code, differential encoding, Power spectral densities of line coding techniques.

Baseband pulses and matched filter detection, probability of error due to noise, intersymbol interference, Nyquist's criterion for distortionless transmission, pulse shaping, raised cosine spectrum, equalization, eye pattern, signal-space analysis: geometric representation of signals, Gram-Schmidt Orthogonalization procedure, conversion of continuous AWGN channel into a vector channel, maximum-likelihood decoding, correlation receiver, equivalence of correlation and matched filter receivers.

Passband data transmission, coherent schemes: binary amplitude shift keying (BASK), frequency shift keying (BFSK), phase shift keying (BPSK), quadri-phase-shift keying (QPSK), M-ary PSK, M-ary QAM, bandwidth requirements, calculation of average probability of error, constellation diagram. Non-coherent orthogonal modulation techniques: differential phase shift keying (DPSK).

Spread spectrum modulation, pseudo-noise sequences, properties of maximal length sequences, spread spectrum systems, direct sequence spread spectrum with coherent BPSK, processing gain, jamming margin. Entropy, mutual information and channel capacity theorem

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	1	1	1	3	1
CO2	3	3	2	2	1	3	1
CO3	3	2	2	1	1	3	1

Textbooks / References

1. Simon Haykin, *Communication Systems*, 4th ed., John Wiley Inc.
2. B. P. Lathi, *Modern Digital and Analog Communication Systems*, 4th ed., Oxford University Press.
3. John G. Proakis and Masoud Salehi, *Fundamentals of Communication Systems*, 2014 ed., Pearson Education.
4. Bernard Sklar and Ray, *Digital Communications—Fundamentals and Applications*, 3rd ed., Pearson Education.
5. Ian A. Glover and Peter M. Grant, *Digital Communications*, 3rd ed., Pearson Education.

ECC314 Microwave Engineering [3-0-0-3]

Course Objectives

- To understand the fundamentals of microwave engineering, transmission lines, and wave propagation.
- To analyze waveguides, microwave networks, and impedance matching techniques used in high-frequency systems.
- To study the basics of noise analysis in microwave circuits.

Course Outcomes

- CO1: Apply transmission line theory and Smith chart techniques for microwave circuit analysis and design.

- CO2: Analyze propagation characteristics of waveguides, microstrip lines, and microwave network parameters.
- CO3: Evaluate the noise performance in microwave communication systems.

Syllabus

Review of electromagnetics, Maxwell's equations, Transmission Lines, Transmission Line Parameters, Transmission Line Equations, Input Impedance, SWR, and Power, Smith Chart, Some Applications of Transmission Lines, Transients on Transmission Lines, Microstrip Transmission Lines, Characteristic impedance, Stub Matching.

Rectangular waveguides, Analysis of modes TE, TM, TEM, circular waveguides, Microstrip lines; Network analysis: scattering matrix, transmission matrix formulations; Matching networks: Lumped element designs and limitations, single and double-stub tuned designs, Quarter wavelength transformers, power dividers and directional couplers.

Noise in microwave circuits: dynamic range and noise sources, equivalent noise temperature, system noise figure considerations, microwave devices—IMPATT diodes, PIN diodes, Klystrons, and Magnetrons.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	1		3	3
CO2	3	3	2	2		3	3
CO3	3	3	2	2	1	3	3

Textbooks / References

1. Matthew N. O. Sadiku and Sudarshan Nelatury, *Elements of Electromagnetics*, Oxford University Press.
2. David M. Pozar, *Microwave Engineering*, 3rd ed., John Wiley & Sons.
3. Thomas H. Lee, *Planar Microwave Engineering: A Practical Guide to Theory, Measurement, and Circuits*, 1st ed., Cambridge University Press.
4. Robert E. Collin, *Foundations for Microwave Engineering*, John Wiley & Sons Inc.

ECL311 Digital Signal Processing Lab [0-0-2-1]

Course Objectives

- To provide practical understanding of discrete-time signal processing operations such as sampling, convolution, correlation, and difference equation analysis.
- To develop skills in spectral analysis of discrete-time signals using DFT techniques and verification of transform-domain properties through simulation.
- To enable the design, implementation, and performance evaluation of FIR and IIR digital filters for real-world signal and audio processing applications.

Course Outcomes

- CO1: Implement and verify sampling, convolution, correlation, and difference equation operations for discrete-time signals.
- CO2: Analyze discrete-time signals using DFT techniques and verify important transform-domain properties.
- CO3: Perform spectral analysis of signals and interpret magnitude and phase characteristics in the frequency domain.
- CO4: Design and implement FIR digital filters using windowing techniques and evaluate their performance.
- CO5: Design and implement IIR digital filters and assess their effectiveness in practical signal processing applications.

Syllabus

- Verification of sampling theorem
- Linear and circular convolution of two given sequences
- Verification of commutative, distributive and associative properties of convolution
- Auto and cross correlation of two sequences and verification of their properties
- Solving a given difference equation
- Computation of N point DFT of a given sequence and to plot magnitude and phase spectrum
- Verification of DFT properties (like Linearity and Parseval's theorem, etc.)
- DFT computation of square pulse and Sinc function.
- Design and implementation of Low pass and High pass FIR filter to meet the given specifications (using different window techniques) and test the filter with an audio file. Plot the spectrum of audio signal before and after filtering.
- Design and implementation of a digital IIR filter (Low pass and High pass) to meet the given specifications and test with an audio file. Plot the spectrum of audio signal before and after filtering.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	1	2	3	3	2
CO2	3	3	1	2	3	3	3
CO3	3	3	1	2	3	3	3
CO4	3	3	3	2	3	3	3
CO5	3	3	3	2	3	3	3

ECL312 Analog and Digital Communication Lab [0-0-2-1]

Course Objectives

- To understand the principles of analog modulation and demodulation techniques used in communication systems.
- To design and analyze modulation and demodulation circuits using integrated circuits such as AD633, IC2206, and IC566.

- To study pulse modulation techniques and their applications in signal transmission.
- To implement and evaluate digital modulation schemes including ASK, FSK, and PSK.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Construct and analyze AM, DSB-SC, and FM modulation/demodulation circuits and evaluate their performance characteristics.
- CO2: Design and verify the operation of pre-emphasis and de-emphasis circuits used in frequency modulation systems.
- CO3: Generate and analyze pulse modulation signals such as PAM, PWM, and PPM and compare their characteristics.
- CO4: Implement and detect digital modulation schemes including Binary ASK, Binary FSK, and Binary PSK, and interpret the corresponding waveforms.

Syllabus

The following experiments shall be implemented on breadboards using ICs and discrete electronic components.

- Amplitude modulation using AD633/BC108 and demodulation using envelope detector
- DSB-SC modulation using IC AD633
- FM generation and demodulation using IC2206/IC566
- Pre emphasis and De emphasis circuits
- PAM, PWM and PPM generation
- Generation and detection of Binary Amplitude Shift Keying signal
- Generation and detection of Binary Phase Shift Keying signal
- Generation and detection of Binary Frequency Shift Keying signal
- Generation of Delta modulated signal

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	2	2	3		3	1
CO2	3	3	3	3		3	1
CO3	2	2	3	3		3	1
CO4	3	3	3	3		3	1

ECC321 Antenna Theory and Design [3-0-0-3]

Course Objectives

- To understand radiation properties of various antennas.
- To understand applications of various types of antennas.
- To design and analyse various types of antennas.

Course Outcomes

- CO1: Understand the necessity and applications of various types of antennas.

- CO2: Analyse the radiation patterns of various antennas.
- CO3: Design and analyse various types of antennas.

Syllabus

Fundamental parameters of Antenna, Radiation mechanism of wire and loop antennas – dipole, monopole, loop antenna; vector potential method for calculation of radiated field, Reciprocity theorem, Broadband antennas, Balance to unbalance transformer, Field equivalence principle. Radiation from Rectangular and Circular apertures, Uniform aperture distribution on an infinite ground plane, Slot antenna, Horn antenna, Reflector antenna, aperture blockage, Frequency independent antennas.

General structure of phased array, linear array theory, variation of gain as a function of pointing direction, effects of phase quantization, frequency scanned arrays, analog beam forming matrices, digital beam forming.

Radiation Mechanism from patch: Excitation techniques, Rectangular patch, Circular patch, Radiation analysis using transmission line model, cavity model, Microstrip arrays.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	2	1	3	1
CO2	3	3	2	2	1	3	1
CO3	3	3	2	2	1	3	2

Textbooks / References

1. Constantine Balanis, *Antenna Theory: Analysis and Design*, 4th ed., Wiley.
2. Hubregt J. Visser, *Antenna Theory and Applications*, 1st ed., John Wiley & Sons Ltd, New York.
3. Zhijun Zhang, *Antenna Design for Mobile Devices*, 1st ed., John Wiley & Sons (Asia) Ltd, New York.
4. Xavier Begaud, *Ultra-Wide Band Antennas*, 1st ed., ISTE Ltd and John Wiley & Sons Ltd, New York.
5. John D. Kraus and Ronald Marhefka, *Antennas for All Applications*, 3rd ed., McGraw Hill, 2003.

ECC322 FPGA Based System Design [3-0-0-3]

Course Objectives

- To understand the architecture and design flow of FPGA.
- To learn case studies using reconfigurable computing.
- To develop digital hardware modules using HDL, HLS and implement them on FPGA.

Course Outcomes

- CO1: Have understood the necessity of reconfigurable architectures and FPGA architectures.
- CO2: Have learned the case studies on FPGA.
- CO3: Develop applications using FPGA using RTL design or HLS.

Syllabus

General Purpose Computing Vs Reconfigurable Computing—Simple Programmable Logic Devices—Complex Programmable Logic Devices—FPGAs—Device Architecture, Resources- Slices, LUT, DSP, BRAM — Zync - PS, PL.

Compute Models—Programming FPGA Applications in HDL—Compiling C for Spatial Computing—Operating System Support for Reconfigurable Computing.

The Design Flow—Technology Mapping—FPGA Placement and Routing—Configuration Bitstream Generation—Case Studies with Appropriate Tools.

High Level Synthesis—Pragma, Pipelining, Initiation Interval

Case Studies of FPGA Applications—System on a Programmable Chip (SoPC) Designs.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	1	1	1	1	3	1
CO2	3	2	2	3	2	3	3
CO3	3	3	3	3	3	3	3

Textbooks / References

1. Maya B. Gokhale and Paul S. Graham, *Reconfigurable Computing: Accelerating Computation with Field Programmable Gate Arrays*, Springer.
2. Scott Hauck and Andre Dehon (Eds.), *Reconfigurable Computing—The Theory and Practice of FPGA-Based Computation*, Elsevier/Morgan Kaufmann.
3. Christophe Bobda, *Introduction to Reconfigurable Computing—Architectures, Algorithms and Applications*, Springer.

ECC323 Wireless and Cellular Communication [3-0-0-3]

Course Objectives

- To introduce the fundamental concepts of wireless communication systems, wireless channel characteristics, and channel modeling techniques.
- To analyze the performance of digital communication systems under fading environments and understand the impact of multipath propagation, Doppler spread, and inter-symbol interference.
- To study various diversity techniques and CDMA systems for improving reliability and capacity in wireless communication networks.
- To understand the principles, architectures, and signal processing techniques used in MIMO wireless communication systems.
- To explore multicarrier communication techniques such as OFDM and SC-FDMA, including their applications, advantages, and implementation challenges in modern wireless systems.

Course Outcomes

- CO1: Explain the fundamentals of wireless communication systems, wireless channel characteristics, statistical channel models, and channel capacity concepts.
- CO2: Analyze the performance of wireless communication systems under fading channels, including detection techniques, outage probability, diversity order, Doppler effects, and inter-symbol interference.
- CO3: Apply diversity techniques and CDMA principles to improve communication reliability and mitigate multi-user interference and near-far effects.
- CO4: Evaluate MIMO system models and receiver techniques such as ZF, MMSE, SVD, beamforming, and space-time block coding for enhanced wireless communication performance.
- CO5: Analyze multicarrier modulation techniques including OFDM, MIMO-OFDM, OFDM-PAPR reduction methods, frequency offset effects, and SC-FDMA systems used in broadband wireless communications.

Syllabus

Introduction to wireless communication: History of wireless communication, physical modelling of wireless channels, input/ output model of wireless channel, time and frequency coherence, statistical channel models, channel capacity of wireless channels.

Detection and performance of wireless communication systems: Detection in Rayleigh fading channel, error performance of digital modulation techniques under wireless fading channels, outage probability, diversity order, doppler spread, inter symbol interference.

Diversity: Introduction to diversity techniques, time diversity, frequency diversity, spatial diversity, receive diversity, receive diversity combining techniques, transmit diversity. CDMA: CDMA codes and its properties, multi-user CDMA, CDMA near-far problem and power control. MIMO: Introduction to MIMO, MIMO system model, MIMO receivers: ZF, MMSE, SVD decomposition of the MIMO channel, Space time block codes, MIMO beamforming, multicarrier modulation techniques, OFDM, MIMO OFDM, OFDM-PAPR, frequency offset in OFDM, SC-FDMA.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	3	3	1		3	
CO2	3	3	3	3		3	
CO3	2	3	3	2	1	3	1
CO4	2	3	3	2	2	3	2
CO5	2	3	3	3	2	3	2

Textbooks / References

1. Aditya K Jagannatham, “Principles of Modern Wireless Communication Systems: Theory and Practice”. McGraw Hill Education, 2016.
2. A Goldsmith, “Wireless Communications”, Cambridge university press, 2005.
3. D. Tse and P. Viswanath, “Fundamentals of Wireless Communications” Cambridge university press, 2005.

IOECXXX Design Thinking and Innovation [3-0-0-3]

Course Objectives

- To introduce students to the principles, processes, and applications of design thinking and innovation.
- To develop skills in user-centered problem identification, empathy building, and user research methods.
- To foster creativity and innovation through brainstorming, ideation, prototyping, and testing activities.
- To equip students with practical skills in interaction design, usability evaluation, and design communication.
- To encourage interdisciplinary collaboration, ethical thinking, and socially responsible innovation through project based learning.

Course Outcomes

- CO1: Explain the concepts, stages, and significance of the design thinking process in solving real-world problems.
- CO2: Apply user research methods, empathy techniques, and problem-definition strategies to understand user needs effectively.
- CO3: Generate innovative ideas using brainstorming and ideation techniques and develop prototypes for testing and evaluation.
- CO4: Design user-centered solutions using interaction design principles, usability evaluation methods, and iterative improvements.
- CO5: Collaborate effectively in interdisciplinary teams and communicate design solutions through presentations, reports, and project demonstrations.

Syllabus

Introduction to Design Thinking - Introduction to design - Characteristics of good and bad design - Importance of design thinking in engineering and problem solving - Human-centered design approach - The five stages of design thinking: Empathize, Define, Ideate, Prototype, and Test - Design thinking versus traditional problem-solving approaches - Individual and team-based problem solving. User Empathy and Problem Identification - Understanding user needs and stakeholder analysis - Empathy development techniques - User research methods including interviews, observations, surveys, and focus groups - Problem definition and reframing techniques - User-centered design principles - Ergonomics and accessibility considerations. Ideation and Concept Development - Brainstorming methods and innovation heuristics - Creativity enhancement techniques - Affinity diagrams, empathy maps, mind maps, and customer journey maps - Concept generation and evaluation - Scenario building and storytelling techniques - Collaborative ideation activities. Prototyping and Usability Testing - Low-fidelity and high-fidelity prototyping methods - Prototype development tools and techniques - Wireframes - Usability testing methods - Gathering and analyzing user feedback - Iterative de-

sign improvement - Heuristic evaluation techniques - Basics of data analysis using SPSS for design evaluation.

Interaction Design and User Experience - Fundamentals of interaction design - History and evolution of interaction design - User Interface (UI) and User Experience (UX) principles - Information architecture - Personas and task flows - Laws and principles of interaction design - Qualitative and quantitative research methods - Usability evaluation techniques.

Innovation, Ethics, and Social Impact - Principles and processes of successful innovation - Modes of design innovation - Social entrepreneurship and innovation for social good - Ethics in design and responsible innovation - Inclusive and sustainable design practices - Industry case studies on innovation and design impact - Familiarization with modern design tools - Design-based mini project development and presentation.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	2			2	1	
CO2		3	2		2	1	1
CO3	2	3	3	2		2	2
CO4	2	3	3	3	2	2	3
CO5	1	2	2	2	3	1	2

Textbooks / References

1. Kumar, K., and Kurni, M. (Eds.), *Design Thinking: A Forefront Insight*, CRC Press, 2022.
2. Vijay Kumar, *101 Design Methods: A Structured Approach for Driving Innovation in Your Organization*, John Wiley and Sons, 2013.
3. Dan Saffer, *Designing for Interaction: Creating Innovative Applications and Devices*, New Riders Publishing, 2010.
4. Lewrick, M., Link, P., and Leifer, L., *The Design Thinking Toolbox: A Guide to Mastering the Most Popular and Valuable Innovation Methods*, Wiley, 2020.
5. Metts, M. J., and Welfle, A., *Writing is Designing: Words and the User Experience*, Rosenfeld Media, 2020.
6. Uebersnickel, F., Jiang, L., Brenner, W., Pukall, B., Naef, T., and Schindlholzer, B., *Design Thinking: The Handbook*, World Scientific, 2020.
7. Gallagher, A., and Thordarson, K., *Design Thinking in Play: An Action Guide for Educators*, ASCD, 2020.
8. Regine Gilber, *Inclusive Design for a Digital World: Designing with Accessibility in Mind*, Apress, 2019.
9. Michael G. Luchs, *Design Thinking: New Product Development Essentials from the PDMA*, Wiley, 2015.
10. D'Source, *Digital Online Environment for Design Learning*, – <https://www.dsourc.in>.

ECL321 Antenna Design Lab [0-0-2-1]

Course Objectives

- To analyse radiation properties of various antennas.
- To understand applications of various types of antennas.
- To design and analyse various types of antennas.

Course Outcomes

- CO1: Design various types of antennas based on applications.
- CO2: Analyse the radiation properties of various antennas.
- CO3: Design and analyse various types of antennas.

Syllabus

Simulations and analyses to be carried out using licensed CST

Microwave Studio. Characterization, modelling and analysis to be done for the following antennas:

- Dipole and Monopole antenna
- Loop antenna
- Rectangular and pyramidal Horn antenna
- Phased array antenna
- Rectangular and Circular Microstrip Patch antenna
- Ring patch antenna
- Inverted-F antenna
- Microstrip Arrays
- MIMO antennas
- Circular polarization and miniaturization techniques in antennas

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	3	1	3	2
CO2	3	3	2	3	1	3	2
CO3	3	3	2	3	1	3	2

ECL322 Makerspace/Innovation Lab [1-0-2-2]

Course Objectives

- To understand the fundamental concepts of PCB design, schematic creation, and component placement.
- To create schematic symbols, footprints, and complete electronic circuits.
- To study modern Electronic Design Automation tool for system or product design
- Design and implement an end-to-end system or product

Course Outcomes

- CO1: Explain the fundamental concepts of PCB design, schematic capture, component selection, and PCB layout methodologies.
- CO2: Develop schematic symbols, component footprints, and complete electronic circuit designs using Electronic Design Automation (EDA) tools.
- CO3: Apply modern EDA tools and design methodologies for electronic system and product development, including simulation, verification, and PCB implementation.
- CO4: Design, implement, and validate an end-to-end electronic system or product by integrating hardware design, PCB fabrication considerations, assembly, and testing procedures.

Syllabus

Fundamentals of PCB design using EDA tool, Generate schematic circuits, Implement component placement and routing (using EDA tools like Altium), focusing on optimal layout and adherence to design rules for a given schematic. Establish and verify design rules, including trace width and clearance, for a PCB layout, and resolve any Design Rule Check (DRC) errors using a suitable PCB design tool.

Use EDA tools for system design, aiming at product development.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	2	2	3	2
CO2	3	2	2	2	2	3	3
CO3	3	2	3	2	2	3	3
CO4	3	3	3	3	3	3	3

Textbooks / References

1. Karl T. Ulrich, Steven D. Eppinger "Product Design and Development", McGraw-Hill Education.
2. Altium Documentation – Altium Designer User Manual.
3. Kraig Mitzner – Complete PCB Design Using OrCAD Capture and PCB Editor.
4. ONLINE REFERENCES
 1. Altium – <https://www.altium.com/>
 2. PCBWay – <https://www.pcbway.com/>
 3. SparkFun – PCB design tutorials

ECE301 Advanced Digital Signal Processing [3-0-0-3]

Course Objectives

- To introduce the concepts of Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT) algorithms for efficient spectral analysis.
- To provide knowledge of finite word length effects in digital signal processing systems.
- To develop understanding of quantization effects, arithmetic operations, and overflow phenomena in fixed-point and floating-point digital implementations.
- To impart the fundamentals of multirate digital signal processing including decimation, interpolation, and sampling rate conversion techniques.
- To introduce polyphase structures, digital filter banks, quadrature mirror filter banks, and subband coding techniques used in modern DSP applications.

Course Outcomes

- CO1: Apply FFT algorithms for efficient computation of frequency domain representation of discrete time signals.
- CO2: Evaluate the effects of finite word length and quantization in digital signal processing systems.
- CO3: Examine arithmetic errors, overflow effects, and limit cycle oscillations in fixed-point and floating-point digital filter implementations.
- CO4: Implement multirate DSP techniques including decimation, interpolation, sampling rate conversion, and polyphase decomposition.
- CO5: Analyze digital filter banks, quadrature mirror filter banks, and subband coding systems used in modern signal processing applications.

Syllabus

Overview of Discrete Fourier transform (DFT), and FFT: Radix-4 DIT/DIF FFT algorithms, Composite radix

FFT algorithms.

Review of FIR and IIR filter design, Finite word length effects in digital filters – Fixed-point and floating-point arithmetic in digital system, Quantization by Truncation and Rounding, Quantization of Input Data, Quantization of Filter Coefficients, Product Quantization Error, Limit Cycles in Recursive Systems Zero Input Limit Cycle, Overflow Limit Cycle, scaling to prevent overflow.

Introduction to Multi-rate Digital Signal Processing, Sampling, Sampling rate conversion, Sampling rate reduction – decimation by integer factors, Sampling rate increase – interpolation by integer factor, Sampling rate conversion by non-integer factor, Multistage Implementation of Sampling Rate Conversion, Implementation of Sampling Rate Conversion in FIR Filters, Polyphase Decomposition—Polyphase Decomposition of FIR Filters, Polyphase Decomposition of IIR Filters, Applications of Multirate DSP – Digital Filter Banks – 2 Channel Quadrature Mirror Filter Bank – perfect reconstruction (PR) or nearly perfect reconstruction (NPR) property, aliasing cancellation condition, perfect reconstruction condition, M Channel Quadrature Mirror Filter Bank—aliasing cancellation condition, perfect reconstruction condition, Subband coding, Subband coding system.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	1	2	3	3	3
CO2	3	3	1	2	3	3	2
CO3	3	3	1	2	3	3	2
CO4	3	3	2	2	3	3	3
CO5	3	3	2	2	3	3	3

Textbooks / References

1. S. K. Mitra, *Digital Signal Processing: A Computer Based Approach*, Tata McGraw Hill.
2. John G. Proakis and Dimitris G. Manolakis, *Digital Signal Processing: Principles, Algorithms and Applications*, 3rd ed., PHI, 2000.
3. P. P. Vaidyanathan, *Multirate Systems and Filter Banks*, Prentice Hall, 1993.
4. N. J. Fliege, *Multirate Digital Signal Processing—Multirate Systems, Filter Banks, Wavelets*, John Wiley, 1994.
5. C. K. Chui, *An Introduction to Wavelets*, Academic Press USA.
6. I. Daubechies, *Ten Lectures on Wavelets*, SIAM, 1990.
7. Lokenath Debnath and Firdous Ahmad Shah, *Wavelet Transforms and Their Applications*, Springer, Birkhäuser, 2002.

ECE302 Statistical Signal Processing [3-0-0-3]

Course Objectives

- To develop an understanding of statistical signal processing techniques for analyzing discrete-time random

signals and processes.

- To provide the knowledge and skills for energy and power spectrum estimation, using both non-parametric and parametric methods.
- To introduce the concepts and implementation of adaptive filtering algorithms, with a focus on LMS and RLS methods.
- To teach the fundamentals and applications of Kalman filtering for optimal state estimation in dynamic systems.

Course Outcomes

- CO1: Understand the fundamental concepts of random variables, random processes, and statistical signal processing.
- CO2: Be proficient in techniques for energy and power spectrum estimation using both non-parametric and parametric methods.
- CO3: Gain practical experience in the design and implementation of adaptive filters, including LMS and RLS algorithms.
- CO4: Apply Kalman filtering techniques for optimal state estimation in discrete and continuous systems.
- CO5: Analyze the performance of various statistical signal processing methods through real-world applications like noise cancellation, system identification, and equalization.

Syllabus

Discrete-Time Random Signal Processing – Introduction, Random Variables, Ensemble Averages of a Random Variable, Two-Dimensional Random Variables, Bias and Consistency for Parameter Estimation, Random Processes, Stationary Process, Autocorrelation and Autocovariance Matrices, Ergodic Processes, Power Spectrum, Filtering Random Processes, Spectral Factorisation Theorem.

Energy and Power Spectrum Estimation—Introduction, Energy Spectrum of Discrete Time Signal, Power Spectrum of a Random Process, Energy Spectral Density, Estimation of the Autocorrelation and Power Spectrum of Random Signals, Periodogram, Use of DFT/FFT in Power Spectrum Estimation, Power Spectrum Estimation: Non-Parametric Methods—Bartlett method, Welch method, Blackman-Tukey method, AR, MA and ARMA Models, Power Spectrum Estimation: Parametric Methods—AR Model (Yule-Walker Method), The Levinson-Durbin Recursion, Optimum Filters.

Adaptive Filtering: Concept of adaptive filtering, Basic Wiener filter theory, Wiener-Hopf Equation, Steepest Descent Algorithm (SDA), Least mean square (LMS) adaptive algorithm—Implementation of LMS adaptive algorithm, Practical limitations of LMS adaptive algorithm, Recursive Least Square (RLS) algorithm—Limitations of RLS adaptive algorithm, Applications of adaptive filters – System Identification, Noise cancellation, Echo cancellation, Equalization. Kalman filtering: State-space model and the optimal state estimation problem, discrete

Kalman filter, continuous-time Kalman filter, extended Kalman filter.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3		2	2	3	2
CO2	3	3	1	2	3	3	3
CO3	3	3	2	2	3	3	3
CO4	3	3	2	2	3	3	3
CO5	3	3	2	3	3	3	3

Textbooks / References

1. D. H. Johnson, *Statistical Signal Processing*, Available as a coursepak at the Rice Campus Store.
2. Monson H. Hayes, *Statistical Digital Signal Processing and Modeling*, John Wiley & Sons, 2002.
3. S. M. Kay, *Fundamentals of Statistical Signal Processing, Vol I: Estimation Theory, Vol II: Detection Theory*, Prentice Hall.
4. Charles W. Therrien, *Discrete Random Signals and Statistical Signal Processing*, Prentice Hall Signal Processing Series.
5. D. G. Manolakis, V. K. Ingle and S. M. Kogon, *Statistical and Adaptive Signal Processing*, McGraw Hill.
6. R. M. Gray and L. D. Davisson, *An Introduction to Statistical Signal Processing*, Cambridge University Press, 2004.

ECE303 Time Frequency Analysis [3-0-0-3]

Course Objectives

- To introduce the fundamental concepts of time-frequency analysis and the limitations of conventional time-only and frequency-only signal representations.
- To study analytic signals, instantaneous frequency, and major time-frequency distributions including the Wigner distribution and Cohen's class of distributions.
- To understand kernel-based methods for interference reduction in time-frequency distributions and explore applications of time-frequency analysis in radar, speech, and sonar signal processing.

Course Outcomes

- CO1: Analyze non-stationary signals using time-frequency representations and interpret signal characteristics in both time and frequency domains.
- CO2: Apply Wigner distribution, ambiguity function, and Cohen's class of time-frequency distributions for signal analysis and interference reduction.
- CO3: Evaluate and utilize time-frequency analysis techniques for practical applications in radar, speech, and sonar signal processing systems.

Syllabus

The need for time-frequency analysis, introduction, simple analytic examples. real signals, why spectra change. the time and frequency description of signals, bandwidth equation, AM and FM contributions to the

bandwidth, duration and mean time in terms of the spectrum, the covariance of a signal, the Fourier transform of the time and frequency densities, nonadditivity of spectral properties, classification of signals.

Instantaneous frequency and the complex signal, reasons for the complex signal, the analytic signal, calculating the analytic signal, physical interpretation of the analytic signal, the quadrature approximation, instantaneous frequency, density of instantaneous frequency. densities and characteristic functions, one-dimensional densities, one-dimensional characteristic functions, two-dimensional densities, local quantities, relation between local and global averages, distribution of a new variable, negative densities.

The need for time-frequency analysis, time and frequency description of signals, bandwidth equation, duration and mean time of signals, analytic signal and instantaneous frequency, fundamentals of time-frequency distributions, the wigner distribution and its properties, interference terms and the ambiguity function, cohen's class of time-frequency distributions, kernel methods for reduced interference distributions, applications of time-frequency distributions in radar, speech and sonar signal processing.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3		2		3	1
CO2	3	3	2	2		3	2
CO3	2	3	2	3		2	3

Textbooks / References

1. Leon Cohen, *Time-Frequency Analysis*, Prentice Hall PTR, 1995.
2. Boualem Boashash (Ed.), *Time-Frequency Signal Analysis and Processing: A Comprehensive Reference*, Elsevier, 2003.
3. Antonia Papandreou-Suppappola (Ed.), *Applications in Time-Frequency Signal Processing*, CRC Press, 2002.

ECE304 Chip Designing [3-0-0-3]

Course Objectives

- To understand full custom and semicustom ASICs.
- To learn about ASIC library design.
- To know the concept of schematic entry hierarchical design and cell library.
- To understand programmable ASIC, and simulation examples.

Course Outcomes

- CO1: Describe full custom and semicustom ASICs.
- CO2: Explain ASIC library design.
- CO3: Describe the concept of schematic entry, hierarchical design and cell library.
- CO4: Know the end-to-end design flow of IC design and simulation examples.

Syllabus

Introduction to Chip designing: Full Custom with ASIC, Semi-custom ASICs, Standard Cell based ASIC, Gate array based ASIC, Channeled gate array, Channel less gate array, structured gate array, Programmable logic device, FPGA design flow, SIC cell libraries. Data Logic Cells: Data Path Elements, Adders, Multiplier, Arithmetic Operator, I/O cell, Cell Compilers. ASIC Library Design: Logical effort: practicing delay, logical area and logical efficiency logical paths, multi stage cells, optimum delay, optimum number of stages, library cell design. Low-Level Design Entry: Schematic Entry: Hierarchical design. The cell library, Names, Schematic, Icons & Symbols, Nets, schematic entry, connections, vectored instances and buses, Edit in place attributes, Netlist, screener, Back annotation. Programmable ASIC logic cell, Introduction to Synthesis and Simulation, Examples. Floor Planning and Placement And Routing: Physical Design, CAD Tools, System Partitioning, Estimating ASIC size, partitioning methods. Floor planning tools, I/O and power planning, clock planning, placement algorithms, iterative placement improvement, Time driven placement methods. Physical Design flow global Routing, Local Routing, Detail Routing, Special Routing, Circuit Extraction and DRC. Recent trends in related areas from Journals like TVLSI, JSSC, TCAS-I etc, Conference proceedings like ISCAS, ISSCC, CICC, DAC, ICCAD etc.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	2	2	3	3
CO2	3	2	2	2	2	3	2
CO3	3	3	3	3	2	3	3
CO4	3	3	3	3	3	3	3

Textbooks / References

1. M. J. S. Smith, *Application-Specific Integrated Circuits*, Pearson Education.
2. Neil Weste and K. Eshraghian, *Principles of CMOS VLSI Design*, 4th ed.
3. D. A. Pucknell, *Basic VLSI Design*, PHI Publication.
4. R. Jacob Baker, Harry W. Li, and David E. Boyce, *CMOS Circuit Design, Layout, and Simulation*, PHI.

ECE305 VLSI Testing and Verification [3-0-0-3]

Course Objectives

- To understand the concepts of faults and automatic test pattern generation.
- To study the concepts of DFT and BIST.
- To understand the concepts of functional verification.
- To understand the concepts of timing verification.
- To understand the concepts of verification.

Course Outcomes

- CO1: Understand design verification and testing techniques in VLSI circuits.
- CO2: Understand fault modelling.
- CO3: Understand scan design.
- CO4: Familiarize EDA testing and verification tools.

Syllabus

Scope of testing and verification in VLSI design. Introduction to VLSI Design testing, VLSI testing needs and challenges, yield, defects. Fault modeling, Fault equivalence, Fault simulation, Automatic test pattern generation—D Algorithm. Design for testability, Scan design, Boundary scan, System test and test for SoCs, Iddq testing, Delay fault testing, BIST for testing logic and memories.

Design verification techniques based on simulation, analytical and formal approaches, Functional verification, Timing verification, Formal verification—Basics of equivalence checking and model checking. Testing and Verification Tools from EDA vendors, Sample lab experiments can be given for testing and verification, as exercises.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	3	3	3	2
CO2	3	3	2			3	3
CO3	3	2	3	3		3	2
CO4	1	3	3	3	3	3	3

Textbooks / References

1. M. L. Bushnell and V. D. Agrawal, *Essentials of Electronic Testing for Digital, Memory and Mixed-Signal VLSI Circuits*, Kluwer Academic Publishers.
2. Laung-Terng Wang, Cheng-Wen Wu, and Xiaoqing Wen, *VLSI Test Principles and Architectures: Design for Testability*, Elsevier, 2006.
3. M. Abramovici, M. A. Breuer and A. D. Friedman, *Digital Systems and Testable Design*, Jaico House, 2002.
4. Pallab Dasgupta, *A Roadmap for Formal Property Verification*, Springer Netherlands, 2006.
5. Chris Spear, *SystemVerilog for Verification: A Guide to Learning the Testbench Language Features*, Springer US, 2006.
6. Jayaram Bhasker and Rakesh Chadha, *Static Timing Analysis for Nanometer Designs: A Practical Approach*, Springer, 2009.

ECE306 CAD for VLSI [3-0-0-3]

Course Objectives

- To introduce the fundamentals of Computer-Aided Design (CAD) methodologies for VLSI systems.
- To develop proficiency in hardware modeling and synthesis techniques.
- To impart knowledge of logic synthesis, physical design, and optimization techniques.
- To familiarize students with verification and validation techniques for VLSI systems.

Course Outcomes

- CO1: Analyze and explain CAD methodologies for VLSI design.
- CO2: Model and synthesize digital hardware systems using Verilog HDL.
- CO3: Apply logic synthesis, placement, routing, graph algorithms, and optimization methods.
- CO4: Evaluate and verify VLSI designs using formal verification techniques.

Syllabus

Introduction to CAD for VLSI; Hardware description language (Verilog); High-level Synthesis: Scheduling, Allocation and Binding; Logic Synthesis: technology mapping, ASIC design methodology, FPGA based system design and prototyping; Physical Design: Placements, routing algorithms; Graph algorithms and their application in IC design; Optimization methods for systems design; Verification: Binary Decision Diagram.

Recent trends from journals and conferences such as TCAD, TVLSI, ICCAD, DAC, DATE etc

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	2	1	3	2
CO2	3	3	3	3	2	3	2
CO3	3	3	3	3	2	3	3
CO4	3	2	2	3	2	3	3

Textbooks / References

1. G. De Micheli, *Synthesis and Optimization of Digital Circuits*, 1st ed., 1994.
2. N. A. Sherwani, *Algorithms for VLSI Physical Design Automation*, 3rd ed., BSP Books Pvt. Ltd., 2005.
3. D. D. Gajski, N. D. Dutt, A. C.-H. Wu and S. Y.-L. Lin, *High-Level Synthesis: Introduction to Chip and System Design*, 1st ed., Springer, 1992.
4. Huth and Ryan, *Logic for Computer Science*.
5. S. H. Gerez, *Algorithms for VLSI Design Automation*.
6. Recent publications from DAC, ICCAD, DATE etc.

ECE307 Micro-Electro-Mechanical-Systems [3-0-0-3]

Course Objectives

- To introduce the basic concepts of micro systems and applications of MEMS.
- To introduce the fundamental sensing and actuating principles used in MEMS based structures.
- To introduce the various materials and their properties used for micromachining techniques.
- To analyze the fundamentals of micromachining and micro fabrication techniques.
- To impart knowledge of the basic concept of packaging of MEMS based microdevices.

Course Outcomes

- CO1: Understand the basic overview of MEMS and Microsystems with broad category of MEMS & Micro system applications.
- CO2: Understand the basic sensing and actuating principles used in MEMS based structures.
- CO3: Give an outline of various materials used for MEMS.
- CO4: Understand the microsystem fabrication processes and analyze the different micro manufacturing processes.
- CO5: Describe the packaging techniques for MEMS devices

Syllabus

MEMS and Microsystems: Introduction – components of a microsystem, Multidisciplinary nature of Microsystem design and manufacture. Modern applications of MEMS. Basic sensing and actuating principles used in MEMS based structures: Micro Sensors - Acoustic wave sensors, Biomedical sensors and Biosensors, Pressure sensors, Thermal sensors. Actuation techniques for microdevices - actuation using: thermal forces, shape memory alloys, piezoelectric crystals, electrostatic forces. MEMS with micro actuators - Micro grippers - micro motors, micro valves, micro pumps, Micro accelerometer, Microfluidics. Materials for MEMS- substrate and wafers, Silicon - properties, silicon compounds- Silicon Dioxide, Silicon carbide, Silicon Nitride and Polycrystalline Silicon, Silicon Piezoresistors, Gallium Arsenide, Quartz, Piezoelectric Crystals, Polymers- SU-8, PMMA, PDMS, Langmuir-Blodgett Films, Packaging materials.

Micro System fabrication – Photolithography, Ion implantation, Diffusion, Oxidation, Deposition- Chemical Vapor Deposition, Physical Vapor Deposition, Deposition by Epitaxy, Etching. Overview of Micro manufacturing – Bulk micro manufacturing, Surface micro machining, LIGA process.

Micro system Packaging: General considerations in packaging design, Levels of Micro system packaging – die level, device level, system level, Interfaces in microsystem packaging, Essential packaging technologies – die preparation, surface bonding, wire bonding, sealing.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	1	1	2	3	2
CO2	3	3	2	2	2	3	3
CO3	3	2	1	1	2	3	2
CO4	3	3	2	2	3	3	3
CO5	2	2	2	1	2	2	3

Textbooks / References

1. Tai-Ran Hsu – Mems & Microsystems Design and Manufacturing – John Wiley & Sons – 2008 – 2nd Edition.
2. Marc J Madou – Fundamentals of Microfabrication – CRC Press – 2002 – 2nd Edition.

3. Chang Liu, Foundations of MEMS, Pearson International Edition, 2006.
4. Introductory MEMS, Thomas M Adams, Richard A Layton, Springer International Publishers.

ECE308 Hardware for AI/ML [3-0-0-3]

Course Objectives

- To introduce the fundamentals of machine learning, neural networks, and AI computing requirements
- To provide knowledge of modern hardware platforms and accelerator architectures for machine learning
- To develop understanding of hardware-software co-design and optimization techniques
- To expose students to emerging AI hardware technologies, benchmarking methodologies, and real-world ML accelerator case studies

Course Outcomes

- CO1: Explain machine learning algorithms, neural network fundamentals, and their computational requirements
- CO2: Analyze and compare different hardware architectures for AI/ML acceleration
- CO3: Apply hardware-software co-optimization techniques and parallel computing strategies
- CO4: Evaluate AI accelerator architectures using benchmarking and profiling methodologies

Syllabus

Basics of Machine Learning and Neural Networks; Mathematical foundations that connect AI/ML algorithms to their hardware implementations; Computing need for machine learning; Overview of hardware platforms for training and inference; Systolic Arrays Architectures for ML in the cloud and at the edge; Memory systems for ML; In-memory or near-memory computing for ML; Computational profiling with Python, benchmarking across hardware platforms, in-memory computing with memristive crossbar arrays; Temporal and spatial parallelism for machine learning; Energy aware architectures for ML; ASIC design for machine learning; GPU based acceleration for ML; FPGA based acceleration for ML; Hardware-software co-optimization for machine learning; Energy, area, delay trade-offs; Case study of ML chips: Google TPU, MIT Eyeriss, emerging AI chips; ML benchmarking (MLPerf); HW/SW Co-design of AI Compute Systems;

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	2	2	3	2
CO2	3	3	3	2	2	3	2
CO3	3	3	3	3	2	3	2
CO4	3	3	3	3	3	3	3

Textbooks / References

1. FPGA Prototyping by Verilog Examples" by Pong P. Chu, John Wiley & Sons
2. Artificial Intelligence and Hardware Accelerators , Ashutosh Mishra, Jaekwang Cha, Hyunbin Park, Shiho Kim, SpringerNature
3. A collection of papers from IEEE/ACM Conferences

ECE309 Electronic Measurements and Instrumentation [3-0-0-3]

Course Objectives

- To understand the basics of measurement systems and the errors associated with it.
- To learn techniques involved in the measurement of physical quantities: displacement, force, torque, vibrations, temperature, pressure etc.

Course Outcomes

- CO1: Understand the basic concepts, types and methods of measurement
- CO2: Understand different types of instruments and their characteristics
- CO3: Understand the principle of commonly used instruments.
- CO4: Understand the basics for converting a physical parameter into an electrical quantity
- CO5: Recognize the principle of measurement of commonly used physical parameters.

Syllabus

Measurement and measurement system: Significance of measurements, method of measurements, Instruments: type of measuring instruments, classification of measuring instruments, functions of measuring instruments and measurement system, Elements of a generalized measurement system, Performance Characteristics - static characteristics and dynamic characteristics, types of errors in measurement.

Electromechanical instruments: Moving coil instruments, Moving iron instruments, DC Voltmeters, DC Ammeters, Ohmmeters, Megger. Electronic instruments: Electronic voltmeters, Digital voltmeters, Electronic multimeter (VOM), Q meter.

AC Bridges: DC bridges: Types of bridges –Wheatstone's bridge, Kelvin's bridge, AC bridges - Maxwell's bridge, Hay's bridge, Wien's bridge.

Transducers: Definition, principle of sensing and transduction, Classification of Transducers, Characteristics of Transducers, Basic requirement of transducers.

Measurement of physical parameters: Displacement measurement – electrical transducers – resistive transducer – potentiometer, Inductive transducers – LVDT, Capacitive transducers – using change in area of plates, using change in distance between plates, using change in dielectric constant, pneumatic transducer – flapper-nozzle arrangement, Pressure measurement using – diaphragm, bourdon tube, bellows, manometers, Temperature measurement using -

thermistors, thermocouples & RTD. Force measurement: Load Cell – types, basic principle, Strain Gauge – types, basic principle. Torque measurement – strain gauge torque meters, Vibration measurement – seismic transducer – piezoelectric accelerometers.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2		1	2	2	2
CO2	3	2		1	2	2	2
CO3	3	2	1	1	2	3	2
CO4	3	3	1	2	3	3	3
CO5	3	3	2	2	3	3	3

Textbooks / References

1. A. K. Sawhney, “A Course in Electrical and Electronic Measurements and Instrumentation, 19th edition, Dhanpat Rai & Co. Publications.
2. David A Bell, Electronic instrumentation and Measurements, 3rd Edition, Oxford, 2017.
3. Kalsi H S, Electronic Instrumentation and Measurements, Mc Graw hill, 4th ed., 2019.
4. A.D. Helfrick and W.D. cooper, Modern Electronic Instrumentation & Measurement Techniques, PHI – 2001.
5. DVS Murthy, Transducers and Instrumentation, PHI 2nd Edition 2013.

ECE310 Quantum Circuits and Quantum Chips [3-0-0-3]

Course Objectives

- To introduce the fundamentals of quantum technologies and quantum computing
- To provide knowledge of physical implementations of qubits and quantum hardware systems
- To develop understanding of quantum hardware design challenges,
- To familiarize students with quantum gate implementation, quantum computing applications, and emerging Quantum Electronic Design Automation (EDA) tools

Course Outcomes

- CO1: Explain the principles of quantum computing and quantum technologies
- CO2: Analyze different physical implementations of qubits and quantum systems
- CO3: Evaluate hardware challenges in scalable quantum computing systems
- CO4: Apply concepts of quantum gate implementation, quantum circuit realization, and Quantum EDA tools

Syllabus

Overview of Quantum Technologies; Quantum Computing; physical implementation of qubits, superconducting circuits; Components; Qubit Types; semiconductor

spin qubits; Semiconductor quantum dots; light-matter interaction; quantum multi-level systems; Hamiltonian engineering; decoherence and noise; control/readout electronics; Circuit QED; Control, Readout, and Fabrication; Hardware aspects of quantum computing; Challenges for scaling up quantum computers from a system-level hardware perspective; Implementation of quantum gates and circuits; Applications; Introduction to Quantum chip EDA tools.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	1	2	3	2
CO2	3	1	2	1	1	3	2
CO3	3	3	3	3	2	3	3
CO4	3	3	3	3	3	3	3

Textbooks / References

1. *Quantum Computation and Quantum Information*, Michael Nielsen and Isaac Chuang, Cambridge University Press, 2011, ISBN No. 9781107002173
2. H. Häffner, C.F. Roos, and R. Blatt, “Quantum computing with trapped ions”, Physics Reports 469 (4) 155203 (2008)
3. Alexandre Blais, Arne L. Grimsmo, S. M. Girvin, and Andreas Wallraff, “Circuit quantum electrodynamics”, Rev. Mod. Phys. 93 (2) 025005 (2021)
4. Relevant IEEE/ACM Conference papers and journal papers

ECE311 Low Power VLSI [3-0-0-3]

Course Objectives

- To understand Power Dissipation Fundamentals
- To apply Low Power Metrics and Design Flow
- To analyze Architectural Low Power Techniques
- To apply Leakage Power Reduction Techniques
- To design and Implement Power Gating Strategies

Course Outcomes

- CO1: Identify the sources of power consumption in a given VLSI Circuit.
- CO2: Calculate Power-Delay Product (PDP) and Energy-Delay Product (EDP) for given circuit configurations
- CO3: Analyze the impact of DVS, DVFS, variable frequency operation, multi-voltage domains, and architectural clock gating on system power and performance
- CO4: Compare and evaluate leakage reduction techniques
- CO5: Apply gate-level low-power implementation techniques

Syllabus

Low Power CMOS VLSI design: Introduction: Sources of Power Dissipation, Static Power Dissipation, Active Power Dissipation. Effects of scaling on power consumption- Low power design

flow- Normalized Figure of Merit – PDP& EDP- Overview of power optimization at various levels.

Architectural Techniques for Low Power: Parameters effecting power dissipation, Variable frequency, Dynamic voltage Scaling, Dynamic Voltage and Frequency Scaling, Reduced VDD, Architectural clock gating, Power gating, Multi-voltage, Optimizing memory power.

Leakage power reduction techniques-stacking techniques, sleepy keeper technique, super cut off CMOS, VTCMOS, MTCMOS, DTCMOS- energy constrained and delay constrained. Sleep Transistor Design- switch efficiency, area efficiency, IR drop, normal Vs reverse body bias.Power gating –coarse grain and fine grain. Isolation, retention, power down and wake up methods. Path balancing and hazard filtering power glitch power minimization.

Low Power Implementation Techniques: Library Selection, Clock Gating, Timing Impact due to Clock gating, Gate-level power optimization techniques, Power Optimization for Sleep Mode.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	3		3	3	3
CO2	3	3	2			3	3
CO3	3	2	3			3	3
CO4	1	2	3	3		3	3
CO5	3	3	3			3	3

Textbooks / References

1. Kaushik Roy, Sharat Prasad, Low-Power CMOS VLSI design, John Wiley, 2000.
2. L. Benini and G. De Micheli, Dynamic Power Management Design Techniques and CAD Tools, Springer, 1998.
3. K.-S. Yeo and K. Roy, Low-Voltage Low-Power Subsystems, McGraw Hill, 2004.
4. Soudris D, Pigué C and Goutis C, Designing CMOS Circuits for Low Power, Kluwer Academic Publishers, 2002.
5. Jan Rabaey, Low Power Design Essentials, Springer, 2009.

ECE312 Semiconductor Memory Devices and Circuits [3-0-0-3]

Course Objectives

- Understand Memory Hierarchy & Architecture
- Analyze SRAM Design & Stability
- Understand DRAM Technology & Scaling
- Explore Emerging Non-Volatile Memory Technologies
- Understand Compute-in-Memory (CIM) Principles

Course Outcomes

- CO1: Analyze the read, write, and hold operations of a 6T SRAM cell and compute Static Noise Margin
- CO2: Compare the storage mechanisms and scalability of emerging NVMs — PCM, RRAM, and MRAM

- CO3: Apply the principles of Compute-in-Memory to design and evaluate synaptic device requirements for analog and digital CIM architectures

Syllabus

Semiconductor Memory Technologies Overview. Introduction to Memory Hierarchy Introduction to Memory Array Architecture. Static Random Access Memory (SRAM).Principles of Hold, Read and Write. SRAM Stability Analysis. Variability and Reliability. SRAM Layout. FinFET-Based SRAM Dynamic Random Access Memory (DRAM)- Overview DRAM Technology. DRAM Scaling Trend, Embedded DRAM.

Emerging Non-volatile Memories, Phase Change Memory (PCM),Resistive Random Access Memory (RRAM),Magnetic Random Access Memory (MRAM). Compute-in-Memory-Principle of CIM, Synaptic Device Properties,CIM Prototype Chips

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	3	3	3	3	3
CO2	3	3	2			3	3
CO3	3	2	3			3	3

Textbooks / References

1. Yu, Shimeng. *Semiconductor memory devices and circuits*. CRC Press, 2022.
2. Li, Hai, and Yiran Chen. *Nonvolatile memory design: magnetic, resistive, and phase change*. CRC Press, 2017.
3. Xie, Yuan, ed. *Emerging memory technologies: design, architecture, and applications*. Springer Science & Business Media, 2013.

ECE313 Analog IC Design [3-0-0-3]

Course Objectives

- Apply gm/ID Methodology for Transistor Sizing
- Understand Biasing and Reference Circuit Design
- Analyze Frequency Response and Noise in Amplifiers
- Stability and frequency compensation in two stage opamp
- Apply Stability Analysis and Frequency Compensation in operational amplifier

Course Outcomes

At the end of the course the student will be able to

- CO1: Understand the design challenges in nanometer design.
- CO2: Understand the fundamentals of bandgap reference circuit
- CO3: Understand frequency compensation techniques of op-amp and its stability analysis

Syllabus

Introduction to gm/ID methodology for transistor sizing. Nanometer design studies- Design methodology of a

basic MOSFET amplifier with biasing. Determining voltage swing, and parameters affecting gain of Amplifier. Introduction to Biasing Circuits- Bandgap Reference circuit. Frequency response of differential amplifier. Noise- Noise Analysis in basic MOSFET Amplifier topologies, Noise in differential amplifier.

Operational amplifiers- Need for Single and Multistage amplifiers, Design of two-stage op-amp. Gain boosting technique in opamp design.

Stability and frequency compensation- Basic Concepts, Instability condition, Effect of Pole Locations on Stability, Frequency Compensation in operational amplifiers. Concepts and Techniques for Frequency Compensation. Dominant pole, Miller Compensation in two stage op-amp. Compensation of Miller RHP Zero.

Oscillator- Design issues, Need of negative resistance in oscillator designs.

Introduction to Analog layout techniques- Common centroid and Interdigitization.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	3	3	3	3	3
CO2	3	3	2	3		3	3
CO3	3	2	3			3	3

Textbooks / References

1. Behzad Razavi, Design of Analog CMOS Integrated Circuits, 2017, Second Edition, McGraw-Hill.
2. Phillip E. Allen and Douglas R. Holberg, CMOS Analog Circuit Design, 2016, Second Edition, Oxford University Press.
3. Jespers, Paul GA, and Boris Murmann. *Systematic design of analog CMOS circuits*. Cambridge University Press, 2017.
4. David Johns and Ken Martin, Analog Integrated Circuit Design, 2012, Second Edition, John Wiley & Sons, Inc.
5. Gray, P. R., Hurst, P. J., Lewis, S. H., & Meyer, 2009 Analysis and Design of Analog Integrated Circuits, 5th Edition John Wiley & Sons.

ECE314 Sensors and Actuators [3-0-0-3]

Course Objectives

- To introduce the fundamentals of sensors and actuators used in engineering systems.
- To provide knowledge of sensing, transduction, and actuation principles.
- To develop understanding of the characteristics and operation of various sensing and actuation devices.
- To impart knowledge of interfacing and performance aspects of sensors and actuators.
- To introduce modern sensing technologies and their engineering applications.

Course Outcomes

- CO1: Explain the principles and characteristics of sensors and actuators.
- CO2: Analyze the operation and performance of various sensing and actuation devices.
- CO3: Select suitable sensors and actuators for engineering applications.
- CO4: Apply sensing and actuation concepts in measurement, monitoring, and control systems.
- CO5: Explain modern trends and applications in sensor and actuator technologies.

Syllabus

Classification of sensors and actuators, sensing and actuating strategies, general requirements for interfacing and actuation, sensing, transduction, actuation. Performance Characteristics of Sensors and Actuators: Input/output characteristics, dynamic characteristics

Temperature Sensors and Thermal Actuators: Thermoresistive sensors: Thermistors, Resistance temperature sensors, Silicon resistive sensors, Thermoelectric sensors, PN junction temperature sensors, Optical and acoustic temperature sensor.

Optical sensors: Photodiodes, phototransistors and photoresistors based sensors, Photomultipliers, light to light detectors, infrared sensors - thermal, PIR, AFIR, thermopiles, CCD sensors and detectors.

Electric and Magnetic Sensors and Actuators: Motors as actuators - linear, rotational, stepping motors), magnetic valves, inductive sensors - eddy current, LVDT, RVDT, Proximity, Hall Effect sensors, Magnetoresistive sensors, Magnetostrictive sensors and actuators, Magnetometers, Voice coil actuators - speakers and speaker like actuators, Bolometers.

Mechanical Sensors and Actuators: Accelerometers - capacitive, piezoelectric, piezoresistive, thermal, Force sensors - strain gauges, tactile sensors, Pressure sensors - semiconductor, piezoresistive, capacitive, VRP, Gyroscopes.

Acoustic Sensors and Actuators: Ultrasonic sensors (piezoelectric, electromagnetic), Piezoelectric actuators, Piezoelectric Resonators, Microphones, hydrophones, speakers, buzzers.

MEMs and Smart Sensors: Micro-Electro-Mechanical (MEMs) Sensors and Actuators, Smart sensors, ASIC based sensors, Wireless Sensors and Issues Associated with Wireless Sensors

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2		1	2	2	2
CO2	3	3	1	2	2	3	3
CO3	3	3	2	2	2	3	3
CO4	3	3	2	2	3	3	3
CO5	2	2	1	1	3	2	3

Textbooks / References

1. Ida, N., Sensors, Actuators, and their Interfaces; Scitech Publishing

2. deSilva, Sensors and Actuators: Control System Instrumentation, CRC Press
3. Sensors and Transducers – D. Patranabis, PHI Learning.
4. Sensor Technology Handbook – Jon S. Wilson, Newnes Publications.
5. Measurement, Instrumentation and Sensors Handbook – John G. Webster, CRC Press.

ECE315 Nanoelectronics [3-0-0-3]

Course Objectives

After completing this course, students will be able to:

- Understand the physical principles governing electronic behavior at the nanoscale.
- Analyze the impact of quantum confinement and transport phenomena on nanoelectronic devices.
- Study fabrication techniques and characterization methods employed in nanoelectronics.
- Evaluate the operation and performance of emerging nanoelectronic devices and systems.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Explain the fundamental concepts of nanotechnology, nanoscale materials, and quantum mechanical effects relevant to nanoelectronics.
- CO2: Analyze carrier transport mechanisms, quantum confinement effects, and electronic behavior in low-dimensional nanostructures.
- CO3: Evaluate the fabrication processes and characterization techniques used in nanoelectronic device manufacturing.
- CO4: Analyze the operation, performance, and limitations of nanoelectronic devices such as CNTFETs, SETs, molecular devices, and quantum-dot-based devices.
- CO5: Assess the applications and future prospects of nanoelectronics in advanced electronic systems.

Syllabus

Introduction (from classical electronics to nanoelectronics) Wave-particle duality, Schrödinger wave equation, Materials for nanoelectronics - Semiconductors - Carbon nanomaterials nanotubes. Electrons in low-dimensional structures Electrons in quantum wells - Electrons in quantum wires - Electrons in quantum dots, Fabrication of nanostructures - Crystal growth - Nanolithography Clusters and nanocrystals - Nanotube growth Characterization of nanostructures Electron transport in semiconductors and nanostructures - Time and length scales of the electrons in solids - Statistics of the electrons in solids and low dimensional structures - Electron transport in nanostructures Nanoelectronic devices - Resonant-tunneling diodes – Field effect transistors - Single-electron transfer devices.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	3	2			2	1
CO2	3	3	2	1		3	1
CO3	3	3	3	2		3	2
CO4	3	2	2	3	1	3	2
CO5	3	2	2	3	1	3	2

Textbooks / References

1. Vladimir V. Mitin, Viatcheslav A. Kochelap, Michael A. Strosio, “Introduction to Nanoelectronics”, Cambridge University Press
2. George W. Hanson, “Fundamentals of nanoelectronics”, Pearson/Prentice. Quantum Mechanics
3. D. J. Griffiths, “Introduction to quantum mechanics (2nd edition)”, Prentice Hall. Solid- state physics
4. R. Turton, “The physics of solids”, Oxford, 2000. Semiconductor physics and devices
5. S. M. Sze, “Physics of Semiconductor Devices (2nd)”, Wiley
6. S. Datta, “Lessons from Nanoelectronics: A New Perspective on Transport (Lessons from Nanoscience: a Lecture Notes Series) World Scientific

ECE316 Advanced Communication Networks [3-0-0-3]

Course Objectives

- To understand the protocol layering and physical level communication.
- To analyze the performance of a network.
- To understand the various components required to build different networks.
- To learn the functions of network layer and the various routing protocols.

Course Outcomes

- CO1: Understand the basic layers and its functions in Communication networks.
- CO2: Design and develop protocols for Communication Networks
- CO3: Understand the basics mechanisms in Quality of Service in networking
- CO4: Analyze and design routing algorithms

Syllabus

Overview of - ATM. TCP/IP Congestion and Flow Control in Internet-Throughput analysis of TCP congestion control. TCP for high bandwidth delay networks. Fairness issues in TCP. Real Time Communications over Internet, Adaptive applications. Latency and throughput issues, Integrated Services Model (intServ), Resource reservation in Internet, RSVP, Characterization of Traffic by Linearly Bounded Arrival Processes (LBAP), Leaky bucket algorithm and its properties. Packet Scheduling Algorithms requirements and choices, Scheduling guaranteed service connections. GPS, WFQ and Rate propor-

tional algorithms. High speed scheduler design. Theory of Latency Rate servers and delay bounds in packet switched networks for LBAP traffic.; Active Queue Management - RED, WRED and Virtual clock. Control theoretic analysis of active queue management. IP address lookup-challenges. Packet classification algorithms and Flow IdentificationGrid of Tries, Cross producting and controlled prefix expansion algorithms. Admission control in Internet. Concept of Effective bandwidth. Measurement based admission control. Differentiated Services in Internet (DiffServ). DiffServ architecture and framework. IPV4, IPV6, IP tunnelling, IPswitching and MPLS, Overview of IP over ATM and its evolution to IP switching. MPLS architecture and framework. MPLS Protocols. Traffic engineering issues in MPLS.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	1	2	2			2	
CO2	2	2	3	1		3	1
CO3	2	2	3			3	
CO4	2	3	3	1		3	1

Textbooks / References

1. Jean Wairand and Pravin Varaiya, “High Performance Communications Networks”, 2nd edition, 2000.
2. Jean Le Boudec and Patrick Thiran, “Network Calculus A Theory of Deterministic Queueing Systems for the Internet”, Springer Veriag, 2001
3. Zhang Wang, “Internet QoS”, Morgan Kaufman, 2001
4. Anurag Kumar, D. Manjunath and Joy Kuri, “Communication Networking: An Analytical Approach”, Morgan Kaufman Publishers, 2004.
5. George Kesidis, “ATM Network Performance”, Kluwer Academic, Research Papers, 2005

ECE317 Error Control Coding [3-0-0-3]

Course Objectives

- To introduce the principles of error control coding for reliable digital communication and storage systems.
- To study the construction, encoding and decoding of linear block codes, cyclic codes, BCH codes, Reed–Solomon codes and convolutional codes.
- To understand modern coding techniques including Turbo codes and LDPC codes, and their decoding algorithms used in contemporary communication systems.

Course Outcomes

- CO1: Analyze the error detection and error correction capabilities of linear block and cyclic codes.
- CO2: Apply encoding and decoding techniques for BCH, Reed–Solomon and convolutional codes using suitable decoding algorithms.
- CO3: Explain the principles of Turbo and LDPC codes and evaluate their role in modern digital communication systems.

Syllabus

Coding for Reliable Digital Transmission and Storage, Introduction to Algebra, Linear Block Codes, Generator Matrices and Parity-Check Matrices, Syndrome Decoding, Important Linear Block Codes, Hamming Codes, Cyclic Codes, Encoding and Decoding of Cyclic Codes. Binary BCH Codes, Nonbinary BCH Codes, Reed–Solomon Codes, Decoding Algorithms for BCH and Reed–Solomon Codes, Distance Properties of Codes, Error-Correcting Capability of Block Codes. Convolutional Codes, State Diagram and Trellis Diagram, Distance Properties of Convolutional Codes, Trellis-Based Decoding Algorithms for Convolutional Codes, The Viterbi Algorithm, Performance of Convolutional Codes. Turbo Coding, Low Density Parity Check Codes, Iterative Decoding, Tanner Graph Representation, Message-Passing Decoding Algorithms, Applications of Error Control Coding in Digital Communication Systems

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	1		3	2
CO2	3	3	2	2		3	3
CO3	2	2	1	2		2	3

Textbooks / References

1. Error Control Coding: Fundamentals and Applications, Shu Lin and Daniel J. Costello Jr., *Error Control Coding: Fundamentals and Applications*, 2nd Edition, Pearson Prentice Hall, 2004.
2. Fundamentals of Error-Correcting Codes, W. Cary Huffman and Vera Pless, *Fundamentals of Error-Correcting Codes*, Cambridge University Press, 2003.
3. Channel Codes: Classical and Modern, William E. Ryan and Shu Lin, *Channel Codes: Classical and Modern*, Cambridge University Press, 2009.
4. Modern Coding Theory, Tom Richardson and Rüdiger Urbanke, *Modern Coding Theory*, Cambridge University Press, 2008.

ECE318 Information Theory and Coding [3-0-0-3]

Course Objectives

- To equip students with the basic understanding of the fundamental concept of entropy and information as they are used in communications.
- To enhance knowledge of probabilities, entropy, measures of information.
- To guide the student through the implications and consequences of fundamental theories and laws of information theory and coding theory with reference to the application in modern communication and computer systems.

Course Outcomes

- CO1: Calculate the information content of a random variable from its probability distribution.

- CO2: Relate the joint, conditional, and marginal entropies of variables in terms of their coupled probabilities.
- CO3: Define channel capacities and properties using Shannon's Theorems.
- CO4: Construct efficient codes for data on imperfect communication channels.
- CO5: Generalize the discrete concepts to continuous signals on continuous channels.

Syllabus

Entropy, Relative Entropy, Mutual information and its properties, entropy for discrete ensembles, Shannon's source coding theorem, Encoding of discrete sources, Kraft Inequality, Huffman Codes, Shannon-Fano-Elias Coding, Arithmetic coding, Lempel-Ziv-Welch Coding. Channel capacity, Capacity of binary symmetric channel, erasure channels and other discrete channels, Shannon's noisy coding theorem and converse for discrete channels, Differential entropy, Application to continuous channels: Gaussian channel, Bandlimited channel, channels with coloured noise.

Introduction to Galois field, addition and multiplication tables of Galois field elements, Linear block codes, Error detection and error correction capabilities of Linear block codes, parity check codes, Hamming codes, encoding and decoding of linear codes, syndrome decoding of linear codes, Cyclic codes, Convolutional codes, Graphical approach: state, tree and trellis diagram, Viterbi decoding; Maximum likelihood decoding; Introduction to iterative codes and its sub-optimal decoding algorithms, BCH codes, RS codes, LDPC Codes.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	1		3	3
CO2	3	3	2	1		3	3
CO3	3	3	2	1		3	3
CO4	3	3	2	1		3	3
CO5	3	3	2	1		3	3

Textbooks / References

1. T. M. Cover and J. A. Thomas, *Elements of Information Theory*, Wiley.
2. R. W. Yeung, *A First Course in Information Theory*, Kluwer Academic Publisher.
3. R. E. Blahut, *Algebraic Codes for Data Transmission*, Cambridge University Press.
4. S. Lin and D. J. Costello, *Error Control Coding*, 2nd ed., Pearson Prentice Hall.

ECE319 Multimedia Communication [3-0-0-3]

Course Objectives

- To introduce the fundamentals of multimedia systems, multimedia data types, and networking requirements for audio and video communication.

- To provide knowledge of multimedia coding, compression techniques, broadband communication technologies, and Quality of Service (QoS) mechanisms for multimedia applications.
- To familiarize students with multimedia transport protocols, multicast communication, VoIP systems, and collaborative multimedia applications such as video conferencing and peer-to-peer systems.

Course Outcomes

- CO1: Analyze the characteristics and networking requirements of multimedia data including text, image, audio, and video.
- CO2: Explain multimedia compression standards, multimedia transport protocols, QoS mechanisms, and multimedia communication technologies used in modern networks.
- CO3: Design and evaluate basic multimedia networking applications involving VoIP, multicast communication, peer-to-peer systems, and video conferencing

Syllabus

Introduction to Multimedia: Digital sound, video and graphics, multimedia characteristics, basic multimedia networking, evolution of Internet service models, network requirements for audio and video. Multimedia Coding and Compression: Text, image, audio and video compression techniques, multimedia coding standards, basics of MPEG standards.

Broadband and Multimedia Communication: Broadband services, ATM and IP fundamentals, IPv6 basics, multimedia traffic characteristics, throughput, delay and jitter performance. Quality of Service for Multimedia: Resource reservation concepts, buffering and traffic shaping, scheduling and caching techniques, storage and media services.

Voice and Video over IP: VoIP fundamentals, video transmission over IP networks, MPEG-2 over ATM/IP, synchronization of multimedia streams, indexing and recording concepts.

Multicast and Multimedia Transport: Multicast communication fundamentals, multicast routing and addressing, RTP and RTCP protocols, MIME support for multimedia applications.

Collaborative Multimedia Applications: Peer-to-peer multimedia systems, shared applications, video conferencing concepts, centralized and distributed conference control, distributed virtual reality fundamentals.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	3	3	3		3	
CO2	2	2	3	3		3	
CO3	2	3	3	3	1	3	1

Textbooks / References

1. Ivan Vidal, Ignacio Soto, Albert Banchs, Jaime Garcia, Ivan Lozano, Gonzalo Camarillo, *Multimedia Network-*

- ing Technologies, Protocols, Architectures, in Artech Publishers, 2019.
2. B O Szuprowicz, "Multimedia Networking", McGraw Hill, Newyork, 1995
 3. K R Rao, Zoran S, Bojkovic and Dragorad A, Milovanovic "Multimedia communication systems", PHI, 2003
 4. Jon Crowcroft, Mark Handley, Ian Wakeman "Internet-working Multimedia" Harcourt, Singapore, 1998
 5. Tay Vaughan, "Multimedia Making it to work", 4th edition Tata McGraw Hill, New Delhi, 2000

ECE320 Digital Signal Processing for Communication [3-0-0-3]

Course Objectives

- To introduce the fundamentals of digital signal processing and its applications in communication systems.
- To understand discrete-time signal analysis, transform techniques, and spectral analysis methods used in communication engineering.
- To study digital filter design and adaptive signal processing techniques for communication applications.
- To analyze DSP techniques used in modulation, synchronization, channel estimation, and equalization.
- To explore modern DSP-based communication system implementation techniques and emerging applications.

Course Outcomes

After successful completion of the course, students will be able to:

- CO1: Explain the principles of discrete-time signal processing, transforms, and spectral analysis techniques used in communication systems.
- CO2: Analyze communication signals using DFT, FFT, and digital spectral estimation techniques for efficient signal processing.
- CO3: Design FIR and IIR digital filters and evaluate their performance in communication signal enhancement and interference mitigation.
- CO4: Apply adaptive filtering, multirate signal processing, synchronization, and equalization techniques in digital communication systems.
- CO5: Analyze and implement DSP algorithms for modern communication applications including OFDM, wireless receivers, software-defined radio, and broadband communication systems.

Syllabus

Fundamentals of DSP for Communication: Introduction to DSP in communication systems, discrete-time signals and systems, sampling theorem and aliasing, discrete-time convolution and correlation, difference equations and system realization, Z-transform and its properties, frequency response of discrete-time systems, DSP hardware and software tools for communication applications. Spectral Analysis and Transform Techniques: Discrete

Fourier Transform (DFT) and its properties, Fast Fourier Transform (FFT) algorithms, spectral leakage and windowing techniques, power spectral density estimation, short-time Fourier transform (STFT), time-frequency analysis methods, DSP implementation of spectral analysis in communication systems.

Digital Filter Design and Adaptive Processing: FIR filter structures and design techniques, window method and frequency sampling method, IIR filter design using Butterworth and Chebyshev approximations, bilinear transformation and impulse invariance methods, finite word-length effects, adaptive filtering principles, LMS and RLS algorithms, noise cancellation and echo suppression techniques.

DSP Techniques in Digital Communication Systems: Matched filtering and optimum detection, synchronization techniques, carrier and timing recovery, channel estimation methods, equalization techniques for wireless channels, multirate signal processing, interpolation and decimation, OFDM signal processing fundamentals, DSP implementation of modulation and demodulation techniques. Modern DSP-Based Communication Applications: DSP in wireless communication systems, MIMO signal processing, software-defined radio (SDR), cognitive radio concepts, DSP for broadband and multimedia communication, DSP implementation issues in real-time systems, FPGA and DSP processor-based communication systems, AI-assisted signal processing, future trends in communication signal processing.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	3	3	3		3	
CO2	3	3	3	3		3	
CO3	2	3	3	3		3	
CO4	2	3	3	2		3	
CO5	2	3	3	2		3	

Textbooks / References

1. Alan V. Oppenheim and Ronald W. Schaffer, Discrete-Time Signal Processing, Pearson Education, 3rd Edition, 2009.
2. John G. Proakis and Dimitris G. Manolakis, Digital Signal Processing: Principles, Algorithms and Applications, Pearson Education, 4th Edition, 2007.
3. Simon Haykin, Adaptive Filter Theory, Pearson Education, 5th Edition, 2013.
4. Richard G. Lyons, Understanding Digital Signal Processing, Pearson Education, 3rd Edition, 2011.
5. Paolo Prandoni and Martin Vetterli, Signal Processing for Communications, CRC Press, 2008.
6. John G. Proakis and Masoud Salehi, Digital Communications, McGraw-Hill Education, 5th Edition, 2008.
7. Travis F. Collins, Robin Getz, Di Pu and Alexander M. Wyglinski, Software Defined Radio for Engineers, Artech House Publishers, 2018.

ECE321 RF and Microwave Circuit Design [3-0-0-3]

Course Objectives

- To understand RF and Microwave design concepts and implement high frequency transceiver systems
- To analyze various components of RF and Microwave communication systems
- To analyze various parameters in RF and Microwave transceiver circuit design

Course Outcomes

- CO1: To design and analyse passive and active RF and Microwave circuits
- CO2: To characterize two-port networks using network parameters
- CO3: To characterize microwave amplifiers, power dividers, filters, and couplers
- CO4: To characterize RF and Microwave oscillators and mixers

Syllabus

Significance of RF and Microwave design, Passive Components: High frequency resistors, capacitors, and inductors. Chip components and circuit board considerations, Surface-Mounted Inductors.

Basic resonator and filter configurations: Filter type and parameters, Low-Pass Filter, High Pass Filter, Bandpass and Bandstop Filters, Insertion Loss, Special Filter Realizations: Butterworth –Type, Chebyshev Filter, Microstrip filter design

Impedance matching using discrete components: Two component matching networks, Frequency response and Quality Factor, Microstrip Line Matching Networks: Discrete Components, Microstrip Lines, Single-Stub Matching Networks, Double-Stub Matching Networks, Amplifier Classes of Operation and Biasing Network: Classes of Operation and Efficiency of Amplifiers.

Characteristics of Amplifiers, Amplifier Power Relations: RF source, Transducer Power Gain, Stability Considerations: Stability Circles, Unconditional Stability, Stabilization Methods, Constant Gain: Unilateral Design, Unilateral Figure of Merit, Available Power Gain Circles, Noise Figure Circles, Constant VSWR Circles.

Basic Oscillator Model: Negative Resistance Oscillator, Feedback Oscillator Design, Quartz Oscillators. High Frequency Oscillator Configuration: Fixed Frequency Oscillators, Dielectric Resonator Oscillators, YIG-Tuned Oscillators, Voltage Controlled Oscillators, Gunn Element Oscillator. Basic Characteristics of Mixers: Basic Concepts, Frequency Domain Considerations, Single-Balanced Mixer.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	1	1	3	2
CO2	3	2	2	1	1	3	1
CO3	3	2	2	1	1	3	1
CO4	3	2	2	1	1	3	1

Textbooks / References

1. Reinhold Ludwig, RF circuit design, 2nd edition, Prentice Hall 2014
2. RF Circuit Design Theory and Application, Reinhold Ludwig and Pavel Bretchko, Ed. 2006, Pearson Education
3. Microstrip Filters for RF/Microwave Applications, Jia-Sheng Hong, M. J. Lancaster, John Wiley & Sons, 2001.
4. David. M. Pozar, Microwave Engineering, 4th edition, John Wiley, 2011

ECE401 Digital Control Systems [3-0-0-3]

Course Objectives

- To introduce the fundamentals of digital control systems and mathematical modelling of discrete-time systems.
- To provide knowledge on sampling, reconstruction techniques, pulse transfer functions, and realization methods of digital filters.
- To develop analytical skills for evaluating the time response, steady-state performance, and stability of discrete-time control systems.
- To impart design methodologies for digital controllers and compensators including lag, lead, lag-lead, and PID controllers.
- To introduce modern state-space methods for modelling, analysis, controllability, observability, and digital controller/observer design of discrete-time systems.

Course Outcomes

- CO1: Explain the concepts of digital control systems, sampling processes, reconstruction techniques, and digital filter realization methods.
- CO2: Derive pulse transfer functions and analyze discrete-time systems using z-transform techniques and mapping between sss-domain and zzz-domain.
- CO3: Analyze the time response, steady-state error, and stability of discrete-time systems using Jury test, bilinear transformation, and Routh stability criterion.
- CO4: Design digital compensators and PID controllers to achieve desired system performance specifications.
- CO5: Develop state-space models of discrete-time systems and design digital controllers and state observers using controllability, observability, and pole placement techniques.

Syllabus

Basic digital control system, Mathematical modelling,

sampling and reconstruction -Zero order and First order hold circuits, the pulse transfer function, realisation of digital filters. Relation between transfer function and pulse transfer function, Mapping between s-domain and z-domain.

Pulse transfer function of different configurations of systems, Modified z-transform, Time response of discrete time system, Order and type of a system, Steady state error and Static error constants.

Stability of Discrete Time System: Jury stability test, Stability Analysis by use of the Bilinear Transformation and Routh Stability criterion.

Modern control approach to digital control: state space modelling of discrete time system, Computation of solution of state equation and state transition matrix, Controllability and observability of discrete time systems, Design of state feedback controller through pole placement – Necessary and sufficient conditions – Ackerman's formula – Design of state observers (Full Order and Reduced Order).

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	1	1	3	3	2
CO2	3	3	1	2	3	3	3
CO3	3	3	2	2	3	3	3
CO4	3	3	3	2	3	3	3
CO5	3	3	3	3	3	3	3

Textbooks / References

1. K. Ogata, Discrete-Time Control systems, 2nd Edition. PHI, 2002.
2. M. Gopal, Digital Control Engineering, Wiley Eastern, 1988.
3. G. F. Franklin, J. D. Powell and M. L. Workman, Digital Control of Dynamic Systems, Addison-Wesley, 1998.
4. B.C. Kuo, Digital Control System, Holt, Rinehart and Winston, 1980

ECE402 Data Compression [3-0-0-3]

Course Objectives

- To impart the fundamental concepts of lossless coding techniques
- To uncover the principles of lossy compression techniques such as vector quantization, differential encoding and transform coding
- To provide knowledge about compression of image, video and audio
- To analyse lossy compression standards for JPEG, MPEG, MP 3 and AAC

Course Outcomes

- CO1: Learn basics of data compression and lossless coding techniques/standards

- CO2: Apply lossy coding techniques such as vector quantization, differential encoding, transform coding and wavelet transforms for data compression
- CO3: Assess image, video and audio compression methods and standards

Syllabus

Need for data compression, Lossy and Lossless compression. Lossless encoding techniques - Run length encoding, Huffman coding, Adaptive Huffman Coding, Arithmetic coding, Arithmetic Coding - Adaptive Arithmetic coding, Dictionary Techniques –Lempel-Ziv coding: LZ77, LZ 78, LZW coding, Lossless Compression standards: gzip, gif, tiff, png.

Scalar quantization, Vector quantization, Advantages of VQ over SQ, Differential encoding: DPCM, Adaptive DPCM, Delta modulation. Transform coding: Discrete Cosine Transform (DCT), MDCT, K-L Transform, Quantization and coding of transform coefficients, JPEG, Wavelet based compression: Fundamentals of wavelets, Various standard wavelet, bases, JPEG 2000

Image and Video compression: Introduction to Video Compression along with video encoder decoder block diagram, Motion Estimation and Compensation, Few Fast Motion Estimation Schemes, Video compression Standards: MPEG-1,2,4

Audio Coding: Human audio perception, Temporal and Spatial Masking, MPEG Audio Coding: MP3, AAC

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	1		3	2
CO2	3	2	2	1		3	2
CO3	3	3	3	1		3	3

Textbooks / References

1. Khalid Sayood, "Introduction to Data Compression", 4/e, Elsevier, 2012
2. David Salomon, "Data Compression: The Complete Reference", Springer.
3. Thomas M. Cover, Joy A. Thomas, "Elements of Information Theory," Wiley

ECE403 Optical Wireless Communication [3-0-0-3]

Course Objectives

- To introduce the concepts of optical wireless channel models.
- To impart knowledge on the modulation techniques used in optical wireless systems.
- To provide insights on concepts of indoor and outdoor optical wireless communication.

Course Outcomes

- CO1: Ability to understand the optical wireless channel models

- CO2: Ability to apply the concepts of communication theory for optical wireless systems
- CO3: Ability to analyze the performance of optical wireless communication systems

Syllabus

Introduction to optical wireless communication systems - OWC/Radio comparison - Optical sources - Optical detectors - Photo-detection noises - Indoor optical wireless channel model - Outdoor optical wireless channel.

Analogue Intensity Modulation – Digital Baseband modulation techniques - Pulse Position Modulation – Subcarrier Intensity Modulation – OFDM for Optical wireless communication - Optical Polarization Shift Keying.

Effect of ambient light sources on Indoor optical wireless channel - Link Performance for Multipath Propagation - Mitigation Techniques - FSO Link Performance under the Effect of Atmospheric Turbulence - Outdoor OWC Links with Diversity Techniques - Aperture Averaging - Visible Light Communications

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	1		3	3
CO2	3	3	3	1	1	3	3
CO3	3	3	3	2		3	3

Textbooks / References

1. Z. Ghassemlooy, W. Popoola, S. Rajbhandari, *Optical Wireless Communications*, Second Edition, Cambridge University Press, 2017.
2. George Karagiannidis, John Barry, Murat Uysal, Robert Schober, Shlomi Arnon, *Advanced Optical Wireless Communication Systems*, Cambridge University Press, 2012.
3. Anthony Boucouvalas, Carlo Capsoni, Eszter Udvary, Murat Uysal, Zabih Ghassemlooy, *Optical Wireless Communications An Emerging Technology*, Springer International Publishing, 2016.
4. Zabih Ghassemlooy, Luis Nero Alves, Stanislav Zvanovec, Mohammad-Ali Khalighi, *Visible Light Communications Theory and Applications*, First Edition, CRC Press, 2019.

ECE404 Adaptive Signal Processing [3-0-0-3]

Course Objectives

- To introduce the fundamental concepts of adaptive signal processing, including stochastic signals and optimal filtering principles
- To develop an understanding of key adaptive algorithms such as LMS and RLS and their convergence behavior
- To expose students to practical applications of adaptive filters in areas such as noise cancellation, system identification, and prediction

Course Outcomes

- CO1: Analyze adaptive filtering problems using Wiener filter theory and statistical signal models
- CO2: Implement and evaluate adaptive algorithms (e.g., LMS, RLS) for given signal processing tasks
- CO3: Apply adaptive filtering techniques to solve real-world problems such as noise reduction and system modeling

Syllabus

Adaptive systems: definition and characteristics, areas of application, general properties, open- and closed-loop adaptation, and example of an adaptive system. The adaptive linear combiner: input signal and weight vectors, desired response and error, the performance function, gradient and minimum mean-square error, and performance surface.

Theory of adaptation with stationary signals: properties of the quadratic performance surface, input correlation matrix, eigenvalues and eigenvectors, and geometric significance. Searching the performance surface: basic ideas of gradient search methods, steepest descent method, stability and rate of convergence, and learning curves.

Adaptive algorithms and structures: the LMS algorithm, derivation, convergence of the weight vector, learning curve, misadjustment, and performance. Overview of other adaptive algorithms and structures: LMS/Newton algorithm, sequential regression algorithm, adaptive recursive filters, and lattice structures.

Applications: adaptive modeling and system identification, adaptive modeling of communication channels, and FIR digital filter synthesis. Inverse adaptive modeling, deconvolution, and equalization. Adaptive interference canceling: concept of adaptive noise canceling and stationary noise-canceling solutions. Introduction to adaptive arrays and beamforming: sidelobe canceling and spatial configurations.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	1		3	2
CO2	3	3	2	3		3	3
CO3	2	3	2	3		3	3

Textbooks / References

1. Adaptive Signal Processing, Bernard Widrow and Samuel D. Stearns, Prentice Hall, 1985.
2. Adaptive Filter Theory, Simon Haykin, Prentice Hall.
3. Adaptive Filters: Theory and Applications, B. Farhang-Boroujeny, Wiley.
4. Theory and Design of Adaptive Filters, J. R. Treichler, C. R. Johnson Jr., and M. G. Larimore, Prentice Hall.
5. Fundamentals of Adaptive Filtering, Ali H. Sayed, Wiley.
6. Adaptive Signal Processing: Theory and Applications, S. Thomas Alexander, Springer.
7. Discrete-Time Signal Processing, A. V. Oppenheim and R. W. Schaffer, Prentice Hall.

ECE405 Neuromorphic Computing [3-0-0-3]

Course Objectives

- To introduce the fundamentals of volatile and non-volatile memory technologies
- To provide understanding of the operating principles, characteristics, and applications of emerging memory technologies
- To develop knowledge of brain-inspired and neuromorphic computing paradigms
- To familiarize students with neuromorphic hardware architectures and real-world applications

Course Outcomes

- CO1: Explain the architecture, operation, and performance characteristics of SRAM, DRAM, and emerging non-volatile memory technologies
- CO2: Analyze the advantages, limitations, and unconventional applications of emerging non-volatile memories
- CO3: Apply concepts of brain-inspired computing, spiking neural networks, and event-based computation
- CO4: Evaluate neuromorphic hardware architectures and emerging memory-based computing platforms

Syllabus

Volatile memory technology: Static Random-Access Memory (SRAM) Dynamic RAM (DRAM); Introduction to Emerging non-volatile memories (eNVM): Resistive RAM (RRAM); Phase change memory (PCM); Ferroelectric RAM (FeRAM); Spin-transfer Torque Magnetic RAM (STT-MRAM); Unconventional Application of eNVMS; Brain-inspired computing principles; Computing with neurons; Learning in neural systems; Event-based sensing and computing; Spiking neural networks; Neuromorphic hardware; Real-world applications; Recent trends from journals and conferences.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	1	1	1	3	2
CO2	3	2	3	1	2	3	1
CO3	3	3	2	2	2	3	2
CO4	3	3	3	3	2	3	3

Textbooks / References

1. Y. Nishi and Magyari-Kope, “Advances in non-volatile memory and storage technology,” Woodhead Publishing, 2019.
2. R. Micheloni, “3D Flash memories,” Springer Netherlands, 2016.
3. IEEE/ACM Publications

ECE406 Mixed Signal Design [3-0-0-3]

Course Objectives

- To understand switched capacitor circuits, sample and hold circuits
- To design comparators
- To analyze ADC architectures
- To analyze DAC architecture
- To study PLL and frequency synthesizers.

Course Outcomes

- CO1: Analyze switched-capacitor building blocks
- CO2: Interpret comparator specifications and design op-amp-based comparator circuits.
- CO3: Compare Flash, SAR, and Pipeline ADC architectures for speed, power, and resolution trade-offs.
- CO4: Implement DAC topologies
- CO5: Study PLL operation and design basic frequency synthesizers for clock generation and communication systems

Syllabus

Switched capacitor circuits-Basic Building Blocks, Basic Operation and Analysis, Noise in Switched-Capacitor Circuits, First-Order Filters, Biquad Filters, Charge Injection, Switched-Capacitor Gain Circuits. Performance of Sample-and-Hold Circuits, Testing Sample and Holds, MOS Sample-and-Hold Basics, Examples of CMOS S/H Circuits.

Introduction to comparators- Comparator Specifications, Using an Opamp for a Comparator

Introduction to data converters- Ideal D/A Converter, Ideal A/D Converter, Quantization Noise, Signed Codes, Performance Limitations, Resolution, Offset and Gain Error, Accuracy and Linearity.

Analog to digital conversion techniques- Pipeline ADC SAR ADC, Flash ADC. Digital to analog conversion techniques. -DAC-R-2R Topologies for DACs, current steering, sigma delta DAC. Phase locked loops and frequency synthesizers.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	3			3	2
CO2	3	3	2		3	3	3
CO3	3	2	3	3		3	2
CO4	2	3	3			3	3
CO5	2	3	3			3	3

Textbooks / References

1. David Johns, Ken Martin, “Analog Integrated Circuit Design”, John Wiley and Sons, 1997.
2. Behzad Razavi, “Design of Analog CMOS Integrated Circuits”, McGraw Hill, 2000.
3. Behzad Razavi, “Principles of Data Conversion System Design”, Wiley-IEEE Press, 1994.
4. Rudy van de Plassche, “Integrated Analog-to-Digital and Digital-to-Analog Converters”, Springer, 2003.
5. Behzad Razavi, “Design of CMOS phase-locked loops: from circuit level to architecture level”. Cambridge University Press, 2020.

ECE407 Quantum Photonic Integrated Circuits [3-0-0-3]

Course Objectives

- Theory, design, fabrication, and application aspects of photonic materials and devices
- Introduction of essential concepts required to understand the operation of various integrated photonic components and draws a parallel with bulk components.

Course Outcomes

After the completion of the course, the student will

- CO1: Be able to design and analyze an integrated optic waveguide
- CO2: Understand the working of various photonic components
- CO3: Be able to choose the technology suitable for the intended device.
- CO4: Be ready to understand current developments.

Syllabus

Integrated Optics: An overview, Integrated Optical circuits, Optical waveguide theory- Slab and channel waveguides, Symmetric and Asymmetric, Analysis techniques: Effective index method, Marcatilli's method, Numerical methods.

Photonic waveguide components "Directional couplers, Coupled mode theory, tapers, bends, gratings, electro-optic, acousto-optic, magneto-optic devices, modulators, switches, polarizers, filters.

Technology: Materials "glass, lithium niobate, silicon, compound semiconductors, polymers, fabrication "lithography, ion-exchange, deposition, diffusion, process and device characterization.

Optical resonators (Fabry Perot cavity, ring resonators, applications etc.), Wave propagation in periodic media (Bragg reflectors, photonic crystals and sub-wavelength structures).

Micro-opto- electro-mechanical systems, Quantum Cryptography.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	3	1	2	2
CO2	2	2	1			3	2
CO3	3	2	2	3	1	3	3
CO4	2	1	3	3	2	1	3

Textbooks / References

1. C. R. Pollock and M. Lip Son, Integrated Photonics, Kluwer Pub., 2003.
2. T. Tamir, (ed), Guided-wave optoelectronics, (2nd edition), Springer-Verlag, 1990
3. H. Nishihara, M. Haruna, and T. Suhara, Optical Integrated Circuits, McGraw-Hill, 1988
4. E. J. Murphy, (Editor), Integrated Optical Circuits and Components: Design and Applications, Marcel and Dekker, 1999.

ECE408 VLSI Signal Processing [3-0-0-3]

Course Objectives

- To introduce the architectural aspects of VLSI implementation of digital signal processing systems.
- To provide knowledge of pipelining, parallel processing, retiming, and unfolding techniques for high speed DSP architectures.
- To develop understanding of fast FIR and IIR filter architectures used in VLSI signal processing systems.
- To impart knowledge of arithmetic architectures and distributed arithmetic techniques for efficient DSP hardware realization.
- To introduce low-power and clocking techniques used in modern VLSI DSP systems.

Course Outcomes

- CO1: Analyze DSP architectures using data flow graphs, critical path analysis, and iteration bound concepts.
- CO2: Apply pipelining, parallel processing, retiming, and unfolding techniques for high-speed VLSI DSP implementations.
- CO3: Implement fast FIR and IIR filter architectures for efficient signal processing applications.
- CO4: Analyze arithmetic architectures including bit-serial, parallel, and distributed arithmetic-based DSP realizations.
- CO5: Explain low-power and clocking techniques used in VLSI signal processing architectures.

Syllabus

Introduction to VLSI DSP Systems: Introduction to DSP systems and typical DSP algorithms, Data flow and dependence graphs, Critical path, Iteration bound, Longest path concepts, Introduction to pipelining and parallel processing, Pipelining and parallel processing of FIR filters, Low-power considerations in DSP systems.

Architectural Transformations: Retiming – definitions and properties, Unfolding – concepts and algorithm, Properties and applications of unfolding, Sample period reduction, Parallel processing applications, Introduction to algorithmic strength reduction in DSP architectures.

Architectural Transformations: Fast FIR filter architectures, Parallel FIR filters, Introduction to fast convolution techniques, Basic DCT architectures, Introduction to rank-order filters and sorting architectures.

Pipelined and Parallel Recursive Filters: Look-ahead pipelining in first-order IIR filters, Power-of-2 decomposition concepts, Parallel processing of IIR filters, Combined pipelining and parallel processing techniques for recursive systems.

Arithmetic Architectures for VLSI DSP: Bit-level arithmetic architectures, Parallel multipliers, Carry-save multiplication concepts, Bit-serial multiplication, CSD representation and multiplication, Distributed arithmetic fundamentals and FIR filter implementation, Introduction to synchronous pipelining and clocking styles.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	1	2	3	3	3
CO2	3	3	3	2	3	3	3
CO3	3	3	3	2	3	3	3
CO4	3	3	2	2	3	3	3
CO5	2	2	2	1	3	3	2

Textbooks / References

1. K. K. Parhi, VLSI Digital Signal Processing Systems, Design and Implementation, John Wiley, 1st Edition, 1999.
2. U. Meyer-Baese, Digital Signal processing with Field Programmable Arrays, Springer, 3rd Edition 2007.
3. S. Ramachandran, Digital VLSI systems design. Springer, 2007.
4. V. K. Madiseti, VLSI Digital Signal Processors: An Introduction to Rapid Prototyping and Design Synthesis, IEEE Press, 1st Edition, 1995.
5. S. Y. Kung, and H. J. Whitehouse, VLSI and Modern Signal Processing, Prentice Hall, 1st Edition, 1985.

ECE409 Characterization and functionalization of nanomaterials for electronic applications [3-0-0-3]

Course Objectives

- The course aims to provide students with a comprehensive understanding of nanomaterial characterization techniques, surface and interface engineering, and functionalization strategies for electronic applications.
- Students will gain knowledge of structural, morphological, spectroscopic, thermal, magnetic, and electrochemical characterization methods used for nanoscale materials and devices.
- The course also introduces advanced microscopy and analytical techniques for correlating nanostructure, properties, and performance in electronic applications.

Course Outcomes

- CO1: Explain the principles of nanomaterial characterization techniques.
- CO2: Analyze the structural, optical, thermal, magnetic, and electrochemical properties of nanomaterials.
- CO3: Interpret characterization data obtained from diffraction, spectroscopy, and microscopy techniques.
- CO4: Evaluate nanomaterial functionalization methods for electronic applications.
- CO5: Correlate nanomaterial properties with the performance of electronic devices.

Syllabus

Nanomaterials and nanoscale phenomena – Classification of nanomaterials – Size-dependent properties – Structure–property relationships – Surface-to-volume ratio – Defects and interfaces in nanomaterials – Overview of materials characterization methods – Spectroscopy,

diffraction, microscopy, thermal and electrochemical analysis.

Crystal structure and crystallography – Principles of X-ray diffraction – Bragg's law – Phase identification – Crystal-lite size and strain analysis – X-ray photoelectron spectroscopy (XPS) – Surface composition and chemical states – UV-Visible spectroscopy – FTIR spectroscopy – Raman spectroscopy – BET surface area analysis and porosity measurements.

Principles of electron microscopy – Scanning Electron Microscopy (SEM) – Transmission Electron Microscopy (TEM) – Electron diffraction – Energy dispersive spectroscopy – Atomic Force Microscopy (AFM) – Scanning Tunneling Microscopy (STM) – Surface morphology and nanoscale imaging – Resolution and contrast mechanisms – Sample preparation techniques.

Thermogravimetric analysis (TGA) – Differential Scanning Calorimetry (DSC) – Thermal stability and phase transitions – Vibrating Sample Magnetometry (VSM) – Magnetic characterization of nanomaterials – Fundamentals of electrochemical characterization – Cyclic Voltammetry (CV) – Chronopotentiometry (CP) – Electrochemical Impedance Spectroscopy (EIS) – Charge transfer and interfacial phenomena in nanomaterials.

Surface modification and functionalization techniques – Chemical and physical functionalization – Self-assembled monolayers – Thin films and nanocoatings – Functionalized carbon nanomaterials and metal oxide nanostructures – Nanomaterials for sensors, energy storage, nanoelectronics, optoelectronics, and flexible electronics – Reliability and performance evaluation of nanomaterial-based electronic devices.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	1		2	1
CO2	3	3	2	2		2	2
CO3	2	3	3	3		2	2
CO4	3	2	2	2		3	2
CO5	3	3	2	2	1	3	2

Textbooks / References

1. Kannan M. Krishnan, Principles of Materials Characterization and Metrology, Oxford University Press, 2021.
2. Ray F. Egerton, Physical Principles of Electron Microscopy: An Introduction to TEM, SEM and AEM, Springer, 2005.
3. Guozhong Cao and Ying Wang, Nanostructures and Nanomaterials: Synthesis, Properties and Applications, World Scientific Publishing, 2011.
4. B. D. Cullity and S. R. Stock, Elements of X-Ray Diffraction, Pearson/Addison-Wesley, 2001.
5. Allen J. Bard and Larry R. Faulkner, Electrochemical Methods: Fundamentals and Applications, Wiley, 2001.

ECE410 Nano Electro Mechanical Systems (NEMS) [3-0-0-3]

Course Objectives

- To introduce the governing principles and scaling laws of nano-electro-mechanical systems
- To study nanoscale materials, transduction mechanisms, and device architectures
- To understand microsystem and nanosystem design methodologies beyond conventional MEMS fabrication concepts
- To analyze nanoscale resonators, sensors, actuators, and micro/nanofluidic elements
- To explore emerging applications of NEMS in biomedical, RF, sensing, and energy domains

Course Outcomes

Upon successful completion of the course, the student will be able to:

- CO1: Explain the physical principles and scaling effects governing NEMS devices.
- CO2: Analyze nanoscale mechanical, electrical, thermal, and surface interaction phenomena.
- CO3: Evaluate nanomaterials and nanoscale transduction techniques for sensing and actuation.
- CO4: Apply microsystem design principles for nano/microscale devices and structures.
- CO5: Identify suitable NEMS technologies for emerging engineering and biomedical applications

Syllabus

Introduction to NEMS – Evolution from MEMS to NEMS – Intrinsic characteristics of micro/nanosystems – Scaling laws and miniaturization effects – Surface-to-volume ratio – Dominance of surface forces – Nanomechanics and size-dependent properties – Electrostatic, capillary, and intermolecular forces at nanoscale – Design implications of scaling laws.

Carbon nanotubes (CNTs) – Graphene and 2D materials – Nanowires and quantum dots – Thin films and nanostructured materials – Residual stress and thin film effects – Mechanical and electrical properties of nanomaterials – Self-assembly and molecular structures – Material selection considerations for NEMS devices.

Electrostatic sensing and actuation – Capacitive transducers – Piezoresistive sensing – Piezoelectric transduction – Thermal and magnetic actuation – Nano-cantilevers and nanoresonators – Resonant frequency, damping, and quality factor – Frequency tuning – Noise and sensitivity limitations in nanoscale systems.

Top-down and bottom-up nanofabrication – Electron beam lithography – Nanoimprint lithography – Focused ion beam techniques – Self-assembly approaches – Thin film processing – Nanoscale metrology and characterization – Microsystem design methodology – Elastic deformation of microstructures – Membranes, beams, vibrations, and capacitive forces – Reliability and integration challenges in NEMS.

NEMS-based pressure, flow, and inertial sensors – RF NEMS and resonant devices – Bio-NEMS and lab-on-chip concepts – Micro/nanofluidic systems – Energy harvesting and nanoactuators – Molecular sensing – Flexible and wearable nanosystems – Emerging trends and future directions in NEMS technology.

Course Outcome – PO/PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	1		2	1
CO2	3	3	2	2		2	1
CO3	3	2	3	2		3	2
CO4	3	3	3	2		3	2
CO5	2	2	2	1	2	2	2

Textbooks / References

1. Chang Liu, Foundations of MEMS, Pearson Education, 2012.
2. Marc J. Madou, Fundamentals of Microfabrication and Nanotechnology, CRC Press, 2011.
3. Sergey Edward Lyshevski, Nano- and Micro-Electromechanical Systems: Fundamentals of Nano- and Microengineering, CRC Press, 2005.
4. Werner Karl Schomburg, Introduction to Microsystem Design, Springer, 2011.
5. Mohamed Gad-el-Hak (Ed.), MEMS: Introduction and Fundamentals, CRC Press, 2006.

ECE411 Optoelectronics [3-0-0-3]

Course Objectives

- This course provides the background in optoelectronics, helps students meet the demand of growing semiconductor optoelectronic industry and prepares them for advanced study and research in the semiconductor optics and optoelectronics devices.
- The course is an introduction to the fundamentals of optoelectronics and principles of the optoelectronic devices operation.
- Topics include a variety of different subjects including a detailed discussion of the design and operation of optical LEDs, the basic physics and operation of lasers and photodetectors
- Familiarize with the most promising silicon optical components and the path to electronic-photonic integration.

Course Outcomes

Upon Completion of the course, the students will be able to

- CO1: Be familiar with recent trends in optoelectronics.
- CO2: Understand the electromagnetic spectrum, wave equation and wave propagation in linear, isotropic media and in anisotropic media.
- CO3: Understand basic principles of light propagation and modal analyses of optical fiber. Understand the basic operating principles of light sources, detectors.

- CO4: Identify trends in optical interconnection and the power of electronic-photonics convergence

Syllabus

Electromagnetic theory: Maxwell's equations, Poynting's theorem, Wave nature of light, Polarization, Interference, Diffraction, Light Source, review of Quantum Mechanical concept, Review of Solid State Physics, Review of Semiconductor Physics and Semiconductor Junction Device.

FIBRE OPTICS AND APPLICATIONS Fibre Bragg Grating, Long- period fibre Bragg Grating. Fabrication of Fibre Gratings. Michelson's interferometer, Mach-Zehnder interferometer, Optical Fibre sensors: Intensity modulation sensors. Phase modulation sensors, Temperature, pressure, chemical and rotation sensors. Fibre optic gyroscopes. Evanescent wave sensors, Optical Biosensors. **SILICON-ON-INSULATOR (SOI) PHOTONICS**, Integration of photonic devices, power and power density issues on-chip, Advanced waveguides and devices, Testing and packaging of Silicon photonic devices: Silicon Photonics Research and Manufacturing using SOI wafer. Silicon Photonics Application. Opto electronic devices, Optical modulators, modulation methods and modulators, transmitters, optical transmitter circuits, LED and laser drive circuits, LED-Power and efficiency, double heterostructure LED, LED structures, LED characteristics, laser modes, strip geometry, gain guided lasers, index guided lasers. Display devices, Photo detectors, photon devices, PMT, photodiodes, photo transistors, noise characteristics, PIN diode, APD characteristics, APD Design of detector arrays, CCD, Solar cells.

OPTICAL COMMUNICATION TECHNOLOGY: Unguided optical communication system, transmission parameters, beam divergence, atmospheric attenuation, guided wave communication, merits of optical fibre communication systems, basic network information rates, time evolution of fibre optic systems, elements of optical fiber transmission link/ repeaters, integrated optics, active and passive components, opto-mechanical switches, all optical switches.

LASER TECHNOLOGY: Laser Emission, Absorption, Radiation, Population Inversion, Optical Feedback, Threshold condition, Laser Modes, Radiative decay of excited states of atoms, spontaneous emission, Planck's law for cavity radiation, Q switching general theory, Mode locking, Laser systems, Classes of Lasers.

LASER APPLICATIONS Industrial applications: Absorption of laser radiation by metals, semi conductors and insulators, laser drilling, welding, cutting and surface cleaning, optical fibre splicing, laser deposition of thin films. Laser pollution monitoring, LIDAR laser gyros, Laser spectroscopy, laser induced fusion reactor, Holography and speckle interferometry

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	2	3	2	1	3	2
CO2	3	3	3	2		3	3
CO3	3	3	3	2		3	3
CO4	2	3	3	3	1	3	2

Textbooks / References

1. Fundamentals of Photonics — Saleh and Teich Wiley Intsc (2007)
2. Opto electronics - An introduction - J Wilson and J F B J iS Hawkers (Prentics-Hall India, 1996)
3. J. Singh, Optoelectronics: an Introduction to Materials and Devices , McGraw Hill 1996.
4. Silicon Photonics: Fueling the Next Information Revolution by Daryl Inness, Roy Rubenstein
5. Introduction to fiber optics — Ajoy Ghatak and K. Thyagarajan, Cambridge Univ Press, 1999
6. Laser Fundamentals - Willaim T Silfvast, Cambridge Univ-Press (1996). (Text) 7. Optical Fibre Communication — G Keiser (3rd Ed), 2000

ECE412 Quantum Optics [3-0-0-3]

Course Objectives

- Understand Quantized Light
- Analyze Light-Matter Interaction
- Master Mathematical Tools essential for quantum optics
- Develop Intuition for Non-Classical Effects

Course Outcomes

- CO1: To quantise the electromagnetic field
- CO2: The various experimental techniques in photonics
- CO3: The various representations of states of light
- CO4: Classical, semi-classical and fully quantum models of light-matter interaction
- CO5: Modelling decoherence through Master equation

Syllabus

Quantization of the electromagnetic field Number states, coherent states, squeezed states Hanbury-Brown and Twiss experiments – Photon bunching, Photon antibunching Hong-Ou-Mandel interference Theory of Optical coherence Young's double slit experiment and first order coherence Coherence functions of arbitrary order Normal ordering, symmetric ordering and anti-normal ordering of operators Interferometry Phase-space representations of states of light Wigner distribution P-function and the notion of non-classicality with some examples of nonclassical states like squeezed states and their applications Husimi Q function Light-matter interaction Classical model of light-matter interaction Semi-classical model of light-matter interaction-Quantum light-matter interaction Rabi Model Jayne's-cummings model

Open quantum systems Fermi golden rule Born-Markov Lindblad Master Equation

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	2		2	1
CO2	2	2	3	2		3	1
CO3	3	2	3	2		3	2
CO4	3	3	3	3		2	1
CO5	2	3	3	2		3	1

Textbooks / References

1. Physics (Oscillations, Waves and Optics), A. B. Bhattacharya & Atanu Nag, Khanna Publishing House (2025)
2. Introductory Quantum Optics, Christopher Gerry and Peter Knight, Cambridge University Press (2004)
3. Quantum Optics, D. F. Walls, Gerard J. Milburn, 2nd Edition, Springer (2008)
4. Quantum Optics: An introduction, Mark Fox, Oxford University Publishers (2006)
5. Quantum Optics for Beginners, Z. Ficek and M. R. Wahiddin, 1st edition, Jenny Stanford Publishing (2014)

ECE413 Detection and Estimation [3-0-0-3]

Course Objectives

- To provide students an understanding of the concepts related to application of statistical hypothesis testing to the detection of signals in noise.
- To explore the detection of multiple signals with unknown parameters.
- To familiarize students on the concepts of statistical parameter estimation methods to extract information from signals in noise.
- To summarize the application of filtering in signal detection and estimation process.

Course Outcomes

- CO1: Use statistical information in basic detection theory to solve the problems that involve detection of signals in the presence of noise
- CO2: Detect multiple types of signals (deterministic signals, random signals, signals with unknown parameters).
- CO3: Analyse optimal estimators for various signal parameters.
- CO4: Understand optimal filtering, linear estimation, and Wiener/Kalman filtering.

Syllabus

Introduction to random variables and processes, review of vector space and vector-space interpretation of random variables, problem formulation and objectives of signal detection and signal parameter estimation in discrete-time domain.

Bayesian, minimax, and Neyman-Pearson decision rules,

likelihood ratio test (LRT), receiver operating characteristics, composite hypothesis testing, locally optimum tests, generalized LRT, detector comparison techniques, asymptotic relative efficiency.

Matched filter detector and its performance, generalized matched filter, detection of sinusoid with unknown amplitude, phase, frequency and arrival time, linear model.

Estimator-correlator, linear model, general Gaussian detection, detection of Gaussian random signal with unknown parameters, weak signal detection.

Detection in the absence of complete statistical description of observations, sign detector, Wilcoxon detector, detectors based on quantized observations, robustness of detectors.

Minimum variance unbiased estimation, Fisher information matrix, Cramer-Rao bound, sufficient statistics, minimum statistics, complete statistics, linear models; best linear unbiased estimation, maximum likelihood estimation, invariance principle; estimation efficiency, Bayesian estimation: philosophy, nuisance parameters, risk functions, minimum mean square error estimation, maximum a posteriori estimation.

Linear Bayesian estimation, Weiner filtering, dynamical signal model, discrete Kalman filtering.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	3	3	2		3	
CO2	2	3	3	2		3	
CO3	3	3	3	2		3	
CO4	2	3	3	2		3	

Textbooks / References

1. H. L. Van Trees, "Detection, Estimation and Modulation Theory", Part I, II, and III, Second Edition, John Wiley, 2012.
2. H.V.Poor, "An Introduction to Signal Detection and Estimation", Second Edition, Springer Verlag, 1994
3. S. M. Kay, "Fundamentals of Statistical Signal Processing: Estimation Theory", First edition, Printice hall, Volume I, 1993.
4. S. M. Kay, "Fundamentals of Statistical Signal Processing: Detection Theory", First edition, Printice hall, Volume II, 1998.

ECE414 Advanced Wireless Communication Techniques [3-0-0-3]

Course Objectives

- To extend the knowledge of digital communication under fading channels
- To study the various communication strategies adapted in MIMO system to increase data rate and reliability of communication systems
- To study the theoretical BER performance of various communication systems

- To understand various advanced communication techniques that are proposed for future communication systems.

Course Outcomes

- CO1: Knowledge on the performance of digital communication systems under fading channels
- CO2: Evaluate BER performance of various communication systems
- CO3: Applying various strategies to improve the BER and data rate of communication systems
- CO4: Knowledge on various advanced communication techniques

Syllabus

SISO: System model – SISO channel – Channel capacity - BER performance of wireless communication systems – Spatial Diversity: Diversity gain – Coding gain – Spatial diversity – Time/ frequency diversity – Transmit antenna diversity – Receive antenna diversity – SIMO: System model – SIMO channel – Channel capacity - Diversity combining techniques - MIMO: System model – MIMO channel – MIMO channel capacity – Spatial Multiplexing – Receiver design: Sphere decoding – V-BLAST – D-BLAST – MIMO detection – Optimum detection – linear detection – Space time block codes – Alamouti Scheme. Cooperative Communications: System model – Cooperation protocols – Amplify and forward (AF) – Decode and forward (DF) – Cooperative communication with single branch: BER performance of AF and DF protocols – Multi-node cooperative communication – Relay selection – Differential modulation for cooperative communication. Definition of Massive MIMO, Correlated Rayleigh Fading, System Model for Uplink and Downlink, Basic Impact of Spatial Channel Correlation, Channel Hardening and Favourable Propagation, Local Scattering Spatial Correlation Model, Massive MIMO propagation channel, Favourable Propagation and Deterministic Channels, Favourable Propagation and Random Channels-Independent Rayleigh Fading-Uniformly Random Line-of-Sight (UR-LoS), Uplink Pilots and Channel Estimation - Orthogonal Pilots-De-Spreading of the Received Pilot Signal-MMSE Channel Estimation, Uplink Data Transmission - Zero - Forcing - Maximum - Ratio, Downlink Data Transmission Linear Precoding - Zero - Forcing - Maximum - Ratio. Index Modulation: Space shift keying – Spatial modulation – Media based modulation – Non-Orthogonal Multiple Access (NOMA) – Simultaneous Wireless Information and Power Transfer (SWIPT) – Reconfigurable intelligent surfaces for wireless communications.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	3		3	
CO2	3	3	3	3		3	1
CO3	3	3	3	3	2	3	3
CO4	3	3	3	3	3	3	2

Textbooks / References

1. Chockalingam and B. Sundarajan, “Large MIMO Systems”, Cambridge University Press, 2014.
2. E. Biglieri, R. Calderbank, A. Constantinides, A. Goldsmith, A. Paulraj and H. V. Poor, “MIMO Wireless Communications”, Cambridge University Press, 2007.
3. A. Goldsmith, “Wireless Communications” Cambridge University Press, 2005.
4. K. J. Ray, Ahmed K Sadek, Weifung Su, and Andres Kwasinski “Cooperative Communications and Networking”. Cambridge University Press, 2009.
5. T. S. Rappaport, “Wireless Communications: Principles and Practices”, Pearson Education, 2002.
6. D. Tse and P. Viswanath, “Fundamentals of Wireless Communication”, Cambridge University Press, 2005.
7. Thomas L. Marzetta, Erik G. Larsson, Hong Yang, Hien Quoc Ngo, “Fundamentals of Massive MIMO”, Cambridge University Press 2016.
8. Emil Björnson, Jakob Hoydis and Luca Sanguinetti, “Massive MIMO Networks: Spectral, Energy, and Hardware Efficiency”, Foundations and Trends, Now, 2017.
9. Miaowen Wen, Xiang Cheng, Liuqing Yang, Index Modulation for 5G Wireless Communications, Springer, 2017.
10. Yuanwei Liu, Zhijin Qin, Zhiguo Ding, Non-Orthogonal Multiple Access for Massive Connectivity, Springer, 2020.
11. Hongliang Zhang, Boya Di, Lingyang Song, Zhu Han, Reconfigurable Intelligent Surface-Empowered 6G, Springer, 2021.

ECE415 Optical Communication [3-0-0-3]

Course Objectives

- To introduce the fundamentals of optical communication systems and optical fiber transmission principles.
- To understand the propagation characteristics of optical fibers and analyze various transmission impairments.
- To study optical sources, detectors, and modulation techniques used in modern fiber optic communication systems.
- To analyze optical communication system design, performance, and multiplexing techniques for high-speed communication networks.
- To explore advanced optical communication technologies and their applications in broadband and next-generation communication systems.

Course Outcomes

- CO1: Explain the fundamentals of optical communication systems, optical fiber structures, propagation mechanisms, and transmission characteristics.
- CO2: Analyze attenuation, dispersion, and other transmission impairments in optical fibers and evaluate their impact on system performance.

- CO3: Evaluate the characteristics and operation of optical sources, photodetectors, and optical receiver systems used in fiber optic communication.
- CO4: Analyze optical communication system architectures, multiplexing techniques, optical amplifiers, and network components for high-speed communication applications.
- CO5: Apply advanced optical communication concepts such as coherent communication, WDM systems, optical networking, and free-space optical communication in modern broadband networks.

Syllabus

Introduction to Optical Communication and Optical Fibers: Evolution of optical communication systems, Advantages and applications of optical communication, Basic components of optical communication systems, Electromagnetic spectrum and optical transmission windows, Ray theory of light propagation, Optical fiber structure and classification, Step-index and graded-index fibers, Single-mode and multimode fibers, Numerical aperture and acceptance angle, Modes in optical fibers and V-number
Optical Fiber Transmission Characteristics: Signal attenuation in optical fibers, Material absorption losses, Rayleigh scattering losses, Bending losses, Dispersion in optical fibers, Intramodal and intermodal dispersion, Chromatic dispersion and polarization mode dispersion, Dispersion management techniques, Optical fiber cables and connectors, Fiber splicing techniques and losses, Optical power budget and rise time budget analysis,
Optical Sources and Optical Detectors: Semiconductor physics for optical devices, LED characteristics and operation, Laser diodes and semiconductor lasers, Optical source characteristics, Modulation of optical sources, Photodetectors: PIN and Avalanche photodiodes, Detector noise and sensitivity, Optical receiver structures, Coherent and non-coherent optical detection, Receiver performance analysis

Optical Communication Systems and Multiplexing: Analog and digital optical communication systems, Intensity modulation and direct detection, Optical modulation formats, Wavelength division multiplexing (WDM), Dense WDM systems, Optical amplifiers: EDFA, Raman amplifiers, Optical repeaters and regenerators, SONET/SDH optical networks, Passive optical networks (PON), Optical access networks and FTTH

Advanced Optical Communication Technologies: Coherent optical communication systems, Optical soliton communication, Free-space optical communication, Fiber nonlinearities and their effects, Optical switching and routing, Optical network architectures, Radio over fiber systems, Visible light communication (VLC), Photonic integrated circuits, Next-generation optical communication systems and applications.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	2		3	3
CO2	3	3	3	3		3	3
CO3	3	3	3	3		3	3
CO4	3	3	3	3		3	3
CO5	3	3	3	3		3	3

Textbooks / References

1. Gerd Keiser, "Optical Fiber Communications", McGraw-Hill Education, 5th Edition, 2021.
2. Govind P. Agrawal, "Fiber-Optic Communication Systems", Wiley India, 4th Edition, 2012.
3. John M. Senior, M. Yousif Jamro, "Optical Fiber Communications: Principles and Practice", Pearson Education, 3rd Edition, 2009.
4. Ajoy Ghatak and K. Thyagarajan, "Introduction to Fiber Optics", Cambridge University Press, 1998.
5. Ajoy Ghatak and K. Thyagarajan, "Optical Fiber Communication", Cambridge University Press, 1998.
6. Joseph C. Palais, "Fiber Optic Communications", Pearson Education, 5th Edition, 2005.
7. R. Ramaswami, N. S. Kumar and G. H. Sasaki, "Optical Networks", Morgan Kaufmann Publishers, 3rd Edition, 2009.

ECE416 Satellite Communication [3-0-0-3]

Course Objectives

- To introduce the fundamentals of satellite communication systems and orbital mechanics.
- To understand the architecture and operation of satellite subsystems and communication links.
- To analyze satellite channel characteristics, multiple access techniques, and modulation methods used in satellite communication.
- To study satellite network design, earth station technologies, and performance analysis of satellite communication systems.
- To explore advanced satellite communication technologies and applications in global communication networks.

Course Outcomes

- CO1: Explain the fundamentals of satellite communication systems, orbital mechanics, satellite orbits, and space environment characteristics.
- CO2: Analyze satellite link design parameters, propagation effects, noise, interference, and system performance metrics in satellite communication systems.
- CO3: Evaluate the operation and performance of satellite subsystems, earth stations, antennas, and transponders used in communication satellites.
- CO4: Analyze multiple access techniques, modulation schemes, and networking concepts employed in satellite communication systems.

- CO5: Apply advanced satellite communication concepts including VSAT networks, mobile satellite systems, GPS, broadband satellite communication, and next-generation satellite networks.

Syllabus

Introduction to Satellite Communication and Orbital Mechanics: Evolution of satellite communication systems, applications of satellite communication, basic components of satellite communication systems, frequency bands used in satellite communication, orbital mechanics, Kepler's laws, satellite orbits: GEO, MEO, and LEO satellites, orbital parameters, satellite launch systems, look angles and orbital perturbations, space environment and its effects on satellites.

Satellite Link Design and Propagation Effects: Satellite link analysis, uplink and downlink design, free-space path loss, antenna gain and G/T ratio, system noise temperature, carrier-to-noise ratio, link budget calculations, propagation impairments, atmospheric absorption, rain attenuation, ionospheric effects, scintillation and depolarization effects, satellite communication reliability and performance analysis.

Satellite Subsystems and Earth Stations: Satellite architecture and subsystems, satellite transponders, power supply systems, attitude and orbit control systems, thermal control, telemetry, tracking and command systems, satellite antennas, earth station design, earth station antennas and tracking systems, high power amplifiers, low noise amplifiers, earth station performance parameters.

Multiple Access Techniques and Satellite Networks: Analog and digital satellite communication systems, modulation and coding techniques for satellite communication, FDMA, TDMA, CDMA, DAMA and random-access techniques, spread spectrum communication, satellite switching systems, VSAT networks, satellite network architectures, satellite internet systems, inter-satellite links, mobile satellite communication systems.

Advanced Satellite Communication Technologies: Global Positioning System (GPS), navigation satellite systems, direct broadcast satellite systems, broadband satellite communication, high throughput satellites (HTS), software-defined satellites, satellite IoT systems, CubeSats and small satellites, deep space communication, 5G and satellite integration, next-generation satellite communication systems and applications.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	2	3	1		3	
CO2	3	3	3	2		3	
CO3	3	3	3	2		3	
CO4	3	3	3	2		3	1
CO5	3	3	3	2		3	2

Textbooks / References

1. Dennis Roddy, "Satellite Communications", McGraw-Hill Education, 4th Edition, 2006.

2. Gerard Maral and Michel Bousquet, Satellite Communications Systems: Systems, Techniques and Technology, Wiley India, 5th Edition, 2009.
3. Tri T. Ha, Digital Satellite Communications, McGraw-Hill, 2nd Edition, 1990.
4. D. C. Agarwal, Satellite Communication, Khanna Publishers, 6th Edition, 2017.
5. K. N. Raja Rao, Fundamentals of Satellite Communication, PHI Learning, 2018.
6. Enrico Del Re, Marina Ruggieri and Luca Pierucci, Satellite Communications and Navigation Systems, Springer, 2008.
7. Bruce R. Elbert, Introduction to Satellite Communication, Artech House Publishers, 3rd Edition, 2008.

ECE417 5G Wireless System Design [3-0-0-3]

Course Objectives

- To introduce the architecture, requirements, and enabling technologies of 5G wireless communication systems.
- To understand the physical layer design and radio access technologies used in 5G networks.
- To analyze advanced techniques such as massive MIMO, beamforming, millimeter-wave communication, and network slicing in 5G systems.
- To study 5G network architecture, resource management, and performance optimization techniques.
- To explore emerging applications, services, and future trends in beyond-5G and 6G communication systems.

Course Outcomes

- CO1: After successful completion of the course, students will be able to:
- CO2: Explain the architecture, requirements, standards, and enabling technologies of 5G wireless communication systems.
- CO3: Analyze 5G radio access technologies, waveform design, channel coding, and multiple access techniques used in next-generation wireless systems.
- CO4: Evaluate advanced 5G technologies including massive MIMO, beamforming, millimeter-wave communication, and ultra-dense networks.
- CO5: Analyze 5G core network architecture, network slicing, virtualization, resource allocation, and quality of service mechanisms.

Syllabus

Introduction to 5G Wireless Systems: Evolution of mobile communication systems, limitations of 4G systems, requirements and use cases of 5G, IMT-2020 standards, spectrum requirements for 5G, key enabling technologies, 5G standardization and ecosystem, performance metrics of 5G systems, challenges in 5G deployment.

5G Radio Access and Physical Layer Design: 5G NR architecture and frame structure, scalable numerology, OFDM

and alternative waveforms, filtered OFDM and FBMC concepts, channel coding techniques: LDPC and polar codes, multiple access techniques, synchronization and reference signals, random access procedures, physical channels and signals in 5G NR.

Advanced 5G Communication Technologies: Massive MIMO systems, hybrid beamforming techniques, millimeter-wave communication, propagation characteristics at mmWave frequencies, ultra-dense networks, device-to-device communication, cooperative communication techniques, non-orthogonal multiple access (NOMA), reconfigurable intelligent surfaces (RIS), energy-efficient communication techniques.

5G Network Architecture and Resource Management: 5G core network architecture, software-defined networking (SDN), network function virtualization (NFV), network slicing concepts, cloud radio access networks (C-RAN), mobility management, radio resource management, interference management, quality of service (QoS) and quality of experience (QoE), security and privacy in 5G networks. **Applications and Beyond-5G Technologies:** Massive machine-type communication (mMTC), ultra-reliable low-latency communication (URLLC), enhanced mobile broadband (eMBB), IoT and industrial automation, vehicular communication systems, edge and fog computing, AI and machine learning in 5G networks, satellite integration with 5G, terahertz communication, 6G vision and future wireless communication technologies.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	3	3	3		3	1
CO2	2	3	3	3	1	3	1
CO3	3	3	3	3	1	3	1
CO4	2	3	3	3		3	
CO5	3	3	3	3	1	3	1

Textbooks / References

1. Erik Dahlman, Stefan Parkvall and Johan Skold, 5G NR: The Next Generation Wireless Access Technology, Academic Press, 2nd Edition, 2020.
2. Jonathan Rodriguez, Fundamentals of 5G Mobile Networks, Wiley, 2015.
3. Afif Osseiran, Jose F. Monserrat and Patrick Marsch, 5G Mobile and Wireless Communications Technology, Cambridge University Press, 2016.
4. Theodore S. Rappaport, Robert W. Heath Jr., Robert C. Daniels and James N. Murdock, Millimeter Wave Wireless Communications, Pearson Education, 2015.
5. Andrea Goldsmith, Wireless Communications, Cambridge University Press, 2005.
6. Aditya K. Jagannatham, Principles of Modern Wireless Communication Systems: Theory and Practice, McGraw-Hill Education, 2016.
7. Maziar Nekovee and Mohammad Dohler, 6G Wireless Systems: Vision and Requirements, Wiley, 2023.

ECE418 MIMO Wireless Communication [3-0-0-3]

Course Objectives

- To provide fundamental knowledge of wireless communication channels, fading models, and MIMO system architectures including SISO, SIMO, MISO, and MIMO systems.
- To develop understanding of diversity techniques, channel capacity analysis, and power allocation methods in MIMO wireless communication systems.
- To analyze space-time coding, spatial multiplexing, signal detection, and channel estimation techniques for reliable and high data-rate wireless communication.
- To enable students to evaluate and apply advanced MIMO techniques such as BLAST architectures, precoding, and SVD-based MIMO systems in modern wireless networks.

Course Outcomes

- CO1: Understand wireless channel characteristics, fading models, and the fundamentals of SISO, SIMO, MISO, and MIMO communication systems.
- CO2: Analyze diversity combining techniques, MIMO channel capacity, and power allocation strategies for improving wireless system performance.
- CO3: Apply space-time coding, spatial multiplexing, and MIMO detection techniques for reliable and high spectral efficiency wireless communication.
- CO4: Design and evaluate channel estimation, precoding, and SVD-based MIMO techniques for modern wireless communication systems.

Syllabus

Introduction to wireless communication systems: Wireless communication systems, wireless channel characteristics, multipath propagation and fading, large-scale and small-scale fading models. Fundamentals of MIMO systems: Introduction to SISO, SIMO, MISO, MIMO and their channel models. Diversity combining techniques: Time, frequency, spatial, diversity techniques, receiver combining techniques, selection combining, equal gain combining, maximum ratio combining. Channel capacity of MIMO systems: Capacity for deterministic MIMO channel, capacity of random MIMO channel, capacity of i.i.d. Rayleigh fading MIMO channels, capacity of separately correlated Rayleigh fading MIMO channel, capacity of keyhole Rayleigh fading MIMO channel. Power allocation in MIMO systems: Uniform power allocation, adaptive power allocation, near optimal power allocation. Space time coding: Space time block codes, space time trellis codes, Alamouti scheme, diversity gain and coding gain. Spatial multiplexing techniques: Spatial multiplexing principles, BLAST architectures – VBLAST, D-BLAST. MIMO detection techniques: Maximum likelihood (ML) detector, linear sub-optimal detectors, sphere

decoding. Channel estimation: Pilot based channel estimation, LS and MMSE channel estimation, precoding techniques, SVD-based MIMO systems.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	1		3	
CO2	3	3	3	2	1	3	1
CO3	3	3	3	3	1	3	1
CO4	3	3	3	3	1	3	

Textbooks / References

1. Rakesh Singh Kshetrimayum, “Fundamentals of MIMO wireless communications”, Cambridge university press, 2017.
2. E. Biglieri, R. Calderbank, A. Constantinides, A. Goldsmith, A. Paulraj and H. V. Poor, “MIMO wireless communications”, Cambridge university press, 2007.
3. A. Paulraj, R. Nabar, and D. Gore, D, “Introduction to space-time wireless communications”, Cambridge university press, 2003.
4. D. Tse and P. Viswanath, “Fundamentals of wireless communication”, Cambridge university press, 2005.

ECE419 Radar Engineering [3-0-0-3]

Course Objectives

- To understand various radar systems
- To examine various applications of Radars
- To analyse the Radar range equation

Course Outcomes

- CO1: To classify various Radar systems
- CO2: To analyze target detection, pulse integration and pulse compression
- CO3: To analyze radar clutter and range ambiguities
- CO4: To analyze moving target indicator, Doppler radar and adaptive array processing

Syllabus

Radar systems, classification of radars, the radar range equation, surveillance radar equation, range measurements, doppler effects, radar cross section, radar equation with jamming, effects of earth’s surface on radar equation, noise figure, atmospheric attenuation.

Target detection, pulse integration and pulse compression, pulse width and range, target detection in presence of noise, probability of false alarm, probability of detection, pulse integration, cumulative probability of detection, constant false alarm rate, time-bandwidth product, pulse compression.

Principle of matched filtering, radar signal resolution and ambiguity, range and doppler uncertainty, target parameter estimation, ambiguity function, nonlinear fm ambiguity, range-doppler coupling, pulse-train codes

Surface and volume clutter, clutter rcs, moving target indicator (mti), prf, delay line cancelers, continuous wave

radar, digital doppler processing, phased arrays, adaptive beamforming, adaptive array processing.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	1	1	1	3	1
CO2	3	3	2	1	1	3	1
CO3	3	3	2	2	1	3	1
CO4	3	3	2	2	1	3	2

Textbooks / References

1. M.I. Skolnik, Introduction to Radar Systems, 3/e, TMH, New Delhi, 2001
2. Nathanson, F. E., Radar Design Principles, New York, McGraw-Hill, 2/E, 1991
3. Toomay, J. C., Radar Principles for the Non-Specialist, New York, Van Nostrand, Reinhold, 1989
4. R.J Sullivan, Radar foundation for imaging & advanced concepts, PHI, 2004.
5. Mark A Richards, Fundamentals of Radar Signal Processing, McGraw Hill, 2005
6. Bassem R. Mahafza, Radar Signal Analysis and Processing Using MATLAB, Chapman and Hall/CRC, 2008

ECE420 Communication System Design [3-0-0-3]

Course Objectives

- To introduce the principles and methodologies involved in the design of modern communication systems.
- To understand the modeling and analysis of analog and digital communication system components and subsystems.
- To study modulation, coding, synchronization, and signal processing techniques used in practical communication system design.
- To analyze wireless and wired communication system performance under realistic channel conditions.
- To explore modern communication system architectures, implementation techniques, and emerging communication technologies.

Course Outcomes

- CO1: Explain the principles, architectures, and functional blocks of analog and digital communication systems.
- CO2: Analyze modulation, coding, multiplexing, and synchronization techniques used in communication system design.
- CO3: Evaluate communication system performance under noise, fading, interference, and bandwidth constraints using appropriate analytical methods.
- CO4: Design communication subsystems including transmitters, receivers, filters, equalizers, and channel coding schemes for practical applications.

- CO5: Apply modern communication system design concepts for wireless, optical, satellite, and broadband communication networks.

Syllabus

Fundamentals of Communication System Design: Communication system architecture, elements of communication systems, analog and digital communication models, signals and spectra, bandwidth and power considerations, noise in communication systems, signal-to-noise ratio, channel capacity, communication standards and design requirements, system performance metrics.

Modulation and Signal Processing Techniques: Amplitude modulation and angle modulation techniques, pulse modulation methods, digital modulation schemes: ASK, FSK, PSK, QAM, OFDM principles, multiplexing techniques: FDM, TDM, CDM and OFDM-based multiplexing, pulse shaping and matched filtering, synchronization techniques, carrier and timing recovery methods.

Communication Channels and Coding Techniques: Wireless and wired channel models, fading and multipath propagation, inter-symbol interference, equalization techniques, source coding principles, channel coding techniques: block codes, convolutional codes, turbo codes and LDPC codes, error probability analysis, diversity techniques and link reliability improvement methods.

Transmitter and Receiver Design: Transmitter architectures and RF front-end design, receiver architectures: superheterodyne and direct conversion receivers, low-noise amplifiers and mixers, frequency synthesizers and oscillators, analog-to-digital and digital-to-analog converters, software-defined radio concepts, adaptive communication system design, communication system simulation and testing methods.

Modern Communication Systems and Applications: Cellular communication systems, satellite communication systems, optical communication systems, broadband communication networks, MIMO communication systems, IoT communication architectures, 5G and beyond communication systems, cognitive radio, green communication systems, communication security and future communication technologies.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	1	1			2	
CO2	3	3	3	1		3	
CO3	3	3	3	2		3	
CO4	3	3	3	3	1	3	2
CO5	2	2	2	2			2

Textbooks / References

1. Simon Haykin, Communication Systems, Wiley India, 5th Edition, 2009
2. B. P. Lathi and Zhi Ding, Modern Digital and Analog Communication Systems, Oxford University Press, 4th Edition, 2009.
3. Herbert Taub and Donald L. Schilling, Principles of

Communication Systems, Tata McGraw-Hill, 2nd Edition, 2008.

4. John G. Proakis and Masoud Salehi, Digital Communications, McGraw-Hill Education, 5th Edition, 2008.
5. Andrea Goldsmith, Wireless Communications, Cambridge University Press, 2005.
6. Aditya K. Jagannatham, Principles of Modern Wireless Communication Systems: Theory and Practice, McGraw-Hill Education, 2016.
7. Edward A. Lee and David G. Messerschmitt, Digital Communication, Springer, 3rd Edition, 2004.

ECE421 Ocean Optics [3-0-0-3]

Course Objectives

- To lay physical and conceptual foundations of ocean optics, the quantitative study of the interactions of light with the Earth's oceans.
- To review concepts, terminology, and measurement techniques used to characterize radiative transfer processes, light fields within and leaving the ocean, and inherent and apparent optical properties of the ocean.
- To study the links and interactions between underwater light field characteristics, ocean optical properties, and various constituents of seawater.

Course Outcomes

- CO1: Understand the electromagnetic spectrum, wave equation and wave propagation in linear, isotropic media and in anisotropic media.
- CO2: Learn the nature of light and solar energy falling upon the Earth and propagation of light through the air-water interface and within the water column
- CO3: Assess the interactions of light with various dissolved and particulate constituents of seawater
- CO4: Examine basic optical phenomena (scattering, absorption, and fluorescence) and the role they play in structuring bio-optical properties of the ocean
- CO5: Apply physical and conceptual foundations of hydrologic optics

Syllabus

Electromagnetic theory: Maxwell's equations, Poynting's theorem, Wave nature of light, Polarization, Interference, Diffraction, Light Source, review of Quantum Mechanical concept.

Basic theory of the light interaction with matter in aquatic environments. Radiative transfer Theory. Optical approaches to ocean biogeochemistry; computation and propagation of measurement uncertainties. Optical constituents of the aquatic water bodies - Phytoplankton, Dissolved organic matter, suspended sediments, bubbles and water itself. Phytoplankton function types

Fundamentals of Inherent and apparent optical properties - Inherent optical properties: absorption and scattering, Apparent optical properties: reflectances and diffuse attenuation coefficient, radiative transfer approximations,

light polarization, classification of water bodies: Case 1 & Case 2, optically deep and shallow, trophic status, ecological regions. Optical Properties: Absorption, Diffuse attenuation (K), Optical scattering (b) & optical backscatter (bb). Relationships Among Optical Properties: Estimating K from chlorophyll, beam attenuation from diffuse attenuation, and wavelength dependence of K

Measuring IOPs, AOPs and biogeochemical constituents - Measuring absorption, attenuation, scattering, water-leaving radiance, chlorophyll, suspended sediments, PAR; sensors for measuring the parameters; buoys, gliders, ARGO, flow-through systems, optical sensor design and function, Fibre Optic sensors

Optical Communication Technology: Unguided optical communication system, transmission parameters, beam divergence, atmospheric attenuation, guided wave communication, merits of optical fibre communication systems, basic network information rates, time evolution of fibre optic systems, elements of optical fiber transmission link/repeaters

Laser Technology: Laser Emission, Absorption, Radiation, Population Inversion, Optical Feedback, Threshold condition, Laser Modes, Radiative decay of excited states of atoms, spontaneous emission, Planck's law for cavity radiation, Q switching general theory, Mode locking, Laser systems, Classes of Lasers. Applications: Industrial applications: Absorption of laser radiation by metals, semi conductors and insulators; optical fibre splicing; Laser spectroscopy;

LED and laser drive circuits, LED-Power and efficiency, double heterostructure LED, LED structures, LED characteristics, laser modes, strip geometry, gain guided lasers, index guided lasers. Display devices, Photo detectors, photon devices, PMT, photodiodes, photo transistors, design of detector arrays

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	2		3	1
CO2	3	3	3	3		3	1
CO3	2	3	3	3		2	1
CO4	2	3	3	2	1	2	1
CO5	3	3	3	2		3	2

Textbooks / References

1. N.G. Jerlov "Marine Optics", Elsevier, 1976.
2. Handbook of Optics, 2nd edition, McGraw-Hill, Inc. 1995.
3. Ocean Optics Web Book: <https://www.oceanopticsbook.info/>
4. Fundamentals of Photonics — Saleh and Teich Wiley Intsc (2007)
5. Introduction to fiber optics — Ajoy Ghatak and K. Thyagarajan, Cambridge Univ Press, 1999
6. Laser Fundamentals - Willaim T Silfvast, Cambridge Univ-Press (1996). (Text)
7. Optical Fibre Communication — G Keiser (3rd Ed), 2000

ECE422 Pattern Recognition [3-0-0-3]

Course Objectives

- To provide a strong foundation in probability theory, stochastic processes, and statistical pattern recognition techniques for analyzing and modeling data.
- To develop understanding of supervised and unsupervised learning methods, including Bayesian classifiers, clustering algorithms, dimensionality reduction, and dictionary learning techniques.
- To introduce modern machine learning and pattern recognition approaches such as support vector machines, neural networks, hidden Markov models, and decision trees for practical classification and recognition tasks.

Course Outcomes

- CO1: Apply probabilistic models, Bayesian decision theory, and statistical estimation techniques for solving pattern recognition and classification problems.
- CO2: Analyze and implement clustering, dimensionality reduction, and nonparametric learning methods for extracting meaningful patterns from data.
- CO3: Examine classical PR techniques such as HMM, K-NN, Parzen window method and dimensionality reduction techniques.
- CO4: Evaluate and compare classical and modern machine learning techniques, including SVMs, neural networks, HMMs, and decision trees, for real-world pattern recognition applications.

Syllabus

Independence of events, conditional and joint probability, Bayes theorem, Stationary and non-stationary processes, Expectation, Autocorrelation, Cross-Correlation, spectra. Bayes Decision Theory: Minimum-error-rate classification. Classifiers, Discriminant functions, Decision surfaces. Normal density and discriminant functions. Discrete features. Parameter Estimation Methods: Maximum-Likelihood estimation: Gaussian case. Maximum Posteriori estimation. Bayesian estimation: Gaussian case. Unsupervised learning and clustering - Criterion functions for clustering. Algorithms for clustering: K-Means, Hierarchical and other methods. Cluster validation. Gaussian mixture models, Expectation-Maximization method for parameter estimation. Maximum entropy estimation.

Sequential Pattern Recognition. Hidden Markov Models (HMMs). Discrete HMMs. Continuous HMMs. Nonparametric techniques for density estimation. Parzen-window method. K-Nearest Neighbour method. Dimensionality reduction: Principal component analysis - relationship to Eigen analysis. Fisher discriminant analysis - Generalized Eigen analysis. Eigen vectors/Singular vectors as dictionaries. Factor Analysis, Total variability space - a dictionary learning methods. Non negative matrix factorization - a dictionary learning method.

Linear discriminant functions: Gradient descent procedures, Perceptron, Support vector machines - a brief introduction. Artificial neural networks: Multilayer perceptron – feed forward neural network. A brief introduction to deep neural networks, convolutional neural networks, recurrent neural networks. Non-metric methods for pattern classification: Non-numeric data or nominal data. Decision trees: Classification and Regression Trees (CART)

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	2		3	3
CO2	3	3	3	2		3	3
CO3	3	3	3	2		3	3
CO4	3	3	3	2		3	3

Textbooks / References

1. R.O.Duda, P.E.Hart and D.G.Stork, Pattern Classification, John Wiley
2. S.Theodoridis and K.Koutroumbas, Pattern Recognition, 4th Ed., Academic Press
3. C.M.Bishop, Pattern Recognition and Machine Learning, Springer
4. C.M.Bishop, Pattern Recognition and Machine Learning, Springer
5. P.A Devijver and J. Kittler, Pattern Recognition: A Statistical Approach, Prentice-Hall International, Englewood Cliffs, NJ
6. K. Fukunaga, Introduction to Statistical Pattern Recognition 2nd Ed. Academic Press, New York

ECE423 Quantum Transformers [3-0-0-3]

Course Objectives

- To understand Transformer architectures and quantum computing principles for intelligent sequence modeling.
- To design hybrid Quantum Transformer models using Quantum Neural Networks and parameterized quantum circuits.
- To explore quantum-inspired attention mechanisms and variational Transformer learning techniques.

Course Outcomes

- CO1: Understand the fundamentals of Transformer architectures and quantum computing principles.
- CO2: Learn hybrid quantum-classical machine learning models and attention mechanisms.
- CO3: Explore Quantum Machine Learning (QML) algorithms and Quantum Neural Networks (QNNs).
- CO4: Implement hybrid AI models using Qiskit, PennyLane, and Transformer frameworks.
- CO5: Analyze scalability, optimization challenges, and limitations in Quantum AI systems.

Syllabus

Introduction to Deep Learning and Quantum Computing: Overview of Artificial Intelligence and Machine Learning, Neural networks, RNNs and LSTMs, Limitations of

sequential models, Introduction to Transformers, Basics of quantum computing, Qubits, superposition, entanglement, Quantum gates and circuits, Classical vs quantum computation

Foundations of Transformer Architectures: Attention mechanisms, Self-attention and multi-head attention, Scaled dot-product attention, Positional encoding, Encoder-decoder architecture, Feed-forward networks, Residual connections and normalization, Training and optimization of Transformer models.

Quantum Machine Learning Fundamentals: Quantum state representation, Data encoding techniques: basis, amplitude, and angle encoding, Quantum feature maps, Swap test and inner product estimation, Parameterized Quantum Circuits (PQCs), Variational Quantum Algorithms (VQAs), Hybrid quantum-classical learning, NISQ systems.

Quantum Neural Networks and Hybrid Architectures: Quantum Neural Networks (QNNs), Quantum perceptrons, Variational quantum classifiers, Quantum convolutional networks, Hybrid quantum-classical architectures, Gradient-based and gradient-free optimization, Parameter shift rule, Barren plateau problem.

Quantum-Inspired Transformer Models: Transformer models for sequence learning, Hybrid Transformer-QNN architectures, Quantum-inspired attention mechanisms, Variational Transformer models, Quantum-enhanced feature representation, Quantum Natural Language Processing (QNLP), Research trends in quantum-inspired Transformers.

Applications and Toolkits: Applications in NLP, image classification, finance, drug discovery, cybersecurity, and healthcare, Introduction to Qiskit Machine Learning, PennyLane, TensorFlow Quantum, Hugging Face Transformers, Practical implementation using simulators and cloud quantum platforms, Challenges: noise, scalability, and hardware limitations.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	1				3	
CO2	3	1				3	1
CO3	3	2				3	2
CO4	2	2		3		2	3
CO5	2	3		1		2	2

Textbooks / References

1. Machine Learning with Quantum Computers, Springer (2021)
2. Natural Language Processing with Transformers, O'Reilly Media
3. Quantum Computation and Quantum Information, Cambridge University Press
4. Attention Is All You Need, NeurIPS (2017)

ECE424 Quantum Machine Learning [3-0-0-3]

Course Objectives

- To understand the fundamentals of Quantum Machine Learning, quantum computing models, and quantum data encoding techniques.
- To study quantum supervised and unsupervised learning algorithms and their advantages over classical machine learning methods.
- To learn the design, implementation, and training of Quantum Neural Networks and hybrid quantum-classical models.
- To gain practical experience using Qiskit Machine Learning, PennyLane, and TensorFlow Quantum for developing Quantum ML applications.
- To explore real-world applications and challenges of Quantum Machine Learning in areas such as finance, image processing, and quantum NLP.

Course Outcomes

- CO1: Understand the intersection of quantum computing and artificial intelligence.
- CO2: Learn quantum-enhanced machine learning algorithms.
- CO3: Explore quantum neural networks and hybrid classical-quantum models.
- CO4: Implement basic quantum machine learning algorithms using simulators.
- CO5: Analyze the advantages and limitations of quantum AI.

Syllabus

Introduction to Quantum ML: Overview of classical ML, Quantum advantages for ML, Quantum computing models: gate-based, annealing, Data encoding in quantum circuits.

Quantum Machine Learning Fundamentals: Quantum state preparation for ML, Inner product estimation, Quantum distance measures, Basics of quantum kernel methods. Quantum Supervised Learning: Quantum classifiers: quantum support vector machines, Variational quantum classifiers, Hybrid quantum-classical training loops.

Quantum Unsupervised Learning: Quantum clustering algorithms. Quantum principal component analysis (Qpca), Quantum Boltzmann machines.

Quantum Neural Networks: Parameterised quantum circuits, Quantum perceptrons and QNNs, Training QNNs: gradient-based and gradient-free.

Applications and Toolkits: Use of Qiskit Machine Learning, PennyLane, TensorFlow Quantum, Application case studies: finance, image classification, quantum NLP, Challenges: scalability, noise, data access

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	1				3	
CO2	3	2				3	1
CO3	3	2				3	2
CO4	2	2		3		2	3
CO5	2	3		1		2	2

Textbooks / References

1. Schuld & Petruccione, Machine Learning with Quantum Computers, Springer (2021).
2. Maria Schuld, Supervised Learning with Quantum Computers, Springer (2018).
3. Qiskit Textbook (<https://qiskit.org/textbook>).
4. M. Benedetti et al., A Review of Quantum Algorithms for Machine Learning, Quantum Sci. Technol. (2019)
5. Quantum Computation and Quantum Information, Michael Nielsen and Isaac Chuang, (Cambridge Univ Press, 2010), <https://link.springer.com/book/10.1007/978-3-030-83098-4>
6. Quantum machine learning Nature, 549, pages 195–202 (2017)
7. A non-review of Quantum Machine Learning: trends and explorations Vedran Dunjko and Peter Wittek Quantum Views 4, 32 (2020)
8. Quantum machine learning: from physics to software engineering, A. Melnikov et. al, Advances in Physics: X, vol 8, page 2165452, (2023).

ECE425 Quantum Sensing [3-0-0-3]

Course Objectives

- Understanding Fundamental Principles of quantum mechanics and their direct application to sensing and imaging.
- To teach the limitations of classical sensors (noise, sensitivity) and how quantum sensing surpasses these limits.
- To understand noise models, Cramer-Rao bounds, and quantum non-demolition measurements.

Course Outcomes

- CO1: The basics of classical sensing
- CO2: Aspects of quantum measurement
- CO3: Ways to quantify quantum sensing
- CO4: About measurements of quantum states of light
- CO5: About the applications of quantum sensing

Syllabus

Classical sensing Photo detection Classical noise Johnson Noise, Telegraph noise, flicker or 1/f noise Sensitivity of classical measurements Classical Fisher information Cramer - Rao bounds (information theory basics may be required here)

Quantum measurements projective/orthogonal measurements Approximate/non-orthogonal measurements Weak

continuous measurements Error-disturbance relations
Standard quantum limits Quantum non-demolition measurements

States of light Fock states Coherent states Squeezed states Tomography Wigner quasi-probability distribution P-distribution Husimi Q function

Quantum photo detection Square-law detectors, Intensity measurements and Photo-detection Linear Detectors and Quadrature Measurements

Quantum Cramer-Rao bounds Single photon-based sensing applications Entanglement based sensing applications Atomic state-based sensing, solid-state spin-based sensing applications (gravimetry, magnetometry)

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	3	3	3		2	3
CO2	2	3	3	3		2	3
CO3	3	3	3	2		2	1
CO4	3	3	3	3		2	3
CO5	2	3	3	3		2	3

Textbooks / References

1. AICTE Prescribed Textbook: Physics (Quantum Mechanics for Engineers), Khanna Publishing House (2025).
2. Quantum Measurement and Control, Howard Wiseman and David Milburn, Cambridge University Press (2014)
3. Quantum Measurement, Vladimir Braginsky and Farid Ya Khalili, Cambridge University Press
4. Quantum Information Science, Manenti R., Motta M., 1st Edition, Oxford University Press
5. Quantum Computing Mechanics, A. B. Bhattacharya, Khanna Publishing House (2025).

ECO301 Speech and Audio Processing [3-0-0-3]

Course Objectives

- To familiarize basic mechanism of speech production and methods for speech analysis.
- To give an overall picture about various applications of speech processing
- To impart ideas of perception of sound, psycho-acoustic analysis, spatial audio perception and rendering
- To analyse audio compression schemes and audio quality

Course Outcomes

- CO1: Understanding of basic mechanism of speech production and methods for speech analysis
- CO2: Develop systems for various applications of speech processing
- CO3: Understanding about human hearing system, psychoacoustic analysis and spatial sound
- CO4: Knowledge about audio compression methods and audio quality analysis

Syllabus

Speech Production: Acoustic theory of speech production. Speech Analysis: Speech signal, Short-Time Speech Analysis, Time domain analysis (Short time energy, short time zero crossing Rate, ACF). LPC Analysis (LPC model, Auto correlation method).

Frequency domain analysis (Filter Banks, STFT, Spectrogram), Cepstral Analysis, MFCC. Probabilistic formulation of speech recognition, Speech coding: fundamentals, Comparison of waveform coding, vocoding and hybrid coding, Speech enhancement: fundamentals, basic types, Speaker verification (block diagram) Signal Processing Models of Audio Perception: Basic anatomy of hearing System. Auditory Filter Banks, Psycho-acoustic analysis: Critical Band Structure, Absolute Threshold of Hearing, Simultaneous Masking, Temporal Masking, MPEG psycho-acoustic model. Audio compression methods: Sampling rate and bandwidth requirement for digital audio, Redundancy removal and perceptual irrelevancy removal, Transform coding of digital audio: MPEG2-AAC coding standard, MDCT and its properties, Loss less coding methods.

Spatial Audio Perception and rendering: The physical and psycho-acoustical basis of sound localization and space perception. Spatial audio standards. Audio quality analysis: Objective analysis methods- PEAQ, Subjective analysis methods - MOS score, MUSHRA score

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	1		3	3
CO2	3	3	3		2	3	3
CO3	3	2	2		1	3	3
CO4	3	2	2		1	3	3

Textbooks / References

1. Douglas O'Shaughnessy, Speech Communications: Human & Machine, IEEE Press, Hardcover 2/e
2. Nelson Morgan and Ben Gold, Speech and Audio Signal Processing: Processing and Perception Speech and Music
3. Lawrence Rabiner, Bing-Hwang Juang and B. Yegnarayana Fundamentals of Speech Recognition, Pearson, 1st Edition
4. Rabiner and Schafer, Digital Processing of Speech Signals, Pearson
5. Thomas F. Quatieri, Discrete-Time Speech Signal Processing: Principles and Practice, Prentice Hall

ECO302 Engineering Foundations of Quantum Technologies [3-0-0-3]

Course Objectives

- To cover topics in electrical, electronics and communication engineering, as well as in computer science that are relevant to Quantum computation, Communications and Sensing.

Course Outcomes

- CO1: Relevant topics from Electrical Networks to design and analyse analog circuits
- CO2: Relevant topics from RF and Microwave Engineering to design systems
- CO3: Relevant topics in Theory of computation to benchmark algorithms
- CO4: Relevant topics in analog and digital communications
- CO5: Basics of cryptography

Syllabus

Analog RLC circuits – resonances, impedances, quality factors Transmission line basics Telegrapher equations, wave impedance, impedance matching, transmission line resonators

Basics of computer architecture Arithmetic Logic Unit Memory Abstract models of Finite State Machine Turing Machines Overview of Hierarchy of languages – Regular, Context-Free, Turing Decidable and Turing Recognisable Complexity Theory Time and Space complexity P vs NP, NP-completeness

Analog Communications Quadrature amplitude modulation Heterodyne and Homodyne demodulation Noise and Signals Characterising Noise Types of Noise Shot Noise Johnson-Nyquist Noise Telegraphic noise or flicker or 1/f noise Signal conditioning and noise mitigation Amplification and Added Noise Linear Amplifier theory Signal-Noise Ratio, Added Noise, Noise Figure of amplification Dynamic Range Noise temperature Quantum limits on noise in linear amplifiers Digital Communications Information entropy Noiseless channel encoding Noisy channel encoding

Basics of cryptography Basics of Number Theory Random Number Generation One time pad, Private key, public key, symmetric and asymmetric cryptography protocols RSA and DH Post Quantum Cryptography (PQC)

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	3		3	3
CO2	3	3	3	3		3	2
CO3	3	3	3	3			1
CO4	3	3	3	3		3	2
CO5	3	3	3	3			1

Textbooks / References

1. Art of Electronics, Paul Horowitz and Winfield Hill, 3rd edition, Cambridge University Press (2015)
2. All-in-One Electronics Simplified, A.K. Maini, Khanna Book Publishing Co. (2022).
3. Digital Design, Morris Mano, Michael D. Ciletti, 6th edition, Pearson Education (2018)
4. Microwave Engineering, David Pozar, 4th edition, Wiley (2013)
5. Information Theory, Robert B. Ash, Dover Publications (2003)

6. Architectures and Cryptographic Universes (The Mechanics of Quantum Computing), A. B. Bhattacharya, Khanna Publishing House (2025).
7. Introduction to the Theory of Computation, Michael Sipser, 3rd edition, Cengage India Pvt. Ltd. (2014)
8. Foundations of Quantum Technologies, A. B. Bhattacharya, Khanna Publishing House (2025).
9. Quantum Computing & Techniques, Dr. R. Chopra, Khanna Publishing House (2025).
10. Protecting Information – From Classical error correction to quantum cryptography, Susan Loepp and William K. Wootters, Cambridge University Press (2006)

ECO303 Biomedical Signal Processing [3-0-0-3]

Course Objectives

- To introduce the characteristics and physiological origins of biomedical signals.
- To study biomedical signal acquisition, conditioning, and artifact removal techniques.
- To analyze biomedical signals using time-frequency and statistical methods for clinical interpretation.
- To understand signal modeling and feature extraction techniques for physiological signals.
- To explore biomedical applications involving ECG, EEG, EMG, and healthcare monitoring systems.

Course Outcomes

- CO1: Explain the characteristics and acquisition of major biomedical signals
- CO2: Analyze biomedical signals using suitable pre-processing and enhancement techniques
- CO3: Interpret physiological information from ECG, EEG, and EMG signals.
- CO4: Apply signal modeling and feature extraction techniques to biomedical signals.
- CO5: Identify biomedical signal processing applications in healthcare and diagnostic systems.

Syllabus

Physiological origin of biomedical signals – Characteristics of bioelectric signals – ECG, EEG, EMG, EOG, PCG and respiratory signals – Biomedical signal acquisition – Electrodes and transducers – Sources of biomedical signal artifacts and noise – Clinical significance of biomedical signals.

Biomedical amplifiers and signal conditioning – Motion artifacts and power-line interference – Baseline wander removal – Signal averaging techniques – Artifact reduction techniques for biomedical signals – Biomedical signal quality assessment–Denoising and smoothing methods.

ECG waveform characteristics – Lead configurations – QRS complex detection – Heart rate and heart rate variability analysis – Arrhythmia detection concepts –

ST-segment analysis – ECG compression techniques– Clinical interpretation of ECG parameters.

EEG rhythms and brain activity – Sleep signals and evoked potentials – EMG signal characteristics and muscle activity analysis – Biomedical signal modeling concepts – Deterministic and stochastic signal models – Time-varying biomedical signals – Physiological interpretation using signal models.

Wearable and ambulatory monitoring systems – Telemedicine and remote healthcare – Biomedical databases and healthcare analytics – Signal processing for intensive care monitoring – Biomedical signal visualization – Computer-aided diagnosis concepts – Emerging trends in biomedical signal processing.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2				2	1
CO2	3	3	2	2		2	2
CO3	3	3	2	2		3	2
CO4	3	3	3	2		3	2
CO5	2	2	2	2	1	2	2

Textbooks / References

1. Rangaraj M. Rangayyan, Biomedical Signal Analysis: A Case-Study Approach, Wiley-IEEE Press, 2002.
2. Eugene N. Bruce, Biomedical Signal Processing and Signal Modeling, Wiley-Interscience, 2001.
3. Willis J. Tompkins, Biomedical Digital Signal Processing, Prentice Hall, 2000.
4. John Enderle and Joseph Bronzino, Introduction to Biomedical Engineering, Elsevier, 2012.

ECO304 Transform Theory [3-0-0-3]

Course Objectives

- To introduce the fundamentals of classical and modern transform techniques used in signal and image processing.
- To provide knowledge of time-domain, frequency-domain, and time-frequency domain signal analysis methods.
- To develop understanding of wavelet and multiresolution analysis techniques for signal representation and processing.
- To impart knowledge of advanced transform methods and their applications in engineering systems.

Course Outcomes

- CO1: Apply various transform techniques for analysis of continuous-time and discrete-time signals.
- CO2: Analyze signals using time-frequency and wavelet-based methods.
- CO3: Implement multiresolution and filter bank approaches for signal processing applications.
- CO4: Explain the principles and applications of advanced transform techniques used in modern signal and image processing.

Syllabus

Review of Laplace Transform, Z transform, Continuous Time Fourier Transform, Discrete Time Fourier transform. Discrete Cosine transform (DCT), Need for time frequency analysis, Short-Time Fourier Transform (STFT), Hilbert Transforms, Introduction to Wavelet, Wavelet Transforms- Continuous Wavelet Transform (CWT), Discrete Wavelet Transform (DWT), Redundant Wavelet Transform (Frames), Multi Resolution Analysis (MRA), DWT using filter banks, Radon Transform, Abel Transform, The Mellin Transform, Hankel Transform, Hartley Transform, Contourlet transform (CTT), Ridgelet and Curvelet transforms.

Discrete Transforms and Applications, Discrete Cosine transform – image compression, DWT – signal compression, signal denoising, discontinuity detection, CTT in image processing,

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	1	2	3	3	3
CO2	3	3	2	3	3	3	3
CO3	3	3	2	2	3	3	3
CO4	3	2	1	1	2	3	2

Textbooks / References

1. Larry Andrews, Bhimsen Shivamoggi, Integral Transforms for Engineers, Prentice Hall of India, New Delhi, 2005.
2. I. N. Sneddon, Fourier Transforms, McGraw Hill, 1951.
3. Alexander D. Poularikas, The Transforms and Applications Handbook, Second Edition, CRC Press
4. Abdul Jerri, Integral and Discrete transforms with applications and error analysis, Marcel Dekker Inc.
5. Lokenath Debnath, Dambaru Bhatta, Integral Transforms and Their Applications, Taylor &
6. Francis Inc.

ECO305 System-On-Chip Design [3-0-0-3]

Course Objectives

- Provide a comprehensive introduction to SoC technology.
- Give an overview to SoC design challenges and design flow.
- Analyze hardware/software tradeoffs, algorithms, and architectures to optimize the system based on requirements and implementation constraints.
- Analyze tradeoffs and explore architecture and microarchitecture design spaces to develop and synthesize custom hardware accelerators

Course Outcomes

- CO1: Design processors keeping area, power, and speed as constraints
- CO2: Analyze memories using reconfigurable architectures for rapid prototyping

- CO3: Analyze system on chip and board based systems.
- CO4: Design custom hardware accelerators

Syllabus

System Approach and Chip Basics: System Architecture, Components of the System, Hardware and Software. An approach for SoC Design, System Architecture and Complexity. Chip Basics. Cycle Time, Die Area and Cost, Ideal and Practical Scaling, Power, Area–Time–Power Trade-Offs in Processor Design, Reliability, Configurability. Memory Design: System-on-Chip and Board-Based Systems – Scratchpads and Cache Memory, Basic Notions, Cache Organization, Cache Data, Write Policies, Strategies for Line Replacement at Miss Time, Other Types of Cache, Split I- and D-Caches and the Effect of Code Density, Multilevel Caches, Virtual-to-Real Translation, SoC (On- Die) Memory Systems, Board-based (Off-Die) Memory Systems, Simple DRAM and the Memory Array, Models of Simple Processor– Memory Interaction. Full custom and semicustom ASIC designs, FPGAs. Architecture and micro-architecture design spaces to develop and synthesize custom hardware accelerators. Recent trends from journals and conferences.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	3	2	3	2
CO2	3	3	3	3	2	3	3
CO3	3	3	2	2	2	3	2
CO4	3	3	3	3	3	3	3

Textbooks / References

1. Micheal J Flynn and Wayne Luk, “Computer System Design: System-on-Chip,” Wiley, First Edition, 2011.
2. M.J.S .Smith, “Application Specific Integrated Circuits” – Pearson Education, 2003.
3. Wayne Wolf, “Modern VLSI Design – System – on – Chip Design”, Prentice Hall, 3rd Edition 2008.

ECO306 Hardware Security [3-0-0-3]

Course Objectives

- To introduce the fundamentals of hardware security, trust, and secure digital system design using FPGA-based platforms.
- To develop understanding of cryptographic hardware implementations, optimization techniques, and hardware security primitives.
- To analyze hardware vulnerabilities along with appropriate countermeasures.
- To explore advanced hardware security concepts for modern computing systems.

Course Outcomes

- CO1: Design and implement secure digital systems and cryptographic accelerators on FPGA platforms while understanding hardware security requirements.

- CO2: Evaluate and optimize cryptographic hardware architectures with respect to performance, area, and security constraints.
- CO3: Analyze the impact of side-channel attacks, fault injection attacks, hardware Trojans, and IP piracy, and apply suitable countermeasures.
- CO4: Investigate modern hardware security challenges in semiconductor supply chains, micro-architectural systems, and post-quantum hardware, and propose secure design methodologies for trusted computing systems.

Syllabus

Introduction to Hardware Security & Trust; Basics of Digital Design on Field-programmable Gate Array (FPGA); Useful Hardware Security Primitives: Cryptographic Hardware and their Implementation, Optimization of Cryptographic Hardware on FPGA, Algorithm to Hardware, Case Study AES or ECC accelerator; Side-Channel Attacks (SCA); Fault Injection Attacks; Hardware Countermeasures: Masking, hiding, and logic obfuscation techniques; Modern IC Design and Manufacturing Practices and Their Implications: Hardware Intellectual Property (IP) Piracy and IC Piracy, Design Techniques to Prevent IP and IC Piracy, Physical Unclonable Functions (PUFs); Hardware Trojans: Hardware Trojan Nomenclature and Operating Modes, Countermeasures Such as Design and Manufacturing Techniques to Prevent/Detect Hardware Trojans, Supply Chain Security: Hardware Trojans detection, reverse engineering, and IC camouflaging; Micro-architectural Security; Advanced topics like post-quantum hardware.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	3	2	3	3
CO2	3	3	3	3	2	3	3
CO3	3	3	2	3	2	3	2
CO4	3	3	2	3	2	3	3

Textbooks / References

1. Ahmad-Reza Sadeghi and David Naccache (eds.): Towards Hardware-intrinsic Security: Theory and Practice, Springer
2. Ted Huffmire et al: Handbook of FPGA Design Security
3. D. Mukhopadhyay and R. S. Chakraborty, “Hardware Security: Design, Threats and Safeguards”, CRC Press

ECO307 Robotics and Automation [3-0-0-3]

Course Objectives

- To introduce the basic concepts of robotics and its specifications.

- To introduce the different sensors and actuators used in robots.
- To introduce the various end effector configuration and their design considerations for a particular application.
- To impart knowledge of the basic kinematic model of robotic manipulators.
- To introduce the basics of control of robotic manipulators

Course Outcomes

- CO1: Describe the anatomy and specifications of Robots.
- CO2: Summarize different sensors and actuators for robots.
- CO3: Choose a robotic configuration and grippers for a particular application.
- CO4: Obtain basic kinematic model of robotic manipulators.
- CO5: Understand the basics of control of robotic manipulators

Syllabus

Introduction: Definition and origin of robotics, Robot Anatomy, Robot specifications, Robot characteristics - accuracy, precision, and repeatability, Classification of robots. Anatomy of a robotic manipulator- links, joints, actuators, sensors, end effector, controller.

Sensors in robotics: Sensor classification: Internal sensors Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors- Range, proximity, touch, force-torque sensing.

Actuators for robots: Actuators classification Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; End effectors classification, Mechanical grippers, Magnetic grippers, Vacuum grippers, Adhesive grippers, selection and design considerations of grippers in robot.

Robotic vision systems: Imaging, Sensing and Digitization, Image processing techniques.

Robot Kinematics: Homogenous coordinates and transfer representations. Kinematics chains. Introduction to forward kinematics and inverse kinematics.

Trajectory Planning, Introduction, Path vs. Trajectory, Joint-Space vs. Cartesian-Space Descriptions Basics of Trajectory Planning, Joint-Space Trajectory Planning: Third-Order Polynomial Trajectory Planning, Fifth-Order Polynomial Trajectory Planning, Linear Segments with Parabolic Blends, Linear Segments with Parabolic Blends and Via Points, Higher-Order Trajectories, Other Trajectories

Robot Programming: Programming methods, Robot language classification, Robot language structure, elements and its functions. Motion, End effector and Sensor commands in VAL programming language.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	1				3	
CO2	3	1				3	1
CO3	2	3				2	3
CO4	3	3				3	2
CO5	3	2				3	2

Textbooks / References

1. Mikell P. Groover et. al., “Industrial Robots Technology, Programming and Applications”, McGraw Hill, New York, 2008.
2. Saeed B. Niku, “Introduction to Robotics”, Pearson, 2011.
3. S K Saha “Introduction to Robotics”, Mc Graw Hill Education.
4. John J. Craig, “Introduction to Robotics”, Pearson, 2009.
5. R K Mittal and I J Nagrath, “Robotics and Control”, Tata McGraw Hill, New Delhi, 2003.
6. Spong & Vidyasagar, Robot Dynamics and Control, Mc Graw Hill.
7. Fu K S, Gonzalez R C, Lee C.S.G, “Robotics : Control, Sensing, Vision and Intelligence”, McGraw Hill, 1987

ECO308 Internet of Things [3-0-0-3]

Course Objectives

- To introduce the concept and the basics of IoT technologies
- To provide knowledge on various applications of IoT based technologies and their associated circuits
- To enable awareness on the different products that were designed based on IoT

Course Outcomes

- CO1: The students will have knowledge on various applications of IoT
- CO2: The students will have knowledge about sensors/actuators and edge-level devices
- CO3: The students will be capable to design or architect IoT based applications

Syllabus

Introduction - Sensor basics, sensing and actuation, basics of networking - wired, wireless, MANET, PAN, wireless and wired protocols. IoT Hardware and devices. IoT Communication Protocols - Higher Layer protocols – MQTT, SMQTT, CoAP, XMPP, AMQP, DDS, IEEE standards – 802.15.4, 802.15.1, Lower Layer protocols – Zigbee, 6LowPAN, Wifi, LoRaWAN, Bluetooth, RFID, NFC, HART, 4G/5G/6G, connectivity challenges.

IoT architectures and programming - basic architectures, IoT cloud platforms, Data processing mechanisms, scalability issues, visualization issues, analytics basics, utility of cloud computing, fog computing, and edge computing,

serverless platforms for IoT. Classification of IoT: Industrial, Commercial, Infrastructure, Transport, Healthcare, Agriculture and Environment, advanced IoT architectures - Golang and Nodejs programming.

IoT Programming and Devices – Arduino programming, Raspbian Micropython programming, Golang and Nodejs programming for services, Devices – IoT Gateways, Edge-level accelerators – Jetson Nano, Jetson Xavier, Raspberry pi, Espressif ESPs, FPGAs, Corel Devs. Applications - IoT for societal upliftment, industrial automation (Industry 4.0, 5.0), smart city, smart home, smart transportation, smart healthcare, smart agricultures, Edge-assisted intelligence, Fog-assisted intelligence. golang based implementation

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	3	1	3	3
CO2	3	2	2	3		3	3
CO3	3	3	3	3	1	3	3

Textbooks / References

1. Misra, Sudip, Anandarup Mukherjee, and Arijit Roy. *Introduction to IoT*. Cambridge University Press, 2021.
2. Pethuru Raj and Anupama C. Raman, *The Internet of Things: Enabling Technologies, Platforms, and Use Cases*, CRC Press, First edition, 2017
3. Honbu Zhou, *The Internet of Things in the Cloud: A Middleware Perspective*, CRC press, First edition, 2012
4. Arshdeep Bahga and Vijay Madisetti, *Internet of Things: A Hands-on Approach*, Universities Press, First edition, 2014
5. Mung Chiang, Bharath Balasubramanian, Flavio Bonomi, *Fog for 5G and IoT (Information and Communication Technology Series, Wiley series*, First edition, 2017
6. Alan A. A. Donovan, Brian W. Kernighan, *The Go Programming Language*, Addison-Wesley Professional Computing Series, First edition, 2015
7. Shajulin Benedict, *Edge Intelligence: Deep Learning-Enabled Edge Computing*, in IOP Publishers, UK (London), Hardback ISBN: 9780750355919, Ebook ISBN: 9780750355933 2024.

ECO309 Real Time Embedded Systems [3-0-0-3]

Course Objectives

- To understand processors and their instruction sets
- To understand hardware platform for embedded systems
- To design and analysis programs for embedded systems
- To understand distributed and multiprocessor embedded systems.

Course Outcomes

- CO1: Have understood the approach to develop assembly code for processors.

- CO2: Have learned to perform platform level performance analysis.
- CO3: Have understood develop applications using RTOS

Syllabus

Introduction to embedded computing - overview of embedded system design process -instruction sets of processors: ARM, PIC, TI C55x, TI C64x - programming I/O - modes and exceptions coprocessors - memory system - CPU performance-CPU power consumption

Basic computing platforms - CPU Bus - memory devices and systems - choosing a platform development environments - debugging consumer electronics architecture - platform-level performance analysis - design example: Audio Player

Components for embedded programs - models of programs - Assembly, linking, and loading compiler optimizations - program-level performance analysis - performance optimization - program-level energy optimization - optimizing program size - program validation and testing design example: Digital Still Camera

System design methodologies - requirements analysis - specifications - architecture design quality assurance - distributed embedded systems, shared-memory multiprocessors - design example: Video accelerator

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	2	2		3	3
CO2	3	3	3	3	1	3	3
CO3	3	3	3	3	1	3	3

Textbooks / References

1. Christopher Hallinan, “Embedded Linux Primer: A Practical Real-World Approach”, Second Edition, Prentice Hall, 2010.
2. Karim Yaghmour et al., “Building Embedded Linux Systems”, O’Reilly, 2008
3. Arnold S. Berger, “Embedded Systems Design: An Introduction to Processes, Tools, and Techniques”, CMP Books, 2001
4. David E. Simon, “An embedded Software Primer”, Addison-Wesley, 1999

ECO310 Wireless Sensor Networks [3-0-0-3]

Course Objectives

- To understand the basic WSN technology and supporting protocols
- Understand the physical layer and MAC layer design issues in WSN and the techniques to address them
- Learn key routing protocols for data dissemination and data gathering
- Learn the significance of localization techniques and its implementation

- Understand the key aspects of hardware design and application environment

Course Outcomes

- CO1: Understand the basic WSN technology, supporting protocols and architecture
- CO2: Identify the design issues in physical and MAC layers of WSN and the techniques to address them
- CO3: Design and develop routing protocols and localization techniques
- CO4: Understand the application environment and OS requirements in WSN

Syllabus

Introduction, WSN Architecture and Protocol Stack: Physical Layer, Data Link Layer, Network Layer, Transport Layer, Application Layer, WSN Applications, Factors Influencing WSN Design: Hardware Constraints, Fault Tolerance, Scalability, Production Costs, WSN Topology, Transmission Media, Power Consumption, Hardware Design Options.

Physical Layer: Physical Layer Technologies, RF, Other Techniques, Overview of RF Wireless Communication, Wireless Channel Effects, Attenuation, Multi-path Effects, Channel Error Rate, Unit Disc Graph vs. Statistical Channel Models, PHY Layer Standards, IEEE 802.15.4. Medium Access Control: Challenges for MAC, CSMA Mechanism, Contention-Based Medium Access-S-MAC, B-MAC, CC-MAC, Other Contention-Based MAC Protocols, Reservation-Based Medium Access, Hybrid Medium Access.

Network Layer: Challenges for Routing, Data-centric and Flat-Architecture Protocols, Hierarchical Protocols, Geographical Routing Protocols, QoS-Based Protocols.

Localization: Challenges in Localization, Ranging Techniques, Range-Based Localization Protocols, Range-Free Localization Protocols, Topology Management: Deployment, Power Control, Activity Scheduling, Clustering.

Application support: Data Aggregation, In-network processing, Security issues, Operating Systems and Execution Environment, Future Trends in Wireless Sensor Networks.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2		1		3	1
CO2	3	3	2	2		3	2
CO3	3	3	2	3		3	3
CO4	2	2	1	2	1	2	2

Textbooks / References

1. Wireless Sensor Networks, Akyildiz, Ian F., and Mehmet Can Vuran, John Wiley & Sons, 2010.
2. Protocols And Architectures for Wireless Sensor Networks, Holger Karl and Andrea Willig, John Wiley & Sons, 2018.
3. Adhoc Wireless Networks: Architectures and Protocols, C Siva Ram Murthy and B S Manoj, Prentice Hall India.

4. Wireless Sensor Networks: A Networking Perspective, Jun Sheng and Abbas Jamalipour, John Wiley & Sons.

ECO311 Optimization Techniques for Communication and Machine Learning [3-0-0-3]

Course Objectives

- To provide fundamental knowledge of convex optimization concepts, including vectors, norms, convex sets, convex functions, and optimization problem formulations.
- To develop the ability to analyze and solve optimization problems using techniques such as LP, QCQP, SOCP, duality, and KKT conditions.
- To apply convex optimization methods to wireless communication and signal processing applications including beamforming, MIMO, OFDM, channel estimation, and sparse signal processing.
- To introduce modern applications of convex optimization in machine learning, and radar systems.

Course Outcomes

- CO1: Understand the mathematical foundations of convex optimization and identify convex sets, convex functions, and optimization constraints.
- CO2: Formulate and solve convex optimization problems using LP, QCQP, SOCP, duality theory, and KKT conditions.
- CO3: Apply optimization techniques to wireless communication and signal processing problems such as beamforming, power allocation, MIMO/OFDM systems, and sparse signal recovery.
- CO4: Analyze and implement convex optimization approaches for applications in machine learning and radar signal processing.

Syllabus

Mathematical foundations: Introduction to vectors and vector spaces, properties of vectors, norms, positive semi-definite matrices, Gaussian random vectors. Introduction to convex optimization – convex sets, affine sets, hyperplanes/ half-spaces, convex hulls, application -power constraints in wireless communication systems.

Convex and concave functions: Definition and properties of convex and concave functions, first-order and second-order conditions for convexity, examples of convex functions, application – beamforming in wireless systems, multi-user wireless, cognitive radio systems.

Convex optimization problems: standard convex optimization formulation, linear programming, feasibility problems, application – power allocation in multi-cell cooperative orthogonal frequency division multiplexing (OFDM). Advanced convex optimization: quadratically constrained quadratic programming, second-order cone programming, application – channel shortening for wireless equalization, robust beamforming in wireless systems.

Duality and Karush–Kuhn–Tucker (KKT) conditions: Duality principle, Lagrangian optimization and KKT condition, application: water-filling power allocation, optimization for multi-input and multi-output (MIMO) systems, OFDM systems and MIMO-OFDM systems. Optimization for signal estimation: Least squares, weighted least squares, regularization techniques, application – wireless channel estimation, image reconstruction and deblurring. Convex optimization in machine learning: Convex optimization for machine learning, principal component analysis, support vector machines. Cooperative communication and geometric programming: Cooperative communication, optimal power allocation for cooperative communication, Geometric Program. Sparse signal processing and compressive sensing: Compressive sensing, sparse signal processing, Orthogonal Matching Pursuit, Least Absolute Shrinkage and Selection Operator for signal estimation. Radar signal processing and array processing: Radar target detection, array processing, multiple signal classification, MIMO-radar schemes for enhanced target detection.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	3	3	1		3	
CO2	3	3	3	2		3	
CO3	3	3	3	3		3	2
CO4	3	3	3	3		3	2

Textbooks / References

1. Stephen Boyd and Lieven Vandenberghe, “Convex optimization”, Cambridge university press, 2004.
2. Chong-Yung Chi, Wei-Chiang Li and Chia-Hsiang Lin, “Convex optimization for signal processing and communications: From fundamentals to applications”, CRC press, 2017.
3. D. P. Palomar and Y. C. Eldar, “Convex optimization in signal processing and communications”, Cambridge university press, 2010.
4. D. P. Bertsekas, A. Nedić and A. Ozdaglar, “Convex analysis and optimization”, Athena scientific, 2003.

ECO312 Applied Linear Algebra for Signal Processing, Data analytics and Machine Learning [3-0-0-3]

Course Objectives

- To provide a strong foundation in linear algebra, optimization, and dynamical systems for engineering and data-driven applications.
- To introduce machine learning and data analytics techniques such as regression, clustering, dimensionality reduction, and probabilistic modeling.
- To develop the ability to apply mathematical and computational tools for solving real-world problems in signal processing, data science, forecasting, and intelligent systems.

Course Outcomes

- CO1: Apply matrix theory, eigenvalue decomposition, SVD, and least squares techniques to engineering and data analytics problems.
- CO2: Analyze and implement machine learning methods including regression, clustering, perceptron learning, SVMs, and probabilistic models.
- CO3: Model and solve problems involving dynamical systems, sparse signal estimation, recommender systems, and Markov processes using computational techniques.

Syllabus

Introduction to vectors, properties and applications, Introduction to matrices and applications – Circuits, Graphs, Social Networks, Traffic flow, Eigenvalue decomposition, properties and applications – Principal Component Analysis (PCA), Eigenfaces for facial recognition, Singular Value Decomposition (SVD) and applications – Dimensionality reduction, data compression, recommendation systems, MUSIC algorithm.

Linear regression and Least Squares. Applications: System identification, linear regression, Support Vector Machines (SVM), kernel SVMs, Optimal linear MMSE estimation, market prediction and forecasting, ARMA models, data analytics: recommender systems, user rating prediction, NETFLIX problem. Structure of FFT/IFFT matrices, properties, applications of FFT in signal and image processing.

Modeling of dynamical systems – examples: robots, chemical plants. Solution of autonomous linear dynamical systems (LDS), solution of systems with inputs and outputs. Unsupervised learning: centroid-based clustering, probabilistic model-based clustering and EM algorithm, linear perceptron, training a perceptron using stochastic gradient methods.

Compressive sensing, orthogonal matching pursuit for sparse signal estimation, discrete-time Markov chains – applications: supply chain management, forecasting, operations research – resource and inventory management.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	2		3	2
CO2	3	3	2	3		2	3
CO3	3	3	3	2		2	3

Textbooks / References

1. Stephen Boyd and Lieven Vandenberghe, *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*, Cambridge University Press
2. Gilbert Strang, *Linear Algebra and Learning from Data*, Wellesley-Cambridge Press
3. Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer
4. Charu C. Aggarwal, *Applied Linear Algebra and Optimization for Machine Learning*, Springer

5. Simon Foucart and Holger Rauhut, *A First Course in Compressed Sensing*, Birkhäuser.

ECO313 Digital Image Processing [3-0-0-3]

Course Objectives

- To study the image fundamentals and selected mathematical transforms for image processing
- To apply image processing techniques for image enhancement, restoration, segmentation and recognition
- To provide understanding about image compression techniques and standards

Course Outcomes

- CO1: Learn image representation and transformation in digital domain
- CO2: Implement techniques for image enhancement, restoration, segmentation and recognition
- CO3: Assess image compression methods and standards

Syllabus

Digital Image Fundamentals: Steps in Digital Image Processing – Components – Elements of Visual Perception – Image Sensing and Acquisition – Image Sampling and Quantization – Relationships between pixels - Colour image fundamentals - RGB, HSI models, Two-dimensional mathematical preliminaries, 2D transforms - DFT, DCT
Image Enhancement: Spatial domain methods: point processing- intensity transformations, histogram processing, image subtraction, image averaging. Spatial filtering- smoothing filters, sharpening filters. Frequency domain methods- low pass filtering, high pass filtering, homomorphic filtering
Image Restoration: Degradation model, Linear Filtering- Inverse and Wiener, Nonlinear filtering- Mean, Median, Max and Min filters, Geometric transformations-spatial transformations
Image Segmentation: Edge detection, Edge linking via Hough transform – Thresholding - Region based segmentation – Region growing – Region splitting and merging – Morphological processing- erosion and dilation, Segmentation by morphological operations
Image Compression and Recognition: Need for data compression, Huffman, Run Length Encoding, Arithmetic coding, JPEG standard, Boundary representation, Boundary description, Fourier Descriptor, Regional Descriptors– Topological feature, Texture - Patterns and Pattern classes – Image Recognition

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	3	2		3	3
CO2	3	3	3	3	2	3	3
CO3	3	3	3	3	2	3	3

Textbooks / References

1. Rafael C. Gonzalez, Richard E. Woods, Steven Eddins, Digital Image Processing using MATLAB', Pearson Education
2. S Jayaraman, S Esakkirajan, T Veerakumar, Digital image processing, Tata Mc Graw Hill
3. Kenneth R Castleman , Digital image processing, Pearson Education, 2/e
4. Pratt William K , Digital Image Processing: , John Wiley, 4/e
5. Jain Anil K , Fundamentals of digital image processing, PHI

ECO314 Transformer Architectures [3-0-0-3]

Course Objectives

- The primary objective of this course is to provide students with the fundamental concepts and skills necessary to understand, analyse and apply transformer-based architectures.
- The course explains the principles of attention mechanisms, self-attention, and multi-head attention, and introduces encoder-only, decoder-only, and encoder-decoder models.
- Covers prompting, fine-tuning, and transformer applications in NLP, vision, and speech through practical assignments and projects.

Course Outcomes

- CO1: Understand the fundamentals of attention mechanisms, self-attention, multi-head attention, and positional encoding.
- CO2: Analyze transformer architectures and compare encoder-only, decoder-only, and encoder-decoder models by scalability, complexity, and task suitability. .
- CO3: Analyze GPT evolution and apply zero-shot, few-shot, and chain-of-thought prompting to domain-specific tasks. .
- CO4: Apply adapter, prefix-tuning, and LoRA fine-tuning; explore Vision Transformer, long-sequence, and multimodal architectures.
- CO5: Evaluate efficiency improvements, ethical considerations, and current research advances in transformer applications.

Syllabus

Introduction to Transformers and Attention Mechanisms: Motivation for Transformers, limitations of recurrent and convolutional neural networks in sequence modelling, need for parallel computation and long-range dependency capture, evolution from sequence-to-sequence models to attention-based architectures. Fundamentals of attention, query-key-value formulation, scaled dot-product attention, SoftMax weighting, and contextual representation learning.
Self-Attention, Multi-Head Attention and Positional Encoding: Self-attention mechanism and contextual embed-

ding generation, mathematical formulation and advantages over traditional architectures, multi-head attention with parallel attention heads, concatenation and linear projection. Positional encoding techniques including sinusoidal and learned embeddings, sequence representation and token embeddings.

Transformer Architecture and Core Components: Overall Transformer architecture, encoder-decoder framework, encoder and decoder block structures, masked self-attention and encoder-decoder (cross) attention, feed-forward neural networks, residual connections and layer normalization.

Types of Transformer Models: Encoder-only Transformer models with bidirectional self-attention and masked language modelling objectives, representative models such as BERT and RoBERTa and their applications. Decoder-only Transformer models with causal self-attention and autoregressive generation, GPT family evolution and applications. Encoder-decoder Transformer models for sequence-to-sequence learning, representative models such as T5 and BART.

Comparative Analysis and Extensions: Comparative study of encoder-only, decoder-only and encoder-decoder architectures with respect to attention directionality, computational complexity, scalability, training cost and task suitability. Fine-tuning strategies including adapters, prefix-tuning and LoRA, long-sequence Transformers, vision Transformers and multimodal Transformer architectures.

Applications and Recent Advances: Applications of Transformers in natural language processing tasks such as classification, sentiment analysis, question answering, text generation, summarization and dialogue systems, along with cross-domain applications in computer vision, speech recognition and scientific modelling. Recent advances including efficiency improvements, mixture-of-experts models, retrieval-augmented Transformers, ethical considerations, bias mitigation and open research challenges in AI systems.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	1				3	
CO2	3	3				3	1
CO3	3	2		1		3	2
CO4	2	2		3		2	3
CO5	2	3				2	1

Textbooks / References

1. Denis Rothman, Transformers for Natural Language Processing and Computer Vision, Second Edition, Packt Publishing, 2024.
2. Lewis Tunstall, Leandro von Werra, Thomas Wolf, Natural Language Processing with Transformers, First Edition, O'Reilly Media, 2022.
3. Palash Goyal, Sumit Pandey, Karan Jain, Generative AI with Large Language Models, First Edition, O'Reilly Media, 2023.

4. Jay Alamar, and Maarten Grootendorst. Hands-On Large Language Models: Language Understanding and Generation. O'Reilly Media, Inc., 2024.
5. Jurafsky, Daniel, and James H. Martin, Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, Draft 3rd Edition, 2024.
6. Amaratunga, Thimira, Understanding large language models: learning their underlying concepts and technologies, Apress Media, 2023.
7. Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, Illia Polosukhin, Attention is All You Need, Advances in Neural Information Processing Systems, 2017.
8. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, Second Edition, O'Reilly Media, 2019.
9. Jay Alamar, The Illustrated Transformer, Online Resource, 2018.
10. Hugging Face Team, Hugging Face Transformers Documentation, Online Resource.

ECO315 Tensor Computations for Data Science [3-0-0-3]

Course Objectives

- To introduce the fundamentals of tensor algebra, tensor operations and multilinear representations used in multidimensional data analysis.
- To study major tensor decomposition techniques including CP decomposition, Tucker decomposition and higher-order singular value decomposition (HOSVD).
- To understand the role of tensor decompositions in low-rank approximation and their applications in image processing and data analytics.

Course Outcomes

- CO1: Apply tensor operations and multilinear algebra concepts for representing multidimensional datasets.
- CO2: Analyze and implement tensor decomposition techniques such as CP, Tucker and HOSVD for dimensionality reduction and feature extraction.
- CO3: Utilize tensor-based methods for applications including image compression, image deblurring and anomaly detection in data science problems.

Syllabus

Basic concepts of matrix properties: norms, rank, trace, inner products, Kronecker product, similarity matrix, diagonalization of matrices, Toeplitz and circulant matrices with their properties. Introduction to Tensors: Tensors and tensor operations, Mode-n product of a tensor, Khatri-Rao product, outer product, tensor rank, symmetric tensor, orthogonal tensor. Tensor Decomposition: Block tensor decomposition,

Canonical Polyadic (CP) decomposition, Tucker decomposition, multilinear singular value (higher-order SVD) decomposition and tensor-train (TT) decomposition.

Applications of Tensor Decompositions: Image compression, image deblurring, background removal with robust principal tensor component analysis, anomaly detection using higher-order statistical moments.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	1	1		3	1
CO2	3	3	2	2		3	2
CO3	2	3	2	3		2	3

Textbooks / References

1. Tensor Decompositions for Data Science, Grey Ballard and Tamara G. Kolda, *Tensor Decompositions for Data Science*, Cambridge University Press, 2025.
2. Tensor Decompositions and Applications, Tamara G. Kolda and Brett W. Bader, *Tensor Decompositions and Applications*, SIAM Review, Vol. 51, No. 3, 2009
3. Tensor Networks for Dimensionality Reduction and Large-scale Optimization, Andrzej Cichocki, Danilo Mandic, Lieven De Lathauwer, Guoxu Zhou, Qibin Zhao, Cesar Caiafa and Anh Huy Phan, *Tensor Networks for Dimensionality Reduction and Large-scale Optimization*, Foundations and Trends in Machine Learning, 2016
4. Tensor Analysis: Spectral Theory and Special Tensors, Liqun Qi and Ziyang Luo, *Tensor Analysis: Spectral Theory and Special Tensors*, SIAM, 2017.

ECO401 Advanced Computer Architecture [3-0-0-3]

Course Objectives

- To understand performance foundations
- To explore ISA design principles
- To study instruction-level parallelism
- To analyze memory hierarchy
- To understand data-level parallelism

Course Outcomes

- CO1: Analyze and Evaluate Performance
- CO2: Design and Compare Instruction Set Architectures
- CO3: Apply ILP Techniques to Processor Design
- CO4: Design and Optimize Memory Hierarchies
- CO5: Classify and Analyze Parallel Architectures

Syllabus

Design Space Exploration :Performance metrics and performance enhancement techniques, Basic concepts of parallel processing and pipelining, Power dissipation in processors, power metrics, and low-power design techniques.

Instruction set architecture design: Instruction set design,

implementation and performance perspectives, relative advantages of RISC and CISC instruction set.

Instruction-level parallelism (ILP): Pipeline data-path, data-dependence. Challenges in ILP realization. Pipeline hazards and their solutions, out-of-order execution, branch prediction, and dynamic scheduling. VLIW and super-scalar processors.

Memory-Hierarchy Design: Multi-level cache design issues , Performance evaluation. Data Level Parallelism: Flynn Processor classification, SIMD, MIMD, GPU architecture

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	3		3	3	3
CO2	3	3	2			3	3
CO3	3	2	3	3		3	3
CO4	1	2	3			3	3
CO5	2	3	3			3	3

Textbooks / References

1. J. L. Hennessy and D. A. Patterson, Computer Architecture: A Quantitative Approach, Morgan Kaufmann, fourth edition, 2006.
2. Kai Hwang, Advanced Computer Architecture: Parallelism, Scalability, Programmability, McGraw-Hill, 3rd Ed, 2015
3. William Stallings, Computer Organization and Architecture, Prentice Hall, 10th edition, 2015, ISBN-10: 013293633X, ISBN-13: 978-0132936330

ECO402 Reconfigurable Computing [3-0-0-3]

Course Objectives

- To instruct students about the possibilities and rapidly growing interest in adaptive hardware and corresponding design techniques
- To provide the students the necessary knowledge for understanding and designing reconfigurable hardware systems
- To study the applications benefiting from dynamic hardware reconfiguration.
- To gain knowledge and understanding of the principles and practice in reconfigurable architecture and computing through class lectures and discussions, reading assignments, homework, and experiments.

Course Outcomes

- CO1: Understand the fundamentals of the reconfigurable computing and reconfigurable architectures
- CO2: Articulate the design issues involved in reconfigurable computing systems with a specific focus on FPGAs both in theoretical and application levels
- CO3: Understand the performance trade-offs involved in designing a reconfigurable computing platform with a specific focus on the architecture of a configurable logic block and the programmable interconnect

- CO4: Discuss the state-of-the-art reconfigurable computing architectures spanning fine grained to coarse grained architectures.
- CO5: Understand both how to architect reconfigurable systems and how to utilize them for solving challenging computational problems.

Syllabus

Introduction- General overview of computing models - History, state of the art, future trends - Basic concepts and related fields of study - Performance, power, Throughput and other metrics, Moore's Law, Technology Scaling, Power wall, memory wall.

Programmable logic devices, CPLD and FPGA

RC Architectures - Device characteristics - Fine-grained architectures - Coarse-grained architectures.

FPGA design flow, FPGA Physical Design Tools - Technology mapping - Placement and Routing - Register Transfer Level, Logic Synthesis - Controller/Data path synthesis - Logic minimization.

AMD Xilinx architecture- LUTs, CLBs, Block RAMs, DSP Slices.

RC Application Design - Parallelism - Pipelining - Optimizations - Bottlenecks - High-level Design - High-level synthesis - High-level languages – HDLs- Design tools, HW/SW Co-design; Implementations of digital systems using HDLs.

Case Studies: AI/ML hardware architectures

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2	1			3	1
CO2	3	2	3	2		3	2
CO3	3	3	2	1		3	3
CO4	3	2	3			3	3
CO5	3	3	3	3	2	3	3

Textbooks / References

1. Scott Hauck and André DeHon, Reconfigurable Computing: The theory and practice of FPGA-Based computation
2. P. Lysaght and W Rosenstiel, New Algorithms, Architectures and Applications for Reconfigurable Computing, Springer
3. Research papers from ISCA, FPT, FPL, FCCM, SC, CASES, ICCAD, DAC etc.
4. AMD Xilinx FPGA data sheets

ECO403 Computer Vision [3-0-0-3]

Course Objectives

- To introduce students the fundamentals of image formation;
- To introduce students the major ideas, methods, and techniques of computer vision and pattern recognition
- To develop an appreciation for various issues in the design of computer vision and object recognition systems

- To provide the student with programming experience from implementing computer vision and object recognition applications.

Course Outcomes

- CO1: Identify basic concepts, terminology, theories, models and methods in the field of computer vision
- CO2: Describe known principles of human visual system
- CO3: Describe basic methods of computer vision related to multi-scale representation, edge detection, detection of other primitives, stereo, motion and object recognition

Syllabus

Introduction to computer vision, geometric camera models, light and shading, local image features: SIFT, HOG, texture and shape descriptors, active contour, segmentation, deformable models, RANSAC, image registration, learning and classification strategies, image classification, object detection and recognition, stereopsis, tracking, applications.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	1				3	
CO2	3	1				3	
CO3	3	2				3	1

Textbooks / References

1. Forsyth and Ponce, *Computer vision: a modern approach*, 2nd Ed., Pearson, 2012.
2. Sonka, Hlavac and Boyle, "Digital image processing and computer vision," Cengage learning, 2008
3. Rick Szeliski, *Computer Vision: Algorithms and Applications*, Springer, 2011

ECO404 Software Defined Radio [3-0-0-3]

Course Objectives

- Understand requirements, benefits and different models for Software Defined Radio
- Understand the design, architecture, and implementation of cognitive radio systems
- Analyze Software Defined Radio architecture for performance optimization

Course Outcomes

- CO1: Analyze functioning of different blocks associated with Software Defined Radio
- CO2: Design and analyze circuits with multirate signalling
- CO3: Analyze Software Defined Radio architecture for performance optimization
- CO4: Design and analyze cognitive radio network and architecture

Syllabus

Software Defined Radio (SDR): Essential functions of SDR, SDR architecture, traditional radio implemented in hardware and SDR, transmitter architecture and its issues, A/D & D/A conversion, data converters, complex ADC and DAC architectures, reconfigurable wireless communication systems, benefits of multi-standard terminals, operational requirements, business models for software defined radio, new base station and network architectures, Cognitive Radio (CR) features and capabilities, CR functions and architecture, dynamic spectrum access, CR architecture for next generation networks, Digital aspects of SDR Spectrum sensing and identification, primary signal detection, energy detector, cyclostationary feature detector, matched filter, cooperative sensing, spectrum opportunity, spectrum opportunity detection, fundamental trade-offs: performance versus constraint, sensing accuracy versus sensing overhead, Flexible RF receiver architecture, implementation of a digital receiver: frequency up conversion using under sampling, achieving processing gain using oversampling, Noise figure, Receiver sensitivity Spectrum management of cognitive radio networks, spectrum decision, spectrum sharing and mobility, mobility management of heterogeneous wireless networks, Multiband Coverage Systems: Multiband Flexible receiver design, Diplexer, Image rejection, Dynamic range enhancement, feedback and feed forward techniques Terminal architecture of CRN, diversity radio access networks, routing in CRN, Control of CRN, Self organization in mobile communication networks, security in CRN, cooperative wireless networks, user cooperation and cognitive systems, Flexible transmitters and Power amplifiers: Analog quadrature up-conversion, Interpolated band pass up conversion, PLL based transmitters, Active All-pass filter

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	1	1	1	1	3	1
CO2	3	2	1	1	1	3	1
CO3	3	2	2	2	1	3	2
CO4	3	2	2	2	1	3	2

Textbooks / References

1. Jeffrey H. Reed “Software Radio: A Modern Approach to Radio Engineering”, Pearson Education Asia
2. P Kenington, “RF and Baseband Techniques for Software Defined Radio”, Artec House, 2005
3. Jouko Vanakka, “Digital Synthesizers and Transmitter for Software Radio”, Springer, 2005
4. Wally H. W. Tuttlebee, “Software Defined Radio: Baseband Technologies for 3G Handsets and Base stations”, John Wiley & sons, 2003.
5. Kwang-Cheng Chen and Ramjee Prasad, “Cognitive Radio Networks”, John Wiley & Sons, Ltd, 2009.
6. Alexander M. Wyglinski, Maziar Nekovee, and Y. Thomas Hou, “Cognitive Radio Communications and Networks - Principles and Practice”, Elsevier Inc., 2010.

ECO405 High Speed Networks [3-0-0-3]

Course Objectives

- To introduce the fundamental concepts, architectures, and protocols of modern communication networks.
- To understand the design and operation of wired, wireless, and mobile communication networks.
- To analyze routing, switching, congestion control, and quality of service mechanisms in advanced communication networks.
- To study emerging networking technologies including software-defined networking, network virtualization, and Internet of Things (IoT) networks.
- To explore next-generation communication networks and their applications in broadband, cloud, and intelligent communication systems.

Course Outcomes

- CO1: Explain the architecture, protocols, and layered models of modern communication networks and analyze their functionalities.
- CO2: Analyze routing, switching, congestion control, and quality of service mechanisms in wired and wireless communication networks.
- CO3: Evaluate the performance of wireless, mobile, sensor, and broadband communication networks using appropriate networking techniques and protocols.
- CO4: Analyze advanced networking technologies including software-defined networking, network virtualization, cloud networking, and IoT communication systems.
- CO5: Apply concepts of next-generation communication networks such as 5G/6G, network security, intelligent networking, and edge computing in modern communication applications.

Syllabus

Evolution of communication networks, network architectures and layered models, OSI and TCP/IP reference models, transmission media and switching techniques, circuit switching and packet switching, multiplexing techniques, network devices and protocols, IPv4 and IPv6 addressing, network performance metrics, network topologies and design principles.

Routing, Transport, and Quality of Service: Routing algorithms and protocols, distance vector and link state routing, intra-domain and inter-domain routing protocols, congestion control mechanisms, transport layer protocols: UDP and TCP, flow control and error control, quality of service (QoS) concepts, traffic engineering, queuing disciplines, MPLS networks, network performance analysis. Wireless and Mobile Communication Networks: Wireless LANs and IEEE 802.11 standards, cellular communication networks, mobile IP and handoff mechanisms, ad hoc and mesh networks, wireless sensor networks, vehicular communication networks, satellite and broadband wireless networks, Bluetooth and ZigBee technologies, network resource management in wireless systems.

Advanced Networking Technologies: Software-defined networking (SDN), network function virtualization (NFV), cloud networking architectures, data center networks, Internet of Things (IoT) communication protocols, machine-to-machine communication, edge and fog computing, network virtualization techniques, optical and high-speed communication networks, intelligent network management.

Next-Generation Communication Networks: 5G and beyond communication networks, 6G vision and enabling technologies, ultra-reliable low-latency communication (URLLC), massive machine-type communication (mMTC), network slicing, AI and machine learning in communication networks, cybersecurity in communication networks, blockchain for networking, green communication networks, future trends and applications.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	2	2				3	
CO2	2	3				3	
CO3	2	3				3	
CO4	2	3				3	
CO5	2	3				3	

Textbooks / References

1. James F. Kurose and Keith W. Ross, Computer Networking: A Top-Down Approach, Pearson Education, 8th Edition, 2021.
2. William Stallings, Data and Computer Communications, Pearson Education, 11th Edition, 2017.
3. Vijay K. Garg, Wireless Communications and Networking, Morgan Kaufmann Publishers, 2007.
4. Alberto Leon-Garcia and Indra Widjaja, Communication Networks: Fundamental Concepts and Key Architectures, McGraw-Hill Education, 2nd Edition, 2004.
5. Thomas D. Nadeau and Ken Gray, SDN: Software Defined Networks, O'Reilly Media, 2013.
6. Erik Dahlman, Stefan Parkvall and Johan Skold, 5G NR: The Next Generation Wireless Access Technology, Academic Press, 2nd Edition, 2020.
7. Andrew S. Tanenbaum and David J. Wetherall, Computer Networks, Pearson Education, 5th Edition, 2011.

ECO316 Natural Language Processing [3-0-0-3]

Course Objectives

- To introduce the fundamental concepts, challenges, and linguistic foundations of Natural Language Processing
- To develop a strong theoretical and practical understanding of statistical and neural approaches to language modeling
- To equip students with the ability to design, analyze, and implement NLP systems for real-world applications, leveraging advanced models such as RNNs,

LSTMs, sequence-to-sequence models, transformers, and large language models.

Course Outcomes

- CO1: Analyze and apply core NLP techniques for text processing, syntactic parsing, and probabilistic modeling using tools such as HMMs, CFGs, CYK algorithms, and PCFGs.
- CO2: Implement and evaluate word representation and language modeling techniques, including TF-IDF, word embeddings, N-gram models, and neural language models based on deep learning architectures.
- CO3: Design and develop end-to-end NLP solutions for real-world applications, such as text classification, machine translation, speech processing, and information extraction, using state-of-the-art transformer-based and large language models.

Syllabus

Introduction to NLP: Definition, goal, challenges, stages of NLP. Words and Tokens: Tokenization, Noisy channel model of communication, Edit distance algorithm for spell correction, Type to Token Ratio, Basics of text processing: Normalization, Stemming and Lemmatization. POS tagging, Hidden Markov Model for POS tagging, Syntax analysis, Context-Free Grammar (CFG) and Cocke-Younger-Kasami (CYK) algorithms, Overview of Probabilistic Context-Free Grammar (PCFG).

Representation of meaning: Lexical semantics and wordnet, Distributional semantics, Count vectorization, co-occurrence matrix, TF-IDF, neural word embedding- skip-gram method and CBOW method.

Language models: Statistical Language Models and N-Grams, Counting n-gram probability, Smoothing techniques. Neural Language Models-Deep learning for NLP, overview of RNN, LSTM, Encoder decoder models, Attention mechanism. Transformers, Masked Language Models and BERT, Causal Language Models and GPT, Introduction to Large Language Models, Transformer models for speech.

Real world Applications: Text classification, entity linking, information/relation extraction, machine translation, Automatic speech recognition, text-to-speech, Report generation from signal logs, Edge and on-device NLP

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3	2	2		2	2
CO2	3	3	2	3		2	3
CO3	2	3	3	3		2	3

Textbooks / References

1. Jurafsky, Dan, and James H. Martin. *Speech and language processing*. 3rd Edition Pearson Education, 2024
2. Manning, Christopher D., and Hinrich Schütze. *Foundations of statistical natural language processing*. MIT Press, 1999
3. Bird, Steven, Ewan Klein, and Edward Loper. *Natural language processing with Python*. O'Reilly Media, 2009

4. Tunstall, Lewis, Leandro von Werra, and Thomas Wolf. *Natural language processing with transformers*. O'Reilly Media, 2022
5. Eisenstein, Jacob. *Introduction to natural language processing*. MIT Press (Indian reprint), 2019
6. Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. *Deep learning*. MIT Press (Indian reprint), 2016

ECO406 Wireless Ad-hoc Networks [3-0-0-3]

Course Objectives

- To understand the architecture, characteristics, and challenges of wireless ad hoc and sensor networks.
- To study routing, MAC, transport, and security mechanisms in infrastructure-less wireless networks.
- To analyze protocols, mobility management, QoS, and energy-efficient communication techniques for ad hoc environments

Course Outcomes

- CO1: Explain the architecture, characteristics, and applications of wireless ad hoc networks.
- CO2: Analyze and compare routing, MAC, transport, and multicast protocols used in MANETs.
- CO3: Evaluate QoS, security, and energy-efficient mechanisms in wireless ad hoc networks

Syllabus

Introduction to ad hoc wireless networks, cellular and ad-hoc wireless networks, applications of ad hoc wireless networks, issues in ad hoc wireless networks, medium access schemes, routing issues, transport layer issues, QoS provisioning, energy management, security issues. Design issues in MAC protocols, goals and classification of MAC protocols, contention-based protocols, IEEE 802.11 MAC protocol, MACA, MACAW, floor acquisition multiple access (FAMA), Bluetooth MAC protocol, MAC protocols using directional antennas, power-aware MAC protocols. Issues in designing routing protocols, classification of routing protocols, table-driven routing protocols: DSDV, WRP, CGSR; on-demand routing protocols: DSR, AODV, TORA; hybrid routing protocols: ZRP; multicast routing protocols, hierarchical routing protocols, QoS-aware routing protocols. Issues in designing transport layer protocols, classification of transport layer solutions, TCP over ad hoc wireless networks, TCP feedback, TCP with explicit link failure notification, ad hoc transport protocol (ATP), QoS in ad hoc wireless networks, QoS routing, ticket-based probing, QoS frameworks.

Need for energy management, classification of energy management schemes, battery management schemes, transmission power management schemes, system power management schemes, security in ad hoc wireless networks, network security requirements, issues and challenges in security provisioning, network security attacks, secure routing protocols.

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	2		1		3	1
CO2	3	3	2	2		3	2
CO3	2	3	2	2	3	2	2

Textbooks / References

1. C. Siva Ram Murthy and B. S. Manoj, *Ad Hoc Wireless Networks: Architectures and Protocols*, Pearson Education.
2. Carlos de Moraes Cordeiro and Dharma Prakash Agrawal, *Ad Hoc and Sensor Networks: Theory and Applications*, World Scientific.
3. Mohammad Ilyas, *The Handbook of Ad Hoc Wireless Networks*, CRC Press
4. Stefano Basagni, Marco Conti, Silvia Giordano, and Ivan Stojmenovic, *Mobile Ad Hoc Networking*, Wiley-IEEE Press.
5. Kazem Sohraby, Daniel Minoli, and Taieb Znati, *Wireless Sensor Networks: Technology, Protocols, and Applications*, Wiley.

ECP411 B.Tech Project Phase I [0-0-6-3]

Course Objectives

- To enable students to identify and formulate a relevant engineering problem through a comprehensive literature survey and problem analysis.
- To develop the ability to design, implement, and validate innovative hardware and/or software solutions using modern engineering tools and techniques.
- To cultivate skills in experimentation, data analysis, project management, teamwork, technical documentation, and professional communication.

Course Outcomes

- CO1: Identify, analyze, and formulate a complex engineering problem by applying knowledge of mathematics, science, and engineering fundamentals.
- CO2: Design, develop, and implement an appropriate hardware/software system or process to address the identified problem using modern engineering tools and methodologies.
- CO3: Conduct experiments, analyze and interpret data, evaluate system performance, and draw valid conclusions based on the obtained results.
- CO4: Prepare technical reports and research documentation, and effectively communicate project outcomes through oral presentations while demonstrating professional and ethical practices

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3		2	1	3	2
CO2	3	3	3	3	1	3	3
CO3	3	3	3	3	1	3	3
CO4	2	1	2	3		2	2

ECP421 B.Tech Project Phase II [0-0-18-9]

Course Objectives

- To enable students to identify and formulate a relevant engineering problem through a comprehensive literature survey and problem analysis.
- To develop the ability to design, implement, and validate innovative hardware and/or software solutions using modern engineering tools and techniques.
- To cultivate skills in experimentation, data analysis, project management, teamwork, technical documentation, and professional communication.

Course Outcomes

- CO1: Identify, analyze, and formulate a complex engineering problem by applying knowledge of mathematics, science, and engineering fundamentals.
- CO2: Design, develop, and implement an appropriate hardware/software system or process to address the identified problem using modern engineering tools and methodologies.
- CO3: Conduct experiments, analyze and interpret data, evaluate system performance, and draw valid conclusions based on the obtained results.
- CO4: Prepare technical reports and research documentation, and effectively communicate project outcomes through oral presentations while demonstrating professional and ethical practices

Course Outcome – PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2
CO1	3	3		2	1	3	2
CO2	3	3	3	3	1	3	3
CO3	3	3	3	3	1	3	3
CO4	3	3	3	3	1	3	3